Where irrigation exists is globally contested

4. Main analysis

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Contents

1 Preliminary

```r
# Load libraries -----------------------------------------------------------

sensobol::load_packages(c("data.table", "scales", "cowplot", "tidyverse", "sensobol",
                          "here", "terra", "rnaturalearth", "sf", "countrycode",
                          "wesanderson", "benchmarkme", "parallel"))

# Create custom theme -----------------------------------------------------

theme_AP <- function() {
  theme_bw() +
    theme(panel.grid.major = element_blank(),
          panel.grid.minor = element_blank(),
          legend.background = element_rect(fill = "transparent", color = NA),
          legend.key = element_rect(fill = "transparent", color = NA),
          legend.key.width = unit(0.4, "cm"),
          legend.key.height = unit(0.5, "lines"),
          legend.key.spacing.y = unit(0, "lines"),
          legend.box.spacing = unit(0, "pt"),
          legend.spacing.y  = unit(0.1, "cm"),
          legend.text = element_text(size = 6),
          legend.title = element_text(size = 7),
          axis.text.x = element_text(size = 7),
          axis.text.y = element_text(size = 7),
          axis.title.x = element_text(size = 7.3),
          axis.title.y = element_text(size = 7.3),
          axis.title = element_text(size = 10),
          plot.title = element_text(size = 8),
          strip.text.x = element_text(size = 7.4),
          strip.text.y = element_text(size = 7.4),
          strip.background = element_rect(fill = "white"),
          strip.text = element_text(margin = margin(t = 1.5, b = 1.5)))
}


# Source all .R files in the "functions" folder -------------------------------

r_functions <- list.files(path = here("functions"),
                          pattern = "\\.R$", full.names = TRUE)

lapply(r_functions, source)

# Set seed ------------------------------------------------------------------

seed <- 123
```

```r
# Load countries as sf (medium scale is a good balance) -----------------------

countries_sf <- ne_countries(scale = "medium", returnclass = "sf")

# Keep only what is needed -----------------------------------------------------

countries_sf <- countries_sf[, c("name", "geometry")]

# Define colors and palettes and plotting datasets -----------------------------

tau_palette <- c("Agreement"= "#2ECC71",
                 "Low disagreement" = "#F7DC6F",
                 "Moderate disagreement" = "#F39C12",
                 "High disagreement" = "#E74C3C",
                 "Extreme disagreement" = "red")

res_palette <- c("0.2deg" = "lightblue",
                 "0.4deg" = "#117A65",
                 "1deg"   = "darkblue")
```

2 Uncertainty and sensitivity analysis

```
# LOAD ALL DATASETS ############################################################

# List all resolution files ----------------------------------------------------

files <- list.files("./datasets/irrigated_areas_regridded",
                    pattern = "irrigated_areas_regridded_.*\\.csv",
                    full.names = TRUE)

# Load them, bind them and label them ------------------------------------------

dt <- rbindlist(lapply(files, function(f) {
  x <- fread(f)
  res <- sub(".*_(\\d+)\\.csv", "\\1", f)
  x[, resolution:= paste0(as.numeric(res) / 10, "deg")]
  x
}))

# Load cropland mask -----------------------------------------------------------

crop_path  <- "./irrigated_area_datasets/GirrEO/asap_mask_crop_v02.tif"
crop_native <- rast(crop_path)

# Retrieve only cells within cropland ------------------------------------------

dt <- filter_long_dt_by_crop_mask_aggregated(dt = dt, crop_native = crop_native,
                                             rule = "any")

# DEFINE TAU GRID ##############################################################

# Vector with the datasets used ------------------------------------------------

dataset_names <- unique(dt$dataset)
new_dataset_names <- c("GAEZ+2015", "GIAM", "GMIA", "GRIPC", "LUH2", "Meier",
                       "MIRCA 2000", "MIRCA OS", "Nagaraj", "SPAM2010")

# Define a dense grid of tau values from 1 ha (10^0) to 1e5 ha (10^5) ----------

log_tau_min <- 0
log_tau_max <- 5
log_step <- 0.1

taus_ha_nonzero <- 10^(seq(log_tau_min, log_tau_max, by = log_step))
taus_ha <- c(0, taus_ha_nonzero)
taus_mha <- taus_ha * 1e-6
```

```r
# Labels: "any_positive", then "1ha", "1.26ha", ..., "1e+05ha"---

tau_labels <- ifelse(taus_ha == 0, "any_positive", sprintf("%gha", signif(taus_ha, 3)))

# Define tau grid ------------------------------------

tau_grid <- data.table(tau_label = tau_labels, tau_ha = taus_ha, tau_mha = taus_mha)

# TRANSFORM THE SAMPLE MATRIX ##################################################

# Define the settings -------------------------------------------------------

N <- 2^13
params <- c("resolution", "tau", "exclusion")
matrices <- c("A", "B", "AB")
R <- 10^3
boot <- TRUE

# Resolution vector ----------------------------------------------------------

resolution_vec <- c("0.2deg", "0.4deg", "1deg")
n_resolution_vec <- length(resolution_vec)

# Tau vector ------------------------------------------------------------------

tau_vec <- tau_grid$tau_ha
n_tau_vec <- length(tau_vec)

# Dataset exclusion vector ---------------------------------------------------

dataset_vec <- c("all", dataset_names)
n_dataset_vec <- length(dataset_vec)

# Create the sample matrix ---------------------------------------------------

mat <- data.table(sobol_matrices(matrices = matrices, N = N, params = params))

# Transform resolution column ------------------------------------------------

mat[, resolution:= pmin(n_resolution_vec, 1L + floor(resolution * n_resolution_vec))]
mat[, resolution:= resolution_vec[resolution]]

# Transform tau column ---------------------------------------------------------

mat[, tau:= pmin(n_tau_vec, 1L + floor(tau * n_tau_vec))]
mat[, tau:= tau_vec[tau]]
```

```r
# Transform exclusion column ----------------------------------------------

mat[, exclusion:= pmin(n_dataset_vec, 1L + floor(exclusion * n_dataset_vec))]
mat[, exclusion:= dataset_vec[exclusion]]

# PRE-COMPUTE ALL COMBINATIONS #################################################

# Compute all possibilities by looping over resolutions and tau values ---------

# Set parallel backend -------------------------

resolutions <- sort(unique(dt$resolution))
ncores <- max(1, parallel::detectCores() - 1)

# loop over resolutions and scenarios in parallel

tau_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]

    # datasets actually present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    rbindlist(
      lapply(dataset_vec, function(scn) {

        # which datasets to use ---------------------------------------------

        if (scn == "all") {

          use_ds <- ds_res

        } else {

          use_ds <- setdiff(ds_res, scn)
        }

        # if only one dataset left, skip --------------------------------------
        if (length(use_ds) < 2L) {

          return(NULL)
        }

        # reshape to wide ------------------------------------------------------

        dt_res_wide <- dcast(dt_res_long[dataset %in% use_ds],
```

```r
          lon + lat + country + code + continent ~ dataset,
          value.var = "mha", fill = 0
        )

        dt_res_wide[, `:=`(A_min = do.call(pmin, c(.SD, list(na.rm = TRUE))),
                           A_max = do.call(pmax, c(.SD, list(na.rm = TRUE)))),
           .SDcols = use_ds]

        # compute tau ----------------------------------------------------------

        out <- tau_grid[, compute_tau_fun(dt_res_wide, tau_mha, use_ds),
                        .(tau_label, tau_ha, tau_mha)]

        # tag with resolution + scenario ---------------------------------------

        out[, `:=`(resolution = res, scenario = scn)]

        out
      }),
      use.names = TRUE, fill = TRUE
    )
  }, mc.cores = ncores),
  use.names = TRUE, fill = TRUE)

# ARRANGE OUTPUT ################################################################

# Rename tau_ha to tau to match mat --------------------------------------------

setnames(tau_results, "tau_ha", "tau")

# Join all three disagreement metrics onto mat ---------------------------------

mat[tau_results, on = .(resolution, tau, exclusion = scenario),
    `:=`(frac_disagree = i.frac_disagree,
         frac_major_disagree = i.frac_major_disagree,
         frac_minor_disagree = i.frac_minor_disagree,
         share_existence = i.share_existence,
         share_marginal = i.share_marginal)]

# Reshape just these three for plotting ----------------------------------------

mat_long <- melt(mat, id.vars = c("resolution", "tau", "exclusion"),
                 measure.vars = c("frac_disagree", "frac_major_disagree",
                                  "frac_minor_disagree", "share_existence",
                                  "share_marginal"),
                 variable.name = "agreement", value.name = "frac_value")
```

2.1 Composite figure

```r
# PREPARE DATA TO PLOT UNCERTAINTY ##############################################

# Calculate mean and quantiles by tau × agreement ------------------------------

mat_sum <- mat_long[, .(frac_mean = mean(frac_value, na.rm = TRUE),
                        frac_median = median(frac_value, na.rm = TRUE),
                        frac_lwr = quantile(frac_value, 0.025, na.rm = TRUE),
                        frac_upr = quantile(frac_value, 0.975, na.rm = TRUE)),
                    .(tau, agreement)]

# Get values at tau = 0, 1, 10, 500, 1000, 1600, 2000, 5000, 10000, 12000 ------

target_tau <- c(0, 1, 10, 500, 1000, 1600, 2000, 5000, 10000, 12000)

mat_sum[, {dt_sub <- .SD
idx <- sapply(target_tau, function(x) {
  which.min(abs(log10(tau + 1e-12) - log10(x + 1e-12)))
      }
    )
    dt_sub[idx]
  },
  agreement
] %>%
  .[, .(agreement, tau, frac_mean, frac_lwr, frac_upr)]
```

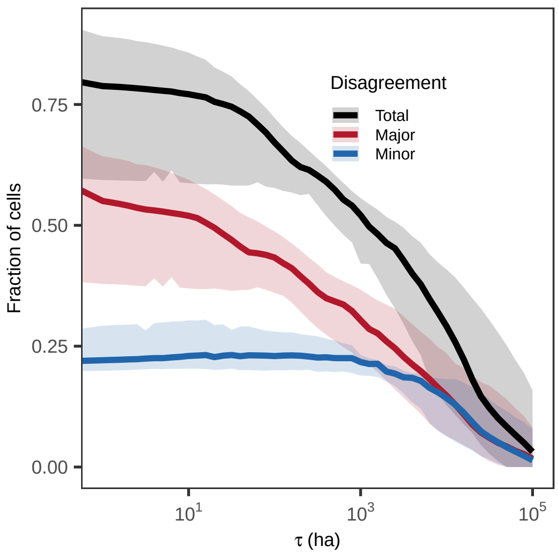
```
##                agreement        tau    frac_mean     frac_lwr     frac_upr
##                   <fctr>      <num>        <num>        <num>        <num>
##  1:        frac_disagree     0.0000 0.7725147940 0.5965542885 0.9048508038
##  2:        frac_disagree     1.0000 0.7641095924 0.5935795736 0.8913352054
##  3:        frac_disagree    10.0000 0.7458169499 0.5872583044 0.8574304934
##  4:        frac_disagree   501.1872 0.5696037035 0.4990357022 0.6046632124
##  5:        frac_disagree  1000.0000 0.5101819636 0.4207578610 0.5563956371
##  6:        frac_disagree  1584.8932 0.4773297398 0.3893680378 0.5323500248
##  7:        frac_disagree  1995.2623 0.4499272243 0.3558259018 0.5173525037
##  8:        frac_disagree  5011.8723 0.3573094252 0.2308683309 0.4640555280
##  9:        frac_disagree 10000.0000 0.2758302271 0.1276267473 0.4086514626
## 10:        frac_disagree 12589.2541 0.2498516315 0.1079319283 0.3922905305
## 11: frac_major_disagree     0.0000 0.5459749341 0.3824987605 0.6639383467
## 12: frac_major_disagree     1.0000 0.5345828870 0.3787803669 0.6428472244
## 13: frac_major_disagree    10.0000 0.5066465761 0.3694843827 0.5942080415
## 14: frac_major_disagree   501.1872 0.3422024147 0.2610007097 0.3931581557
## 15: frac_major_disagree  1000.0000 0.2967305803 0.2158021415 0.3670054536
## 16: frac_major_disagree  1584.8932 0.2702110404 0.1959761780 0.3446950917
## 17: frac_major_disagree  1995.2623 0.2537648037 0.1781096677 0.3362667328
## 18: frac_major_disagree  5011.8723 0.1922679985 0.1100610979 0.2802429351
## 19: frac_major_disagree 10000.0000 0.1453529051 0.0641605209 0.2356222112
```

```
## 20: frac_major_disagree 12589.2541 0.1288693376 0.0554201253 0.2146752603
## 21: frac_minor_disagree     0.0000 0.2265398599 0.1985622211 0.2863424569
## 22: frac_minor_disagree     1.0000 0.2295267055 0.1991819534 0.2922131021
## 23: frac_minor_disagree    10.0000 0.2391703737 0.2035200793 0.3034221002
## 24: frac_minor_disagree   501.1872 0.2274012888 0.1963311849 0.2613710001
## 25: frac_minor_disagree  1000.0000 0.2134513833 0.1893901834 0.2318943438
## 26: frac_minor_disagree  1584.8932 0.2071186994 0.1860436292 0.2213102108
## 27: frac_minor_disagree  1995.2623 0.1961624206 0.1772688000 0.2109764073
## 28: frac_minor_disagree  5011.8723 0.1650414267 0.1208072330 0.1941001487
## 29: frac_minor_disagree 10000.0000 0.1304773220 0.0634662264 0.1818294497
## 30: frac_minor_disagree 12589.2541 0.1209822939 0.0525118030 0.1823252355
## 31:     share_existence     0.0000 1.0000000000 1.0000000000 1.0000000000
## 32:     share_existence     1.0000 0.9998842092 0.9993735644 1.0000000000
## 33:     share_existence    10.0000 0.9987395427 0.9958634953 0.9994211288
## 34:     share_existence   501.1872 0.9265090021 0.8442199776 0.9606578698
## 35:     share_existence  1000.0000 0.8878074738 0.7835751053 0.9334183673
## 36:     share_existence  1584.8932 0.8549958612 0.7312409812 0.9107164472
## 37:     share_existence  1995.2623 0.8301866190 0.6886404631 0.8912397072
## 38:     share_existence  5011.8723 0.7100854672 0.4888273553 0.7785060526
## 39:     share_existence 10000.0000 0.5726446689 0.2750825646 0.6224474129
## 40:     share_existence 12589.2541 0.5234532434 0.2116215677 0.5748619682
## 41:      share_marginal     0.0000 0.0000000000 0.0000000000 0.0000000000
## 42:      share_marginal     1.0000 0.0001157908 0.0000000000 0.0006264356
## 43:      share_marginal    10.0000 0.0012604573 0.0005788712 0.0041365047
## 44:      share_marginal   501.1872 0.0734909979 0.0393421302 0.1557800224
## 45:      share_marginal  1000.0000 0.1121925262 0.0665816327 0.2164248947
## 46:      share_marginal  1584.8932 0.1450041388 0.0892835528 0.2687590188
## 47:      share_marginal  1995.2623 0.1698133810 0.1087602928 0.3113595369
## 48:      share_marginal  5011.8723 0.2899145328 0.2214939474 0.5111726447
## 49:      share_marginal 10000.0000 0.4273553311 0.3775525871 0.7249174354
## 50:      share_marginal 12589.2541 0.4765467566 0.4251380318 0.7883784323
##           agreement        tau    frac_mean     frac_lwr     frac_upr
##              <fctr>      <num>        <num>        <num>        <num>
```

```r
# Plot uncertainty ------------------------------------------------------------

agreement_cols <- c(frac_disagree = "black",
                    frac_major_disagree = "#B2182B",
                    frac_minor_disagree = "#2166AC"  )

plot_uncertainty1 <- mat_sum[!agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(.,
  aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_colour_manual(values = agreement_cols,
                      breaks = c("frac_disagree",
```

```r
                                  "frac_major_disagree",
                                  "frac_minor_disagree"),
                       labels = c("Total",
                                  "Major",
                                  "Minor"),
                       name = "Disagreement") +
  scale_fill_manual(values = agreement_cols,
                     breaks = c("frac_disagree",
                                "frac_major_disagree",
                                "frac_minor_disagree"),
                     labels = c("Total",
                                "Major",
                                "Minor"),
                     name = "Disagreement") +
  labs(x = expression(tau~"(ha)"), y = "Fraction of cells") +
  theme_AP() +
  theme(legend.position = c(0.65, 0.77))

plot_uncertainty1
```

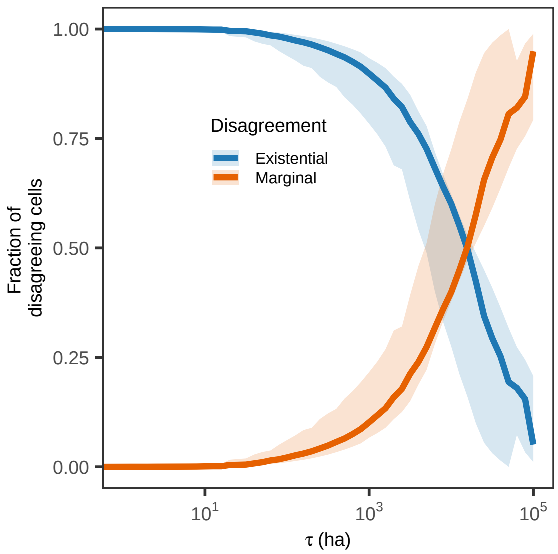
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```




```r
# Set colors for existential and marginal disagreement -------------------------

exist_cols <- c("share_existence" = "#1f78b4", "share_marginal" = "#e66101")

plot_agreement1 <- mat_sum[agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(., aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = exist_cols,
```

```
                      labels = c(share_existence = "Existential",
                                 share_marginal = "Marginal"),
                      name = "Disagreement") +
  scale_fill_manual(values = exist_cols,
                    labels = c(share_existence = "Existential",
                               share_marginal = "Marginal"),
                    name = "Disagreement") +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"), y = "Fraction of \ndisagreeing cells") +
  theme_AP() +
  theme(legend.position = c(0.37, 0.7))


plot_agreement1
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

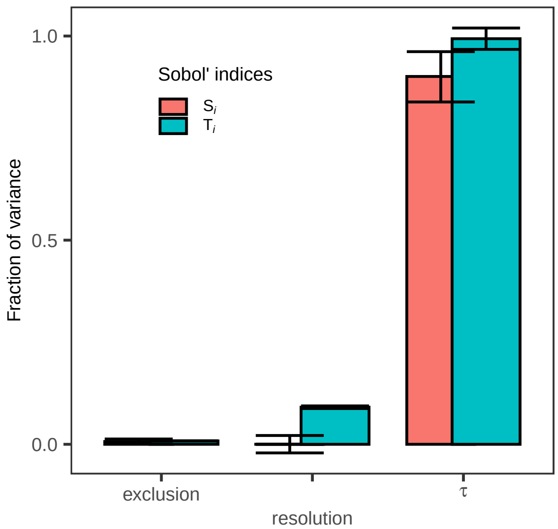



```
# Around tau > 10^4 ha, the existential and marginal components intersect. That
# intersection has a clear conceptual meaning:
# -It is the tolerance level at which disagreement about whether irrigation exists
# at all becomes less dominant than disagreement about how much irrigation exists.
# -Still small relative to grid-cell area. 1000 ha is only 2-3% of a 0.2º grid cell.
# That makes it a conservative tolerance as we are not allowing massive within-cell
# divergence.

# SENSITIVITY ANALYSIS ##########################################################

# Sobol' indices ----------------------------------------------------------------

ind <- sobol_indices(matrices = matrices, N = N, params = params, Y
                     = mat[, frac_disagree], R = R, boot = TRUE)
```

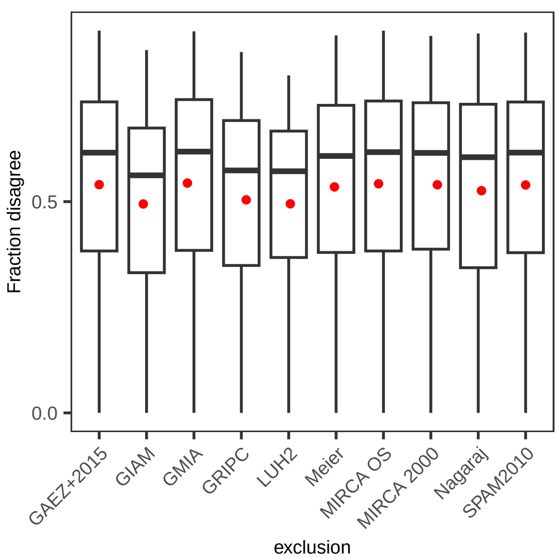
```r
print(ind)
```

```
##
## First-order estimator: saltelli | Total-order estimator: jansen
##
## Total number of model runs: 40960
##
## Sum of first order indices: 0.9079082
##        original          bias    std.error        low.ci     high.ci
##           <num>         <num>        <num>         <num>       <num>
## 1: 0.0001506532 -3.275755e-05 0.0108794512 -0.0211399217 0.021506743
## 2: 0.9010276594  9.404230e-04 0.0314407547  0.8384644897 0.961709983
## 3: 0.0067298400 -2.710929e-06 0.0031939823  0.0004724605 0.012992641
## 4: 0.0907715303  2.194818e-06 0.0016776382  0.0874812249 0.094057446
## 5: 0.9935216113  6.024186e-05 0.0133459914  0.9673037070 1.019619032
## 6: 0.0083001208  8.542363e-06 0.0001909142  0.0079173935 0.008665763
##    sensitivity parameters
##         <char>     <char>
## 1:          Si resolution
## 2:          Si        tau
## 3:          Si  exclusion
## 4:          Ti resolution
## 5:          Ti        tau
## 6:          Ti  exclusion
```

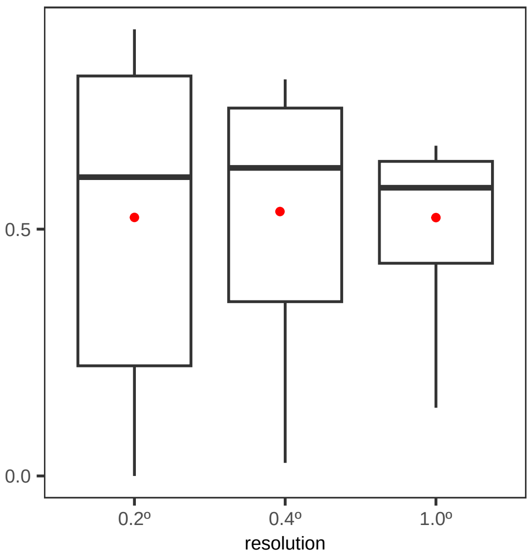
```r
plot_indices1 <- plot(ind) +
  labs(x = "", y = "Fraction of variance") +
  theme_AP() +
  scale_x_discrete(labels = c(exclusion = "exclusion",
                              resolution = "resolution",
                              tau = expression(tau)),
                   guide = guide_axis(n.dodge = 2)) +
  theme(legend.position = c(0.3, 0.8))

plot_indices1
```




```r
# SCATTERPLOTS OF OUTPUT AGAINS INTPUT ##########################################

label_map <- setNames(new_dataset_names, dataset_names)

a1 <- ggplot(mat[!exclusion == "all", c("exclusion", "frac_disagree")],
             aes(exclusion, frac_disagree)) +
  geom_boxplot() +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  scale_x_discrete(labels = label_map) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "exclusion", y = "Fraction disagree")

a1
```

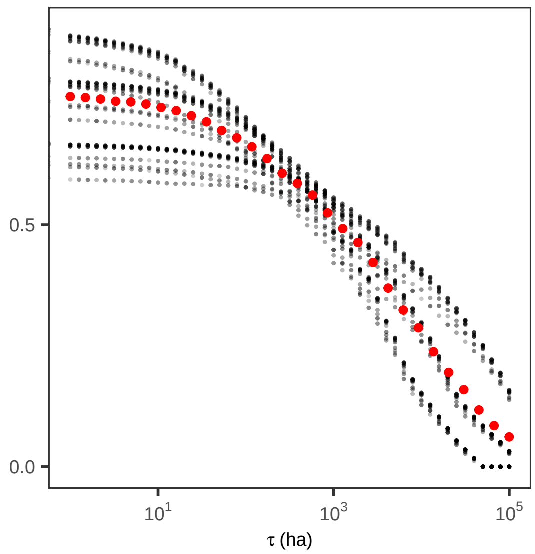


```r
b1 <- ggplot(mat[1:N, c("resolution", "frac_disagree")],
       aes(resolution, frac_disagree)) +
  geom_boxplot() +
  scale_x_discrete(labels = c(`0.2deg` = "0.2º",
                              `0.4deg` = "0.4º",
                              `1deg` = "1.0º")) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  theme_AP() +
  labs(x = "resolution", y = "")

b1
```




```r
c1 <- ggplot(mat[1:N, c("tau", "frac_disagree")],
       aes(tau, frac_disagree)) +
  geom_point(size = 0.1, alpha = 0.1) +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"), y = "")

c1
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```



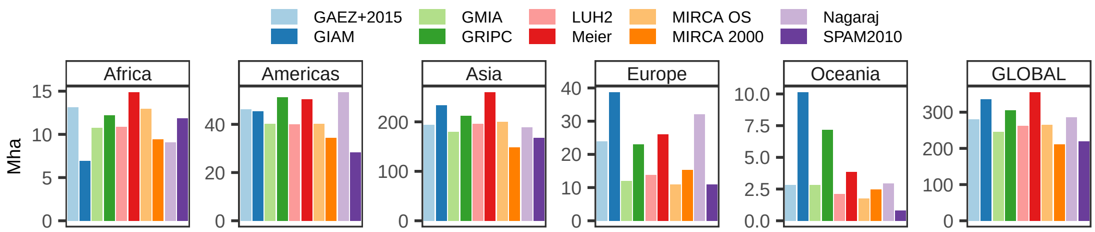

```r
# PLOT DATASETS ##################################################################

tmp <- dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), dataset] %>%
  .[, continent:= "GLOBAL"]

dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), .(continent, dataset)] %>%
  rbind(., tmp)
```

```
##      continent  dataset         mha
##         <char>   <char>       <num>
##   1:  Americas  gaez_v4  46.2015416
##   2:    Europe  gaez_v4  23.8489672
##   3:    Africa  gaez_v4  13.1101914
##   4:      Asia  gaez_v4 194.2821806
##   5:   Oceania  gaez_v4   2.8450580
##   6:  Americas     giam  45.5089223
##   7:    Europe     giam  38.6557015
##   8:    Africa     giam   6.9337475
##   9:      Asia     giam 233.9728781
##  10:   Oceania     giam  10.1185604
##  11:  Americas     gmia  40.2113502
##  12:    Europe     gmia  11.9504917
##  13:    Africa     gmia  10.7517031
##  14:      Asia     gmia 180.0792101
##  15:   Oceania     gmia   2.8368249
##  16:  Americas    gripc  51.2475536
##  17:    Europe    gripc  23.0350045
##  18:    Africa    gripc  12.2038719
##  19:      Asia    gripc 212.3461235
##  20:   Oceania    gripc   7.1657891
```

```
## 21:  Americas      luh2  40.0523677
## 22:    Europe      luh2  13.7826118
## 23:    Africa      luh2  10.8576084
## 24:      Asia      luh2 195.5457561
## 25:   Oceania      luh2   2.1157916
## 26:  Americas     meier  50.4449965
## 27:    Europe     meier  26.0018526
## 28:    Africa     meier  14.8522635
## 29:      Asia     meier 259.4777712
## 30:   Oceania     meier   3.8551223
## 31:  Americas mirca2000  34.4909117
## 32:    Europe mirca2000  15.3232002
## 33:    Africa mirca2000   9.4163731
## 34:      Asia mirca2000 148.8331835
## 35:   Oceania mirca2000   2.4615961
## 36:  Americas  mirca_os  40.1775798
## 37:    Europe  mirca_os  10.9262841
## 38:    Africa  mirca_os  12.9762031
## 39:      Asia  mirca_os 199.6637856
## 40:   Oceania  mirca_os   1.7670877
## 41:  Americas   nagaraj  53.2348690
## 42:    Europe   nagaraj  32.0035173
## 43:    Africa   nagaraj   9.0617581
## 44:      Asia   nagaraj 188.5240888
## 45:   Oceania   nagaraj   2.9588799
## 46:  Americas      spam  28.3824799
## 47:    Europe      spam  10.9216630
## 48:    Africa      spam  11.8448877
## 49:      Asia      spam 167.5769329
## 50:   Oceania      spam   0.8283964
## 51:    GLOBAL   gaez_v4 280.2879388
## 52:    GLOBAL      giam 335.1898098
## 53:    GLOBAL      gmia 245.8295799
## 54:    GLOBAL     gripc 305.9983426
## 55:    GLOBAL      luh2 262.3541356
## 56:    GLOBAL     meier 354.6320061
## 57:    GLOBAL mirca2000 210.5252647
## 58:    GLOBAL  mirca_os 265.5109403
## 59:    GLOBAL   nagaraj 285.7831131
## 60:    GLOBAL      spam 219.5543599
##     continent   dataset         mha
##        <char>    <char>       <num>
```

```
dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), .(continent, dataset)] %>%
  rbind(., tmp) %>%
  .[, continent:= factor(continent, levels = c("Africa", "Americas",
                                               "Asia", "Europe", "Oceania",
```

```r
                                                         "GLOBAL"))] %>%
  ggplot(., aes(dataset, mha, fill = dataset)) +
  geom_bar(stat = "identity") +
  scale_fill_brewer(palette = "Paired", label = label_map, name = "") +
  facet_wrap(~continent, ncol = 6, scales = "free_y") +
  theme_AP() +
  labs(x = "", y = "Mha") +
  theme(legend.position = "top") +
  theme(axis.text.x = element_blank(),
        axis.ticks.x = element_blank())
```




```r
# MERGE UA/SA PLOTS ##################################################################

top <- plot_grid(plot_uncertainty1, plot_agreement1, plot_indices1,
                 ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

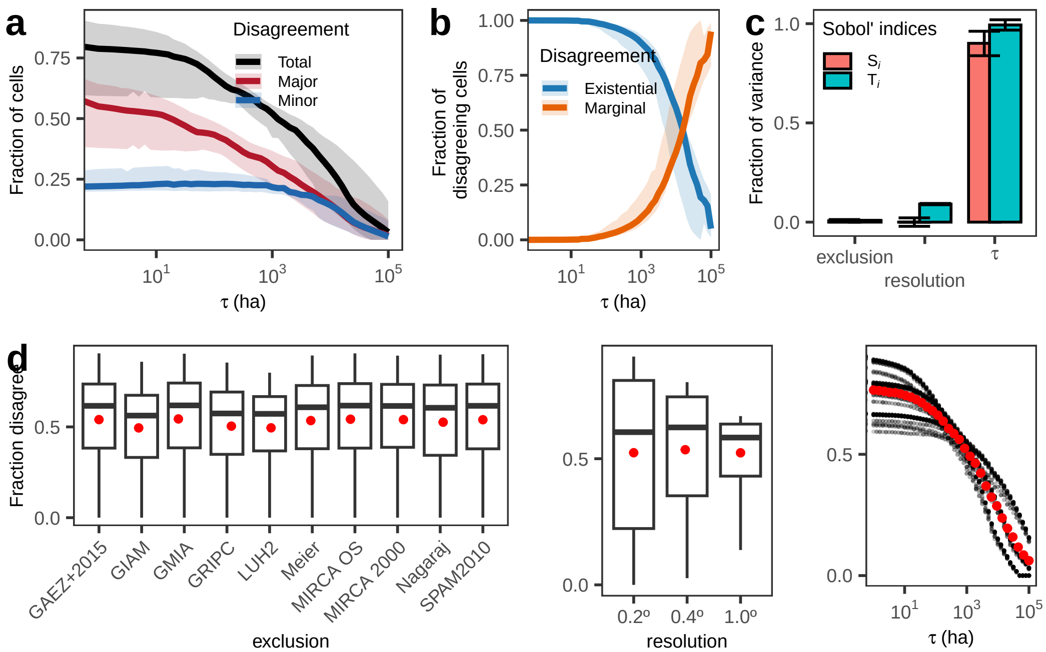
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```r
bottom <- plot_grid(a1, b1, c1, ncol = 3, rel_widths = c(0.5, 0.25, 0.25), labels = "d")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```r
plot_grid(top, bottom, ncol = 1)
```

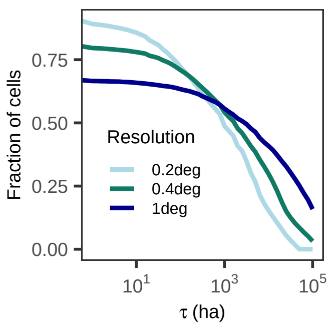


2.2 Does coarsening reduce disagreement?

```
# DOES ZOOMING OUT REDUCE DISAGREEMENT? ##########################################

effect_resolution_dt <- tau_results[scenario == "all", .(tau, frac_disagree, resolution)]

# Order resolution -------------------------------------------------------------

effect_resolution_dt[, resolution:= factor(resolution,
                                           levels = c("0.2deg", "0.4deg", "1deg"))]

# To wide table ----------------------------------------------------------------

wide_dt <- dcast(effect_resolution_dt, tau ~ resolution, value.var = "frac_disagree")

# Rename -----------------------------------------------------------------------

setnames(wide_dt, old = c("0.2deg", "0.4deg", "1deg"), new = c("res_0.2", "res_0.4", "res_1"))

# Compute absolute change (fine - coarse) --------------------------------------

wide_dt[, abs_change:= res_0.2 - res_1]

# Compute elasticity (log-log) -------------------------------------------------

# Area ratio from 0.2 to 1
area_ratio <- 25
wide_dt[, elasticity:= ifelse(res_0.2 > 0 & res_1 > 0,
```

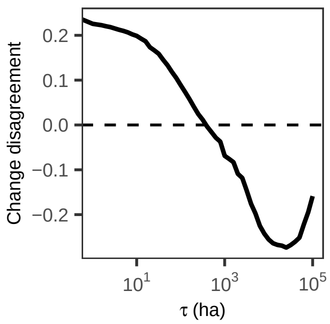
```
                                         log(res_1 / res_0.2) / log(area_ratio), NA_real_)]

# Plot disagreement curves -----------------------------------------------------

p1 <- ggplot(effect_resolution_dt, aes(tau, frac_disagree, colour = resolution)) +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_color_manual(values = res_palette) +
  labs(x = expression(tau~"(ha)"),
       y = "Fraction of cells",
       colour = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.3, 0.35))

p1
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```



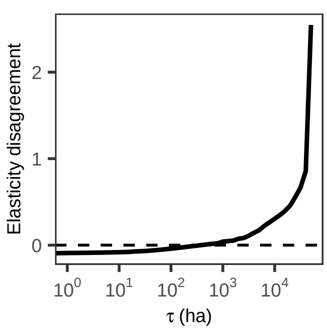

```
# Plot absolute change ---------------------------------------------------------

p2 <- ggplot(wide_dt, aes(tau, abs_change)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"),
       y = "Change disagreement") +
  theme_AP()

p2
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```



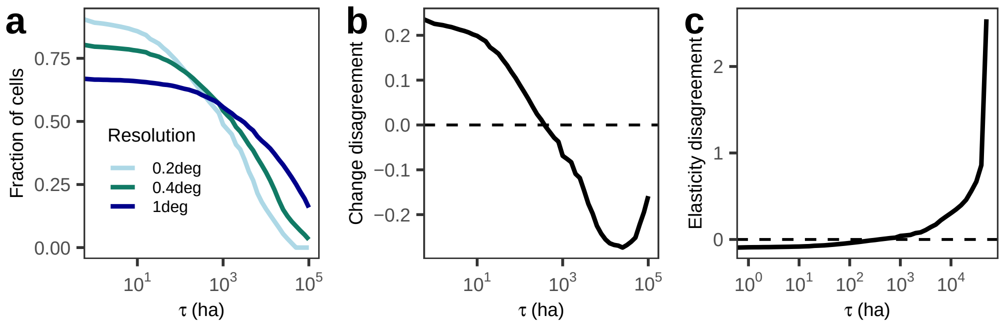

```
# Plot elasticity ------------------------------------------------------------

p3 <- ggplot(wide_dt[!is.na(elasticity)],
       aes(x = tau, y = elasticity)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"),
       y = "Elasticity disagreement") +
  theme_AP()

p3
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```




```
# Merge ----------------------------------------------------------------------

plot_grid(p1, p2, p3, ncol = 3, labels = "auto")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```



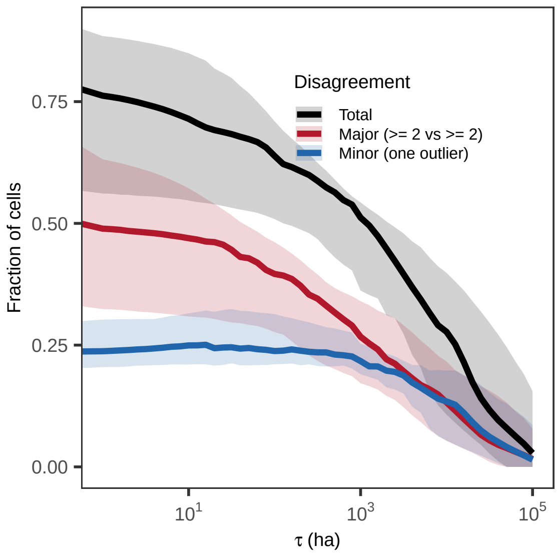

```
# Increasing spatial aggregation from 0.2 to 1 (25x coarsening) reduces
# existential disagreement by c20 percentage points at near-zero
# thresholds. this effect vanishes around tau = 300-500 ha, beyond which
# aggregation increases disagreement. At higher thresholds disagreement is
# greater at 1 than at 0.2. Coarse-scale aggregation can
# amplify rather than resolve structural spatial inconsistencies between datasets.

# These results refut the criticism of "if you zoom out, the maps converge".

# ALSO: Increasing cell area by a factor of 25 (0.2 to 1) yields an elasticity
# of approximately -0.07 at tau =  0, implying that a 1% increase in spatial
# aggregation reduces disagreement by only 0.07%. Elasticity approaches zero
# at moderate thresholds and becomes positive at higher thresholds.
```

2.3 UA/SA by excluding mapping paradigms

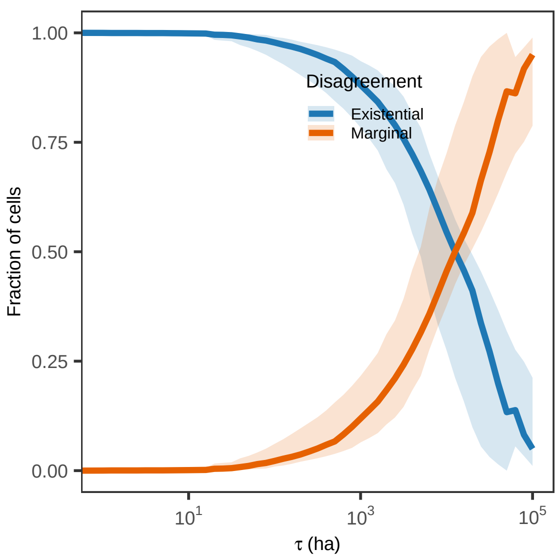
```
##               agreement        tau    frac_mean  frac_median     frac_lwr
##                  <fctr>      <num>        <num>        <num>        <num>
##  1:       frac_disagree     0.0000 0.7430530248 0.7755168771 0.5669310858
##  2:       frac_disagree     1.0000 0.7333800619 0.7621964390 0.5613534953
##  3:       frac_disagree    10.0000 0.7137910743 0.7146835560 0.5469757065
##  4:       frac_disagree   501.1872 0.5389292065 0.5637516586 0.4301616935
##  5:       frac_disagree  1000.0000 0.4806388477 0.5117712159 0.3618199772
##  6:       frac_disagree  1584.8932 0.4489805921 0.4738165904 0.3457740612
##  7:       frac_disagree  5011.8723 0.3349236070 0.3436038030 0.2032739840
##  8:       frac_disagree 10000.0000 0.2572035781 0.2772682201 0.1073456352
##  9: frac_major_disagree     0.0000 0.5031937420 0.4994215001 0.3299454636
## 10: frac_major_disagree     1.0000 0.4896659751 0.4891594145 0.3238720873
## 11: frac_major_disagree    10.0000 0.4602356204 0.4695407214 0.3087506197
## 12: frac_major_disagree   501.1872 0.3058621258 0.3168921978 0.2007899528
## 13: frac_major_disagree  1000.0000 0.2646720627 0.2671663565 0.1717144444
## 14: frac_major_disagree  1584.8932 0.2411082172 0.2412093164 0.1586694233
## 15: frac_major_disagree  5011.8723 0.1711008105 0.1681171085 0.0920017280
## 16: frac_major_disagree 10000.0000 0.1291880616 0.1322249610 0.0532755269
## 17: frac_minor_disagree     0.0000 0.2398592828 0.2369856222 0.2027764006
## 18: frac_minor_disagree     1.0000 0.2437140868 0.2374814080 0.2047595439
## 19: frac_minor_disagree    10.0000 0.2535554539 0.2493837718 0.2100892414
## 20: frac_minor_disagree   501.1872 0.2330670807 0.2306643530 0.2063708478
## 21: frac_minor_disagree  1000.0000 0.2159667850 0.2171889934 0.1901055328
## 22: frac_minor_disagree  1584.8932 0.2078723749 0.2062469013 0.1787816289
## 23: frac_minor_disagree  5011.8723 0.1638227966 0.1626339353 0.1112722560
## 24: frac_minor_disagree 10000.0000 0.1280155164 0.1342340109 0.0540701083
## 25:     share_existence     0.0000 1.0000000000 1.0000000000 1.0000000000
## 26:     share_existence     1.0000 0.9997672523 0.9998374407 0.9993735644
## 27:     share_existence    10.0000 0.9982965766 0.9988512349 0.9958634953
## 28:     share_existence   501.1872 0.9168285618 0.9332604417 0.8442199776
## 29:     share_existence  1000.0000 0.8703566888 0.8809219223 0.7835751053
## 30:     share_existence  1584.8932 0.8326718403 0.8422081094 0.7312409812
## 31:     share_existence  5011.8723 0.6665122263 0.6842837274 0.4888273553
## 32:     share_existence 10000.0000 0.5232850394 0.5448511469 0.2750825646
## 33:      share_marginal     0.0000 0.0000000000 0.0000000000 0.0000000000
## 34:      share_marginal     1.0000 0.0002327477 0.0001625593 0.0000000000
## 35:      share_marginal    10.0000 0.0017034234 0.0011487651 0.0004279575
## 36:      share_marginal   501.1872 0.0831714382 0.0667395583 0.0382878568
## 37:      share_marginal  1000.0000 0.1296433112 0.1190780777 0.0644334054
## 38:      share_marginal  1584.8932 0.1673281597 0.1577918906 0.0856892626
## 39:      share_marginal  5011.8723 0.3334877737 0.3157162726 0.2163691902
## 40:      share_marginal 10000.0000 0.4767149606 0.4551488531 0.3759911779
##               agreement        tau    frac_mean  frac_median     frac_lwr
##                  <fctr>      <num>        <num>        <num>        <num>
##           frac_upr
##              <num>
```

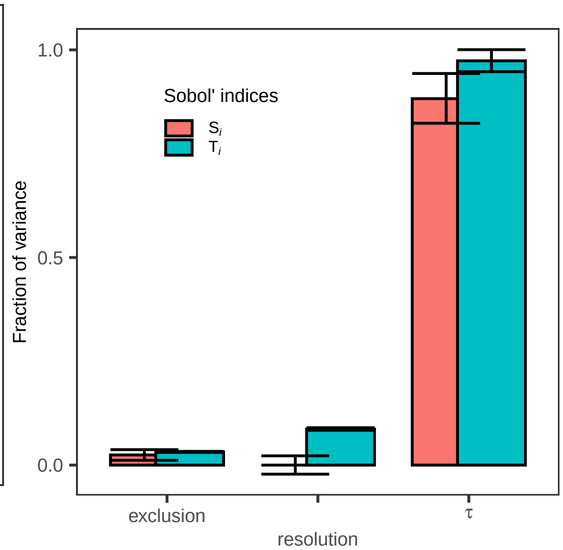
```
##  1: 0.8997747400
##  2: 0.8846468356
##  3: 0.8495463943
##  4: 0.5901705317
##  5: 0.5430094199
##  6: 0.5178482895
##  7: 0.4510411502
##  8: 0.3983639068
##  9: 0.6584379918
## 10: 0.6317153701
## 11: 0.5720677631
## 12: 0.3672533466
## 13: 0.3403569658
## 14: 0.3210213188
## 15: 0.2595438770
## 16: 0.2165344571
## 17: 0.2993566205
## 18: 0.3024038016
## 19: 0.3153176783
## 20: 0.2842595735
## 21: 0.2525534607
## 22: 0.2416298948
## 23: 0.2086018840
## 24: 0.1979424888
## 25: 1.0000000000
## 26: 1.0000000000
## 27: 0.9995720425
## 28: 0.9617121432
## 29: 0.9355665946
## 30: 0.9143107374
## 31: 0.7836308098
## 32: 0.6240088221
## 33: 0.0000000000
## 34: 0.0006264356
## 35: 0.0041365047
## 36: 0.1557800224
## 37: 0.2164248947
## 38: 0.2687590188
## 39: 0.5111726447
## 40: 0.7249174354
##         frac_upr
##            <num>
```

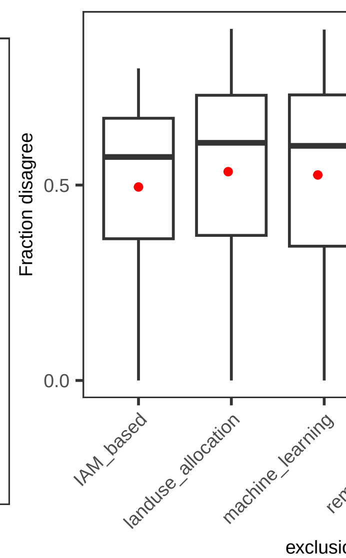
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

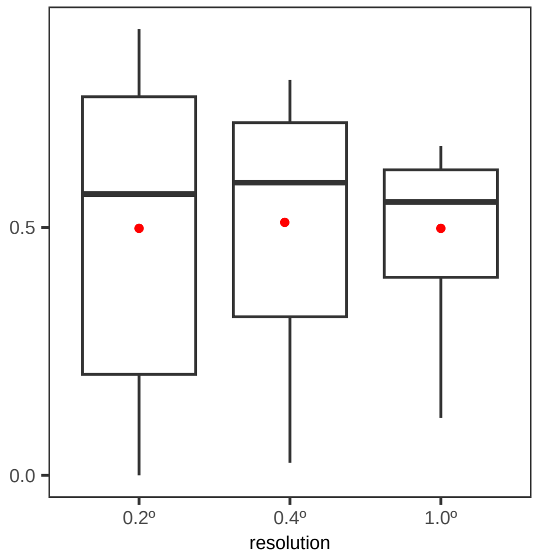



```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```



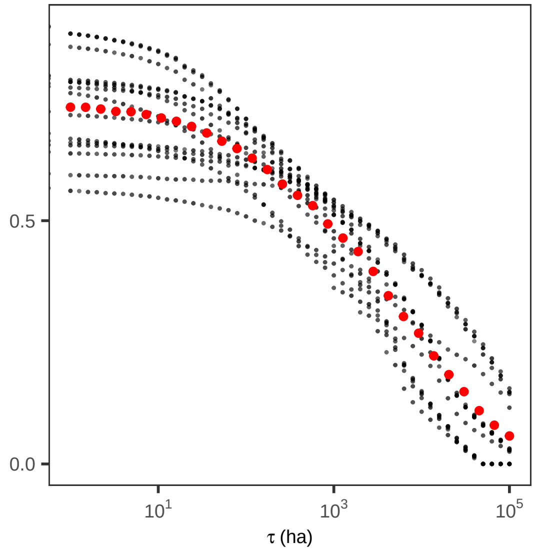







```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```




```
# MERGE UA/SA PLOTS ################################################################

top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices2,
                 ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

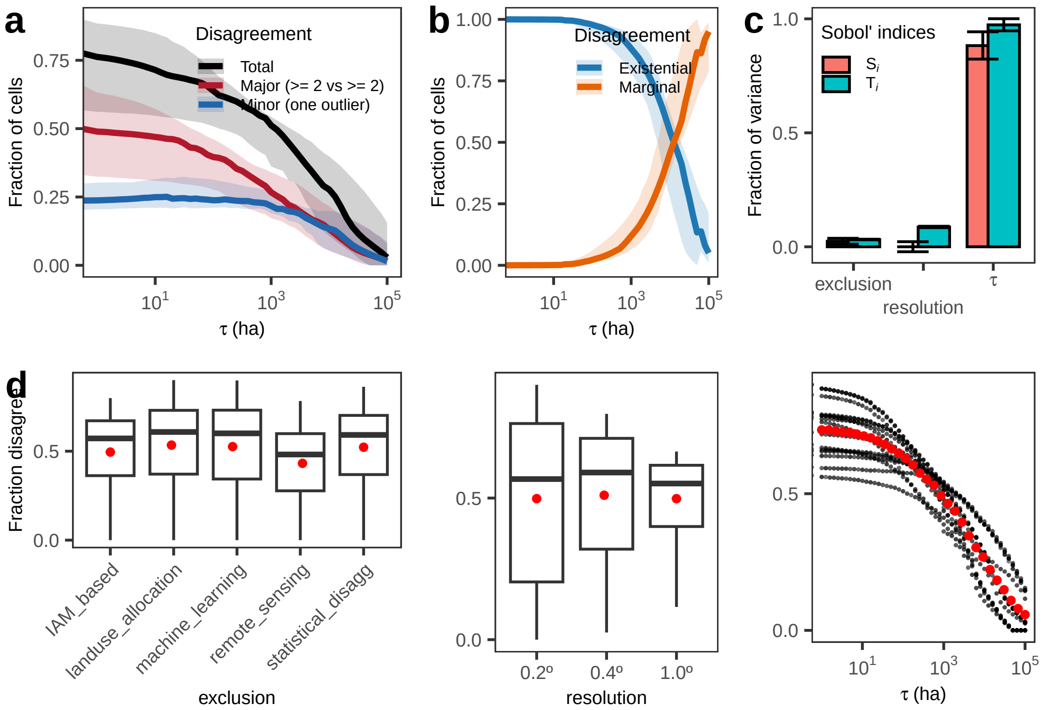
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
bottom <- plot_grid(a2, b2, c2, ncol = 3, rel_widths = c(0.4, 0.3, 0.3), labels = "d")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```r
plot_grid(top, bottom, ncol = 1)
```

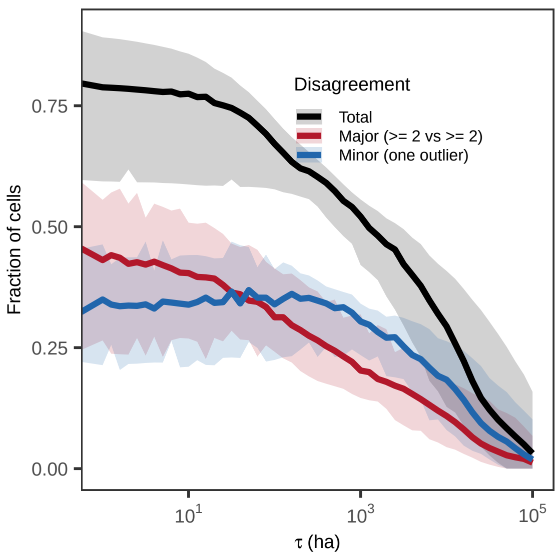


2.4 UA/SA by weighting newer datasets more

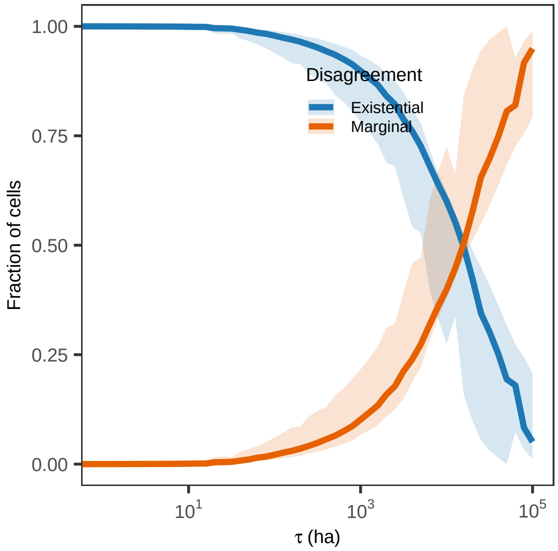
```
##       tau           agreement    frac_mean  frac_median     frac_lwr
##     <num>              <fctr>        <num>        <num>        <num>
##  1:    10       frac_disagree 0.7447800994 7.748378e-01 0.5872583044
##  2:   100       frac_disagree 0.6666677100 6.712284e-01 0.5772186415
##  3:     1       frac_disagree 0.7650881937 7.880427e-01 0.5935795736
##  4:     0       frac_disagree 0.7716544945 7.962423e-01 0.5965542885
##  5:  1000       frac_disagree 0.5103718881 5.209518e-01 0.4207578610
##  6:    10 frac_major_disagree 0.3947745529 4.040948e-01 0.2688398612
##  7:   100 frac_major_disagree 0.3177585471 3.128409e-01 0.2415218934
##  8:     1 frac_major_disagree 0.4230902508 4.310257e-01 0.2649138572
##  9:     0 frac_major_disagree 0.4405374142 4.551409e-01 0.2451660882
## 10:  1000 frac_major_disagree 0.2096229915 2.026456e-01 0.1460315287
## 11:    10 frac_minor_disagree 0.3500055465 3.388696e-01 0.2105850273
## 12:   100 frac_minor_disagree 0.3489091629 3.395040e-01 0.2245909767
## 13:     1 frac_minor_disagree 0.3419979429 3.495649e-01 0.2138076351
## 14:     0 frac_minor_disagree 0.3311170803 3.235827e-01 0.2205658156
## 15:  1000 frac_minor_disagree 0.3007488966 3.040855e-01 0.2345066931
## 16:    10     share_existence 0.9987395006 9.990102e-01 0.9958634953
## 17:   100     share_existence 0.9749187906 9.785986e-01 0.9388543231
## 18:     1     share_existence 0.9998888226 9.999368e-01 0.9993735644
## 19:     0     share_existence 1.0000000000 1.000000e+00 1.0000000000
## 20:  1000     share_existence 0.8881529264 8.982820e-01 0.7835751053
## 21:    10      share_marginal 0.0012604994 9.897843e-04 0.0006042296
## 22:   100      share_marginal 0.0250812094 2.140140e-02 0.0104125963
## 23:     1      share_marginal 0.0001111774 6.324311e-05 0.0000000000
## 24:     0      share_marginal 0.0000000000 0.000000e+00 0.0000000000
## 25:  1000      share_marginal 0.1118470736 1.017180e-01 0.0665816327
##       tau           agreement    frac_mean  frac_median     frac_lwr
##     <num>              <fctr>        <num>        <num>        <num>
##         frac_upr
##            <num>
##  1: 0.8574304934
##  2: 0.7225445120
##  3: 0.8913352054
##  4: 0.9047505169
##  5: 0.5563956371
##  6: 0.5082584548
##  7: 0.4138286634
##  8: 0.5555861863
##  9: 0.5927423088
## 10: 0.3187902826
## 11: 0.4413451381
## 12: 0.4123195332
## 13: 0.4634183973
## 14: 0.4544157127
## 15: 0.3409766981
```

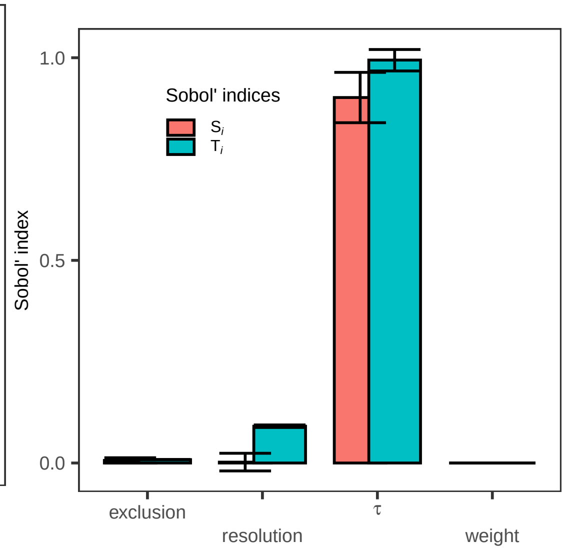
```
## 16: 0.9993957704
## 17: 0.9895874037
## 18: 1.0000000000
## 19: 1.0000000000
## 20: 0.9334183673
## 21: 0.0041365047
## 22: 0.0611456769
## 23: 0.0006264356
## 24: 0.0000000000
## 25: 0.2164248947
##         frac_upr
##            <num>
```

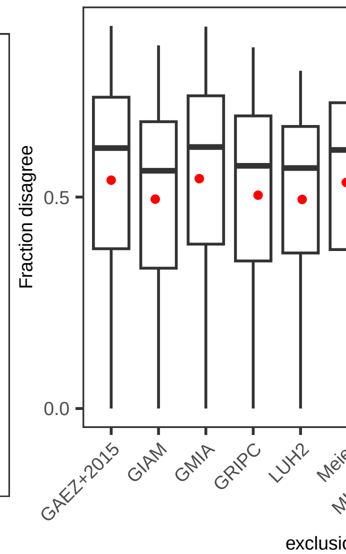
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced
## log-10 transformation introduced infinite values.
```

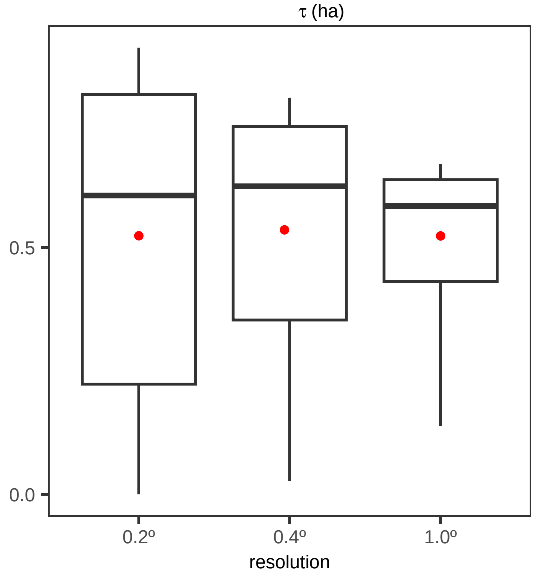



```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced
## log-10 transformation introduced infinite values.
```



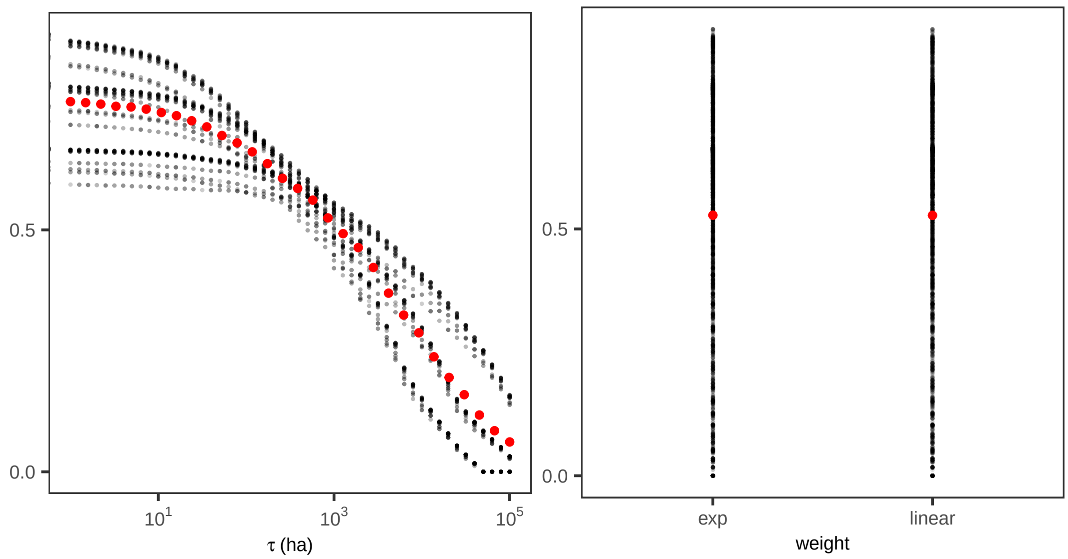







```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```




```
# MERGE UA/SA PLOTS ##############################################################

top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices3,
                 ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

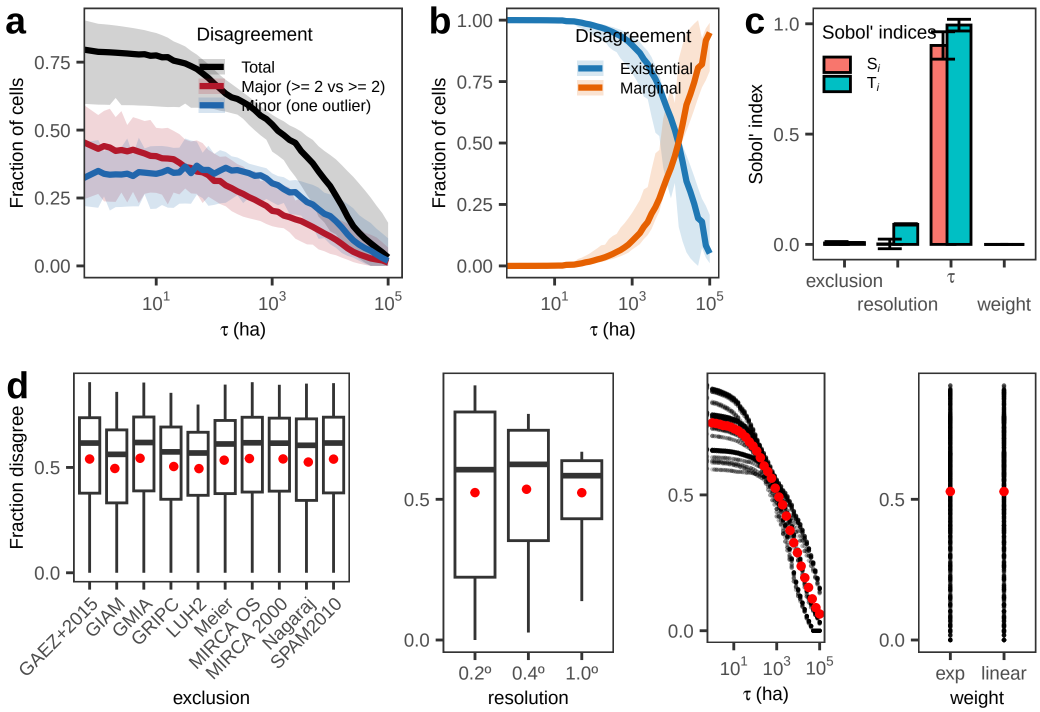
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
bottom <- plot_grid(a3, b3, c3, d3, ncol = 4, rel_widths = c(0.35, 0.25, 0.2, 0.2), labels = "c
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
plot_grid(top, bottom, ncol = 1)
```



2.5 UA/SA on mapping variables

```
# MAPPING PARADIGM EXCLUSION UA/SA ##############################################

ds_groups <- list(AEI = c("meier", "nagaraj", "gaez_v4", "spam", "luh2"),
                  AAI = c("giam", "gmia", "gripc", "mirca2000", "mirca_os"))

mapping_paradigm_vec <- names(ds_groups)
n_mapping_paradigm_vec <- length(mapping_paradigm_vec)

# Create the sample matrix -----------------------------------------------------

params <- c("resolution", "tau", "exclusion")
mat <- data.table(sobol_matrices(matrices = matrices, N = N, params = params))

# Transform resolution column --------------------------------------------------

mat[, resolution:= pmin(n_resolution_vec, 1L + floor(resolution * n_resolution_vec))]
mat[, resolution:= resolution_vec[resolution]]

# Transform tau column ---------------------------------------------------------

mat[, tau:= pmin(n_tau_vec, 1L + floor(tau * n_tau_vec))]
mat[, tau:= tau_vec[tau]]

# Transform exclusion column ---------------------------------------------------
```

```r
mat[, exclusion:= pmin(n_mapping_paradigm_vec, 1L + floor(exclusion * n_mapping_paradigm_vec))]
mat[, exclusion:= mapping_paradigm_vec[exclusion]]

# PRE-COMPUTE ALL COMBINATIONS #################################################

# Compute all possibilities by looping over resolutions and tau values ---------

# Set parallel backend --------------------------

resolutions <- sort(unique(dt$resolution))
ncores <- max(1, parallel::detectCores() - 1)

# ds_groups: named list of dataset keys to exclude by mapping category
# mat: data.table with columns resolution, tau (in ha), exclusion (category name)

idcols <- c("lon","lat","country","code","continent")

tau_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]
    if (nrow(dt_res_long) == 0L) return(NULL)

    # scenarios to run for this resolution
    mat_res <- mat[resolution == res]
    if (nrow(mat_res) == 0L) return(NULL)

    # datasets present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    # ---- CAST ONCE per resolution  ----

    dt_res_wide_all <- dcast(dt_res_long,
                             lon + lat + country + code + continent ~ dataset,
                             value.var = "mha", fill = 0)

    # loop exclusion groups (few) instead of mat rows (many)
    excl_levels <- unique(mat_res$exclusion)

    rbindlist(lapply(excl_levels, function(excl_grp) {

      # datasets to exclude for this group
      excl_ds <- ds_groups[[excl_grp]]
      if (is.null(excl_ds)) excl_ds <- character(0)

      use_ds <- setdiff(ds_res, excl_ds)
      if (length(use_ds) < 2L) return(NULL)
```

```r
      # taus used for THIS (resolution, exclusion-group)
      taus_ha <- unique(mat_res[exclusion == excl_grp, tau])
      taus_mha <- taus_ha * 1e-6

      # ---- compute A_min/A_max ONCE per exclusion group ----
      dt_w <- dt_res_wide_all[, c(idcols, use_ds), with = FALSE]

      dt_w[, `:=`(A_min = do.call(pmin, c(.SD, list(na.rm = TRUE))),
                  A_max = do.call(pmax, c(.SD, list(na.rm = TRUE)))),
           .SDcols = use_ds]

      # ---- now only loop over taus
      out_grp <- rbindlist(lapply(seq_along(taus_mha), function(k) {

        tau_ha  <- taus_ha[k]
        tau_mha <- taus_mha[k]

        out <- compute_tau_fun(dt_w, tau_mha, use_ds)

        out[, `:=`(
          tau_label = paste0(format(tau_ha, scientific = FALSE, trim = TRUE), "ha"),
          tau_ha = tau_ha,
          tau_mha = tau_mha,
          resolution = res,
          exclusion  = excl_grp
        )]

        out
      }), use.names = TRUE, fill = TRUE)

      out_grp
    }), use.names = TRUE, fill = TRUE)

  }, mc.cores = ncores),
  use.names = TRUE, fill = TRUE
)

# ARRANGE OUTPUT ###############################################################

# Rename tau_ha to tau to match mat ---------------------------------------------

setnames(tau_results, "tau_ha", "tau")

# Join all three disagreement metrics onto mat ---------------------------------

mat[tau_results, on = .(resolution, tau, exclusion = exclusion),
    `:=`(frac_disagree = i.frac_disagree,
```

```r
        frac_major_disagree = i.frac_major_disagree,
        frac_minor_disagree = i.frac_minor_disagree,
        share_existence = i.share_existence,
        share_marginal = i.share_marginal)]

# Reshape just these three for plotting --------------------------------------

mat_long <- melt(mat, id.vars = c("resolution", "tau", "exclusion"),
                 measure.vars = c("frac_disagree", "frac_major_disagree",
                                  "frac_minor_disagree", "share_existence",
                                  "share_marginal"),
                 variable.name = "agreement", value.name = "frac_value")

# PREPARE DATA TO PLOT UNCERTAINTY #############################################

# Calculate mean and quantiles by tau × agreement -----------------------------

mat_sum <- mat_long[, .(frac_mean = mean(frac_value, na.rm = TRUE),
                        frac_median = median(frac_value, na.rm = TRUE),
                        frac_lwr = quantile(frac_value, 0.025, na.rm = TRUE),
                        frac_upr = quantile(frac_value, 0.975, na.rm = TRUE)),
                    .(tau, agreement)]

mat_sum[, {dt_sub <- .SD
idx <- sapply(target_tau, function(x) {
  which.min(abs(log10(tau + 1e-12) - log10(x + 1e-12)))
}
)
dt_sub[idx]
},
agreement
]
```

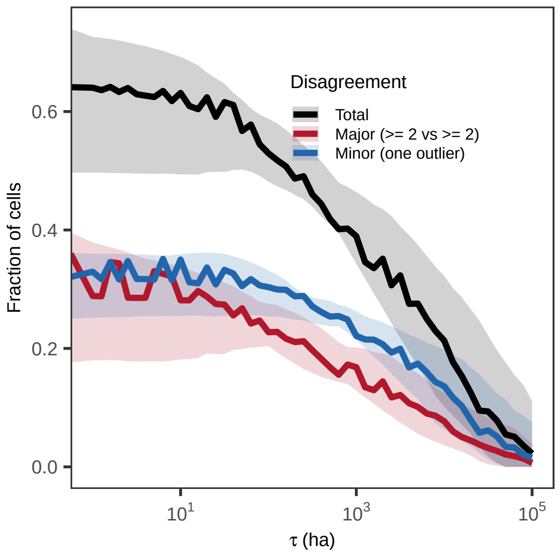
```
##                agreement        tau   frac_mean  frac_median     frac_lwr
##                   <fctr>      <num>       <num>        <num>        <num>
##  1:        frac_disagree     0.0000 0.632763385 0.6411539816 0.4969013386
##  2:        frac_disagree     1.0000 0.626999490 0.6402736556 0.4967773922
##  3:        frac_disagree    10.0000 0.606632456 0.6313949394 0.4939266237
##  4:        frac_disagree   501.1872 0.442056757 0.4188292653 0.4095025766
##  5:        frac_disagree  1000.0000 0.390105738 0.3894397620 0.3426651649
##  6:        frac_disagree  1584.8932 0.360526361 0.3355802606 0.2924753911
##  7:        frac_disagree  1995.2623 0.339223484 0.3515121468 0.2679050822
##  8:        frac_disagree  5011.8723 0.264034661 0.2756426380 0.1703027124
##  9:        frac_disagree 10000.0000 0.201851184 0.2132652548 0.1015444194
## 10:        frac_disagree 12589.2541 0.181480564 0.1767443030 0.0863702287
## 11: frac_major_disagree     0.0000 0.315958611 0.3597766487 0.1761279127
## 12: frac_major_disagree     1.0000 0.308318422 0.2884702450 0.1798463064
```

```
## 13: frac_major_disagree    10.0000 0.284691982 0.2813270285 0.1813336639
## 14: frac_major_disagree   501.1872 0.182193883 0.1683201160 0.1477149998
## 15: frac_major_disagree  1000.0000 0.159155914 0.1683192861 0.1287839047
## 16: frac_major_disagree  1584.8932 0.143161615 0.1294833744 0.1065587682
## 17: frac_major_disagree  1995.2623 0.134430265 0.1443976202 0.0949177647
## 18: frac_major_disagree  5011.8723 0.100141504 0.1012877911 0.0557904157
## 19: frac_major_disagree 10000.0000 0.075214507 0.0770410986 0.0366356034
## 20: frac_major_disagree 12589.2541 0.066654489 0.0598118618 0.0314283951
## 21: frac_minor_disagree     0.0000 0.316804774 0.3207734259 0.2500000000
## 22: frac_minor_disagree     1.0000 0.318681069 0.3295744083 0.2516113039
## 23: frac_minor_disagree    10.0000 0.321940474 0.3500679109 0.2547099653
## 24: frac_minor_disagree   501.1872 0.259862874 0.2538357060 0.2374814080
## 25: frac_minor_disagree  1000.0000 0.230949824 0.2211204760 0.2138812602
## 26: frac_minor_disagree  1584.8932 0.217364746 0.2147375567 0.1859166230
## 27: frac_minor_disagree  1995.2623 0.204793220 0.2071145265 0.1729873176
## 28: frac_minor_disagree  5011.8723 0.163893157 0.1743548468 0.1145122967
## 29: frac_minor_disagree 10000.0000 0.126636677 0.1362241561 0.0635819422
## 30: frac_minor_disagree 12589.2541 0.114826075 0.1169324413 0.0526969482
## 31:     share_existence     0.0000 1.000000000 1.0000000000 1.0000000000
## 32:     share_existence     1.0000 0.997429207 0.9996449844 0.9927644711
## 33:     share_existence    10.0000 0.990364277 0.9854812471 0.9796737767
## 34:     share_existence   501.1872 0.848206846 0.9383355414 0.7107206519
## 35:     share_existence  1000.0000 0.778994714 0.6961660551 0.6343482906
## 36:     share_existence  1584.8932 0.725210004 0.8147204317 0.5797872340
## 37:     share_existence  1995.2623 0.686576017 0.5903933691 0.5496726445
## 38:     share_existence  5011.8723 0.516795763 0.4410986404 0.3880684907
## 39:     share_existence 10000.0000 0.363638334 0.3178324366 0.2300267719
## 40:     share_existence 12589.2541 0.315040530 0.2801972062 0.1763865546
## 41:      share_marginal     0.0000 0.000000000 0.0000000000 0.0000000000
## 42:      share_marginal     1.0000 0.002570793 0.0003550156 0.0002296211
## 43:      share_marginal    10.0000 0.009635723 0.0145187529 0.0009438414
## 44:      share_marginal   501.1872 0.151793154 0.0616644586 0.0468247000
## 45:      share_marginal  1000.0000 0.221005286 0.3038339449 0.0996180777
## 46:      share_marginal  1584.8932 0.274789996 0.1852795683 0.1437288136
## 47:      share_marginal  1995.2623 0.313423983 0.4096066309 0.1815937941
## 48:      share_marginal  5011.8723 0.483204237 0.5589013596 0.3539786710
## 49:      share_marginal 10000.0000 0.636361666 0.6821675634 0.5255298651
## 50:      share_marginal 12589.2541 0.684959470 0.7198027938 0.5815244826
##               agreement        tau   frac_mean  frac_median     frac_lwr
##                  <fctr>      <num>       <num>        <num>        <num>
##        frac_upr
##           <num>
##  1: 0.739917302
##  2: 0.726077699
##  3: 0.692334989
##  4: 0.498038131
##  5: 0.464055528
##  6: 0.442736738
```

```
##  7: 0.435423897
##  8: 0.375061973
##  9: 0.322508676
## 10: 0.301685672
## 11: 0.395709260
## 12: 0.379223625
## 13: 0.337519672
## 14: 0.222772775
## 15: 0.201660882
## 16: 0.193232524
## 17: 0.191497273
## 18: 0.157535944
## 19: 0.124938027
## 20: 0.114154685
## 21: 0.361087111
## 22: 0.359783380
## 23: 0.359166230
## 24: 0.280490828
## 25: 0.262394646
## 26: 0.249504214
## 27: 0.243926624
## 28: 0.217526029
## 29: 0.197570649
## 30: 0.187530987
## 31: 1.000000000
## 32: 0.999770379
## 33: 0.999056159
## 34: 0.953175300
## 35: 0.900381922
## 36: 0.856271186
## 37: 0.818406206
## 38: 0.646021329
## 39: 0.474470135
## 40: 0.418475517
## 41: 0.000000000
## 42: 0.007235529
## 43: 0.020326223
## 44: 0.289279348
## 45: 0.365651709
## 46: 0.420212766
## 47: 0.450327356
## 48: 0.611931509
## 49: 0.769973228
## 50: 0.823613445
##         frac_upr
##            <num>
```

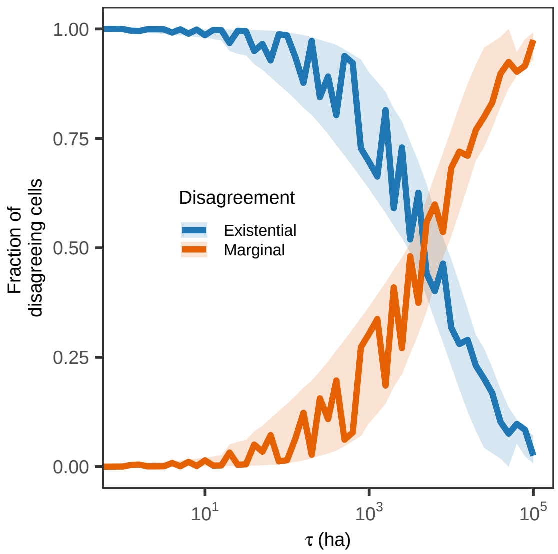
```r
# Plot uncertainty -------------------------------------------------------------

agreement_cols <- c(frac_disagree = "black",
                    frac_major_disagree = "#B2182B",
                    frac_minor_disagree = "#2166AC"  )

plot_uncertainty <- mat_sum[!agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(.,
         aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_colour_manual(values = agreement_cols,
                      breaks = c("frac_disagree",
                                 "frac_major_disagree",
                                 "frac_minor_disagree"),
                      labels = c("Total",
                                 "Major (\u2265 2 vs \u2265 2)",
                                 "Minor (one outlier)"),
                      name = "Disagreement") +
  scale_fill_manual(values = agreement_cols,
                    breaks = c("frac_disagree",
                               "frac_major_disagree",
                               "frac_minor_disagree"),
                    labels = c("Total",
                               "Major (\u2265 2 vs \u2265 2)",
                               "Minor (one outlier)"),
                    name = "Disagreement") +
  labs(x = expression(tau~"(ha)"), y = "Fraction of cells") +
  theme_AP() +
  theme(legend.position = c(0.65, 0.77))

plot_uncertainty
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

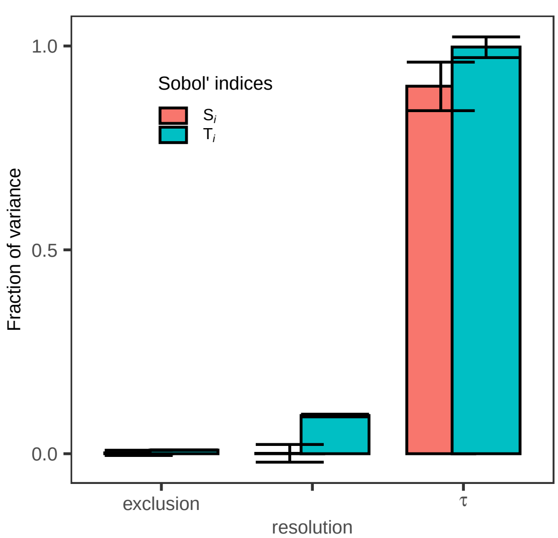



```r
plot_agreement <- mat_sum[agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(., aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = exist_cols,
                      labels = c(share_existence = "Existential",
                                 share_marginal = "Marginal"),
                      name = "Disagreement") +
  scale_fill_manual(values = exist_cols,
                    labels = c(share_existence = "Existential",
                               share_marginal = "Marginal"),
                    name = "Disagreement") +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"), y = "Fraction of \ndisagreeing cells") +
  theme_AP() +
  theme(legend.position = c(0.3, 0.55))


plot_agreement
```

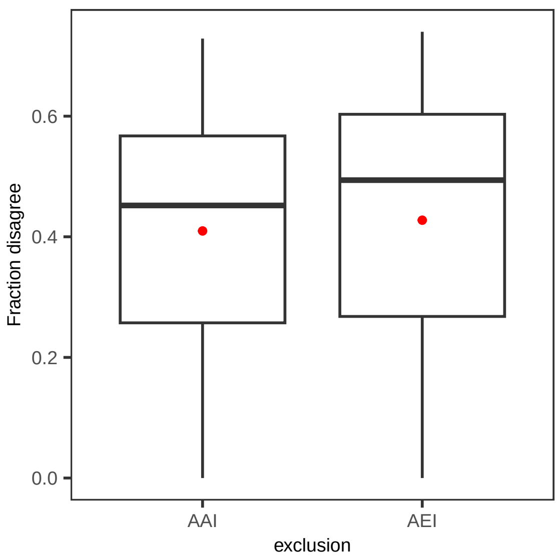
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```



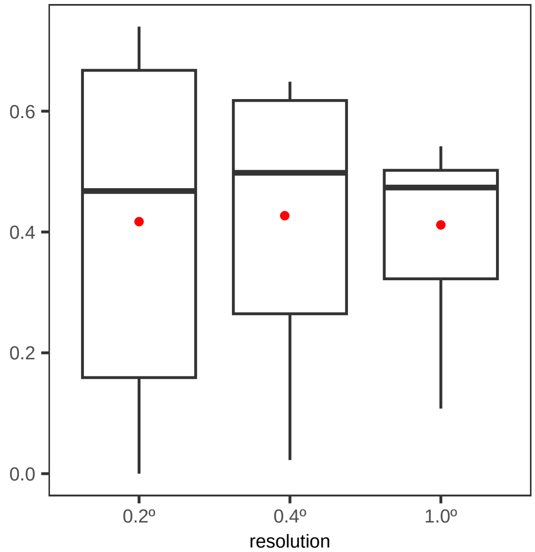

```
# SENSITIVITY ANALYSIS ########################################################

# Sobol' indices -------------------------------------------------------------

ind <- sobol_indices(matrices = matrices, N = N, params = params, Y
                     = mat[, frac_disagree], R = R, boot = TRUE)

plot_indices2 <- plot(ind) +
  theme_AP() +
  labs(x = "", y = "Fraction of variance") +
  scale_x_discrete(labels = c(exclusion = "exclusion",
                              resolution = "resolution",
                              tau = expression(tau)),
                   guide = guide_axis(n.dodge = 2)) +
  theme(legend.position = c(0.3, 0.8))

plot_indices2
```



```r
# SCATTERPLOTS OF OUTPUT AGAINS INTPUT ########################################

label_map <- setNames(new_dataset_names, dataset_names)

a2<- ggplot(mat[!exclusion == "all", c("exclusion", "frac_disagree")],
            aes(exclusion, frac_disagree)) +
  geom_boxplot() +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  scale_x_discrete(labels = label_map) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  labs(x = "exclusion", y = "Fraction disagree")

a2
```



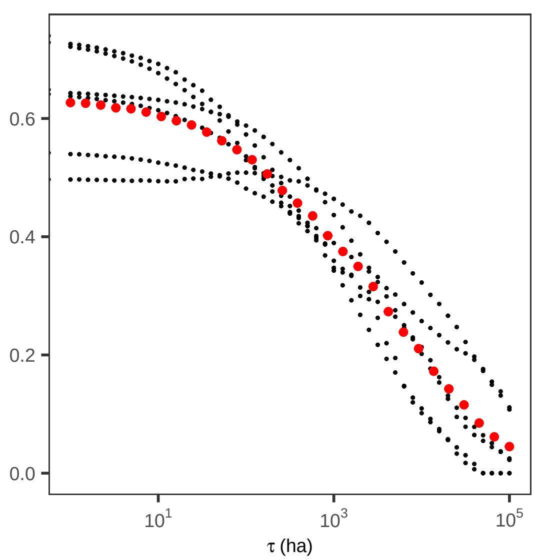

```r
b2 <- ggplot(mat[1:N, c("resolution", "frac_disagree")],
             aes(resolution, frac_disagree)) +
  geom_boxplot() +
  scale_x_discrete(labels = c(`0.2deg` = "0.2º",
                              `0.4deg` = "0.4º",
                              `1deg` = "1.0º")) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  theme_AP() +
  labs(x = "resolution", y = "")

b2
```




```
c2 <- ggplot(mat[1:N, c("tau", "frac_disagree")],
             aes(tau, frac_disagree)) +
  geom_point(size = 0.1, alpha = 0.1) +
  stat_summary_bin(fun = "mean", geom = "point",
                   colour = "red", size = 1) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"), y = "")

c2
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```



```r
# MERGE UA/SA PLOTS ##############################################################

top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices2,
                 ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

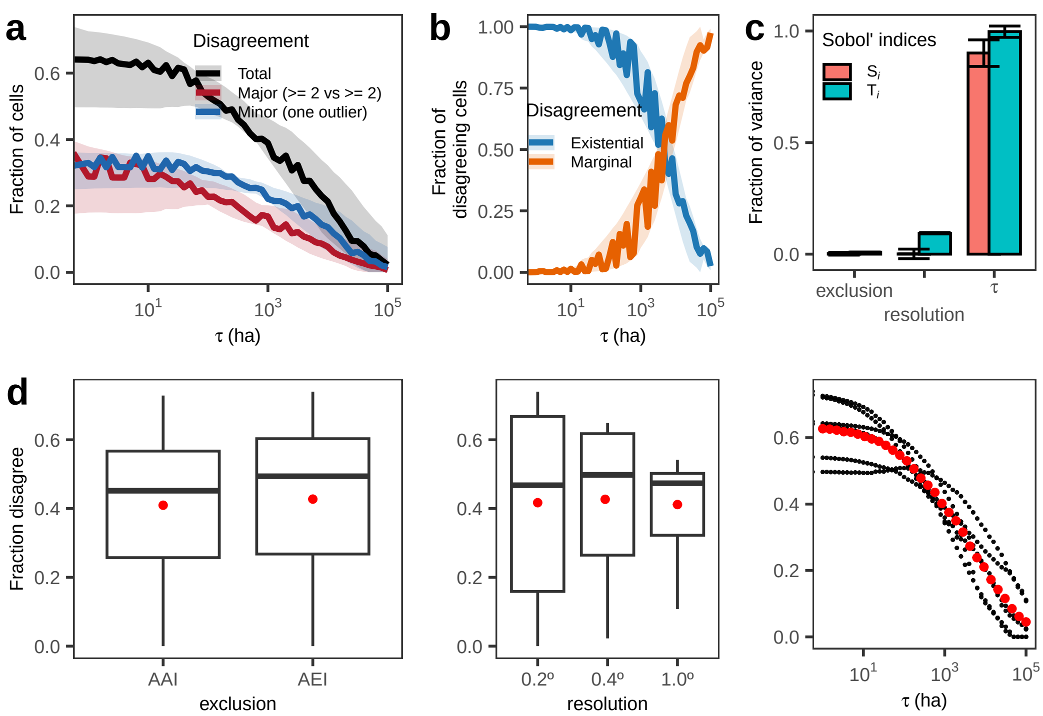
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```r
bottom <- plot_grid(a2, b2, c2, ncol = 3, rel_widths = c(0.4, 0.3, 0.3), labels = "d")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```r
plot_grid(top, bottom, ncol = 1)
```



3 The maximum threshold for existence disagreement: τ_{max}

```r
# COMPUTE TAU MAX ##############################################################
################################################################################

# Tau max is the largest irrigation threshold at which at least one dataset
# classifies the cell as essentially unirrigated  while another classifies
# it as irrigated. Tau max thus quantifies how large the ontological disagreement
# on irrigation is. We also set NA to two cases:

# - Universally non-irrigated cells: all datasets report areas <= threshold.
# - Universally irrigated cells: all datasets report areas >threshold.

# NA cells are cells for which there is agreement irrigation exists or it is absent.

## Compute tau_max_results -----------------------------------------------------

tau_max_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]

    # datasets present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    # resolution-specific detectability thresholds (in ha)----------------------

    zero_tol_ha <- switch(res,
      "0.2deg" = 500, # 1% of cell at 0.2°
      "0.4deg" = 2000, # 1% of cell at 0.4°
      "1deg" = 12000, # 1% of cell at 1°
      stop("Unknown resolution: ", res))

    rbindlist(
      lapply(dataset_vec, function(scn) {

        # which datasets to use ----------------------------------------------

        if (scn == "all") {
          use_ds <- ds_res
        } else {
          use_ds <- setdiff(ds_res, scn)
        }

        # if only one dataset left, skip --------------------------------------

        if (length(use_ds) < 2L) return(NULL)
```

```r
        # reshape to wide ---------------------------------------------------


        dt_res_wide <- dcast(
          dt_res_long[dataset %in% use_ds],
          lon + lat + country + code + continent ~ dataset,
          value.var = "mha", fill = 0
        )

        tau_max_dt <- tau_grid[,
          compute_tau_max_existential_fun(dt_res_wide, use_ds, tau_mha, tau_ha,
                                          zero_tol_ha = zero_tol_ha)
        ]

        tau_max_dt[, `:=`(resolution = res, scenario = scn)]

        tau_max_dt
      }),
      use.names = TRUE, fill = TRUE
    )
  }, mc.cores = ncores),
  use.names = TRUE, fill = TRUE
)


# Add the countries ---------------------------------------------------------

chunk_size <- 500000L
idx <- split(seq_len(nrow(tau_max_results)),
             ceiling(seq_len(nrow(tau_max_results)) / chunk_size))

tau_max_results[, country:= NA_character_]
for (i in idx) {
  tau_max_results[i, country:= lonlat_to_country(lon, lat, countries_sf)]
}

# Add the continents --------------------------------------------------------

country_to_continent(tau_max_results)

# Retrieve the round with all datasets and exclude the rest --------------------

tau_max_all <- tau_max_results[scenario == "all"]
```

```r
# DEFINE PLOTS TAU MAX ##########################################################

# Bin the tau max results -----------------------------------------------------
```

```r
tau_max_results[, det_thresh:= fcase(
  resolution == "0.2deg", 500,
  resolution == "0.4deg", 2000,
  resolution == "1deg", 12000)]

tau_max_results[, tau_bin:= fcase(
  is.na(tau_max_ha), "Agreement",
  tau_max_ha <= 2 * det_thresh, "Low disagreement",
  tau_max_ha <= 5 * det_thresh, "Moderate disagreement",
  tau_max_ha <= 10 * det_thresh, "High disagreement",
  default = "Extreme disagreement"
)]

tau_max_results[, tau_bin:= factor(tau_bin,levels = c("Agreement",
                                                       "Low disagreement",
                                                       "Moderate disagreement",
                                                       "High disagreement",
                                                       "Extreme disagreement"),
                                    ordered = TRUE)]

# Proportion of cells disagreeing at 1% threshold-------------------------------

tau_max_results[, .N, .(tau_bin, resolution)] %>%
  .[, fraction := N / sum(N), by = resolution] %>%
  .[!tau_bin == "Agreement", sum(fraction), resolution]
```

```
##    resolution        V1
##        <char>     <num>
## 1:     0.2deg 0.5508904
## 2:     0.4deg 0.4590500
## 3:       1deg 0.3799635
```

```r
thresh_df <- unique(tau_max_results[!is.na(det_thresh), .(resolution, det_thresh)])

# ANALYSIS OF DATA -------------------------------------------------------------

# Fraction of high and extreme ontological disagreement
# across resolutions in each continent -----------------------------------------

dt_all <- tau_max_results[scenario == "all"]
dt_all[, N_total := .N, by = .(resolution, continent)]
frac_by_rcb <- dt_all[, .(frac = .N / unique(N_total)),
                      .(resolution, continent, tau_bin)]

frac_by_rcb[, .(min = min(frac), max = max(frac)), .(continent, tau_bin)] %>%
  .[tau_bin %in% c("Extreme disagreement", "High disagreement")] %>%
  .[order(-min)] %>%
```

```r
  .[, .(min = min(min), max = max(max)), .(continent)]
```

```
##    continent        min       max
##       <char>      <num>     <num>
## 1:      Asia 0.10417315 0.3561106
## 2:   Oceania 0.10869565 0.2256921
## 3:    Europe 0.08386916 0.1890844
## 4:  Americas 0.05267170 0.1500000
## 5:    Africa 0.03516442 0.1117143
```

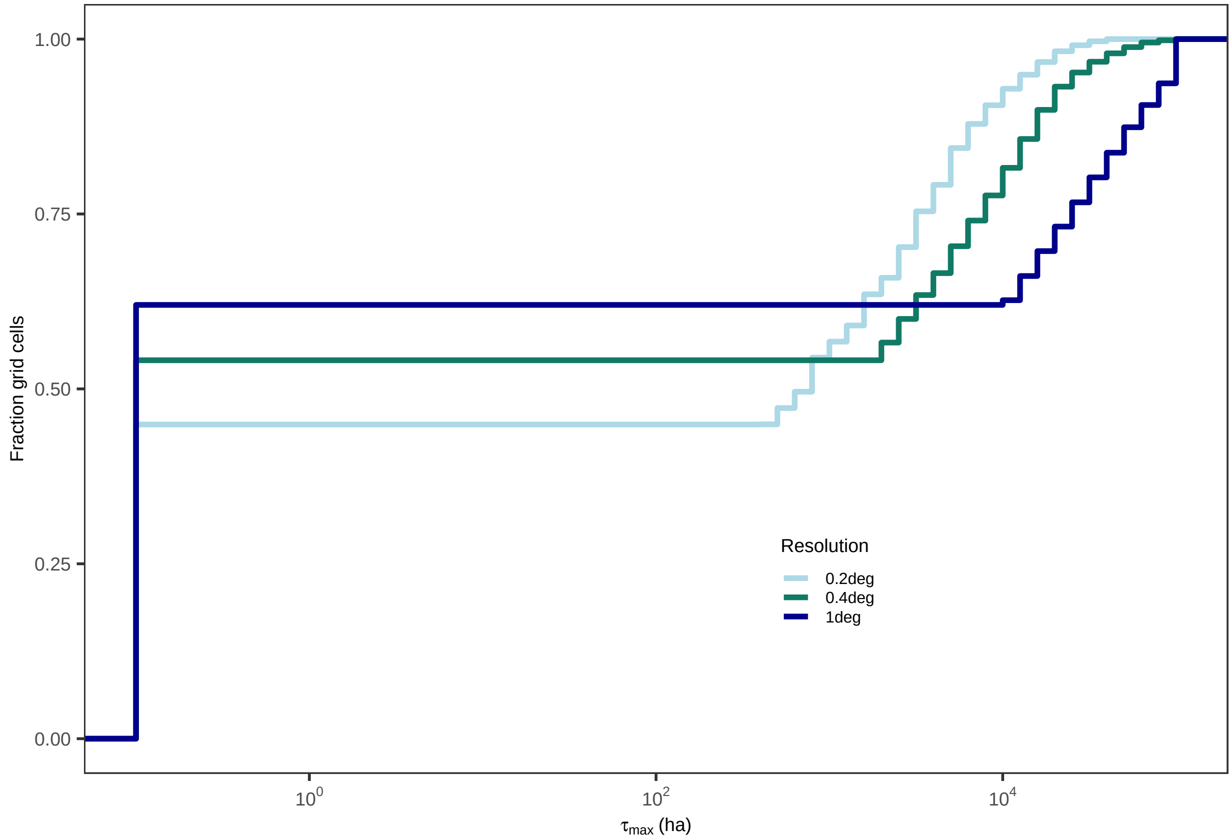
```r
# distribution of detectability limits

tau_max_results[, .N, .(tau_max_ha, resolution, scenario)]
```

```
##      tau_max_ha resolution scenario     N
##           <num>     <char>   <char> <int>
##   1:         NA     0.2deg      all 55250
##   2:   1584.893     0.2deg      all  5919
##   3:   1000.000     0.2deg      all  2969
##   4:   5011.872     0.2deg      all  7187
##   5:   3981.072     0.2deg      all  5088
##  ---
## 588:  31622.777       1deg     spam   291
## 589:  25118.864       1deg     spam   281
## 590: 100000.000       1deg     spam   533
## 591:  63095.734       1deg     spam   259
## 592:  79432.823       1deg     spam   257
```

```r
tmp <- copy(tau_max_results)
tmp[is.na(tau_max_ha), tau_max_ha:= 0.1]

plot_ecdf <- ggplot(tmp, aes(x = tau_max_ha, color = resolution)) +
  stat_ecdf(size = 1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_color_manual(values = res_palette) +
  labs(x = expression(tau[max]~"(ha)"), y = "Fraction grid cells",
       color = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.65, 0.25))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

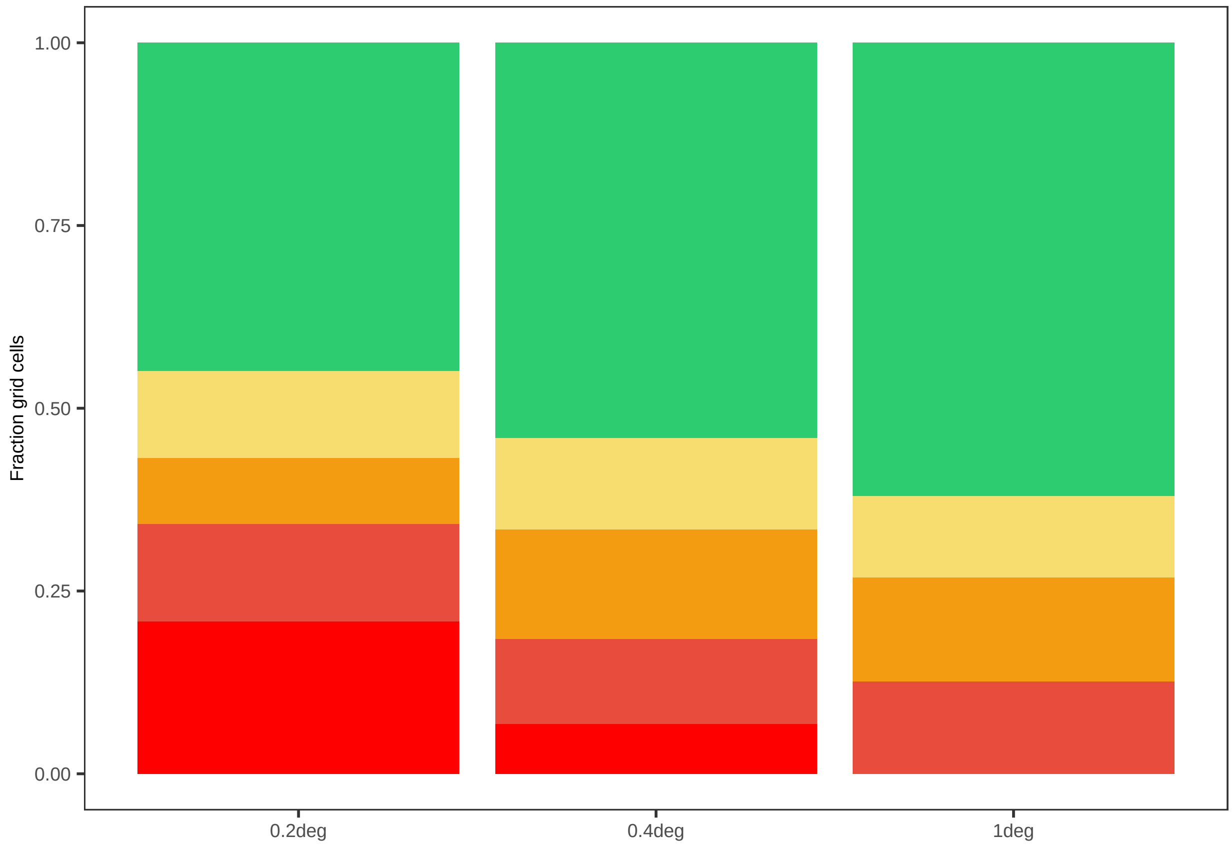
```r
plot_ecdf
```




```r
# How much of the world is perfectably detectable vs fundamentally ambiguous?

tau_max_results[, .N, by = .(resolution, tau_bin)] %>%
.[, frac:= N / sum(N), by = resolution] %>%
  print()
```

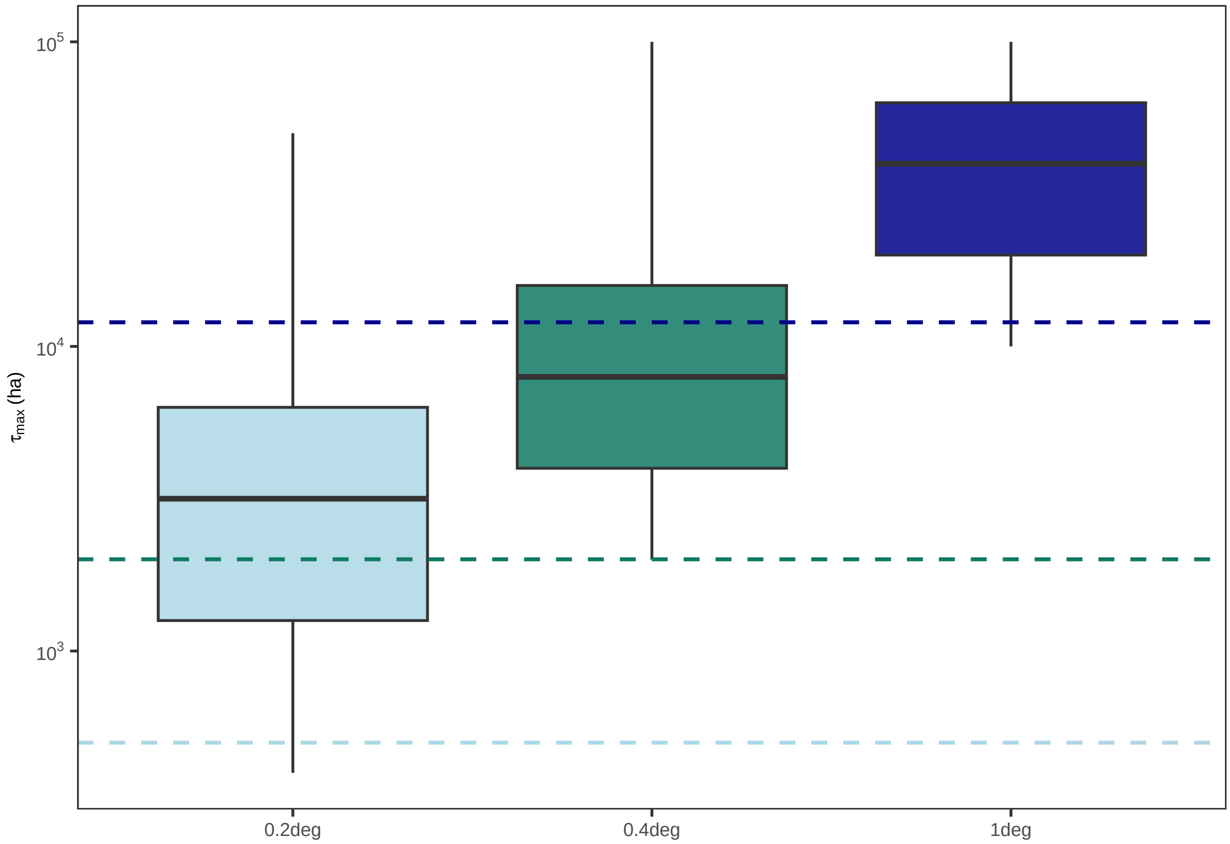
```
##     resolution                tau_bin      N       frac
##         <char>                  <ord>  <int>      <num>
##  1:     0.2deg              Agreement 640389 0.44910962
##  2:     0.2deg Moderate disagreement 130164 0.09128499
##  3:     0.2deg      Low disagreement 168896 0.11844803
##  4:     0.2deg  Extreme disagreement 297053 0.20832550
##  5:     0.2deg     High disagreement 189406 0.13283185
##  6:     0.4deg              Agreement 236578 0.54095002
##  7:     0.4deg Moderate disagreement  65807 0.15047172
##  8:     0.4deg      Low disagreement  54457 0.12451925
##  9:     0.4deg     High disagreement  50769 0.11608641
## 10:     0.4deg  Extreme disagreement  29727 0.06797260
## 11:       1deg              Agreement  55027 0.62003651
## 12:       1deg      Low disagreement   9935 0.11194618
## 13:       1deg Moderate disagreement  12605 0.14203137
## 14:       1deg     High disagreement  11181 0.12598594
```

```r
plot_stacked <- tau_max_results[, .N, by = .(resolution, tau_bin)] %>%
  .[, frac := N / sum(N), by = resolution] |>
  ggplot(aes(resolution, frac, fill = tau_bin)) +
  geom_col() +
  scale_y_continuous() +
  scale_fill_manual(values = tau_palette, name = expression(tau[max])) +
  labs(y = "Fraction grid cells", x = NULL, fill = expression(tau[max])) +
  theme_AP() +
  theme(legend.position = "none")

plot_stacked
```

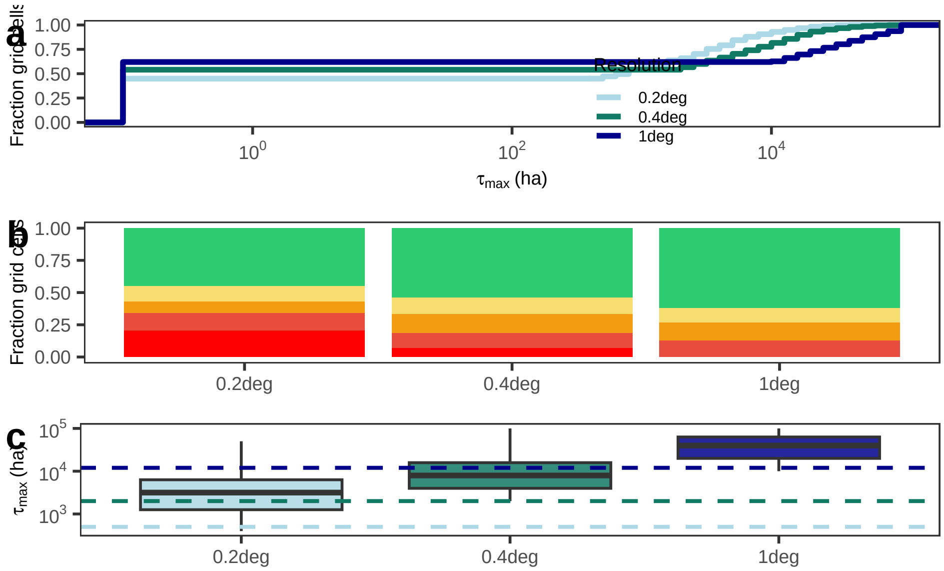



```r
# Boxplots of tau max per resolution -------------------------------------------

plot_box <- ggplot(tau_max_results[!is.na(tau_max_ha) & tau_max_ha > 0],
                   aes(resolution, tau_max_ha, fill = resolution)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.85) +
  geom_hline(data = thresh_df, aes(yintercept = det_thresh, color = resolution),
             linetype = "dashed", linewidth = 0.7, show.legend = FALSE) +
  scale_y_log10(labels = scales::label_log(digits = 2),
                name    = expression(tau[max]~"(ha)")) +
  scale_fill_manual(values = res_palette, guide = "none") +
  scale_color_manual(values = res_palette, guide = "none") +
  labs(
```

```r
    x = NULL,
  ) +
  theme_AP() +
  theme()

plot_box
```

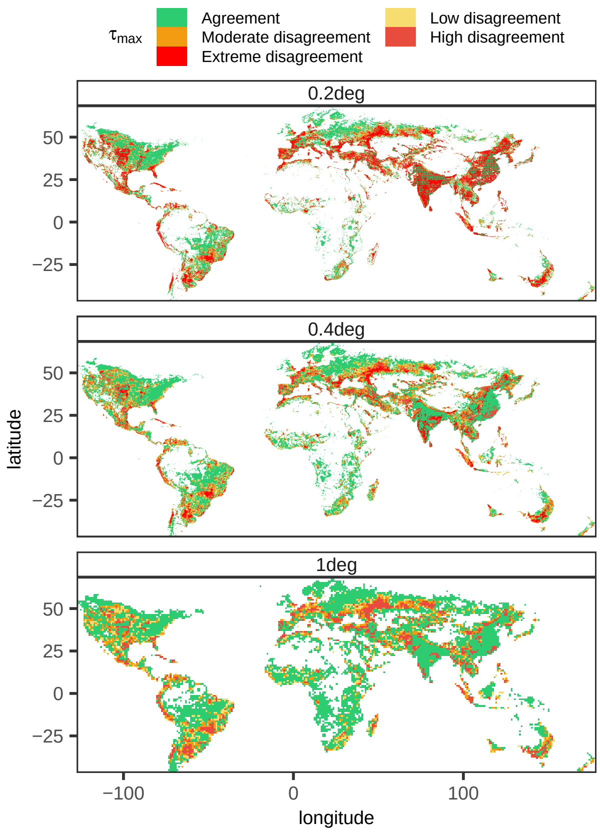



```r
# MERGE TAU MAX PLOTS ##########################################################

top <- plot_grid(plot_ecdf, plot_stacked, plot_box, ncol = 1,
                 rel_heights = c(0.35, 0.35, 0.3),
          labels = "auto")

top
```




```
# The map ####################################################################

plot_raster <- ggplot(tau_max_results, aes(lon, lat, fill = tau_bin)) +
  geom_raster() +
  coord_equal(expand = FALSE) +
  facet_wrap(~ resolution, ncol = 1) +
  scale_fill_manual(values = tau_palette, name = expression(tau[max])) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  theme_AP() +
  theme(panel.grid = element_blank(),
        strip.text = element_text(size = 11),
        legend.position = "top") +
  labs(x = "longitude", y = "latitude") +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

plot_raster
```

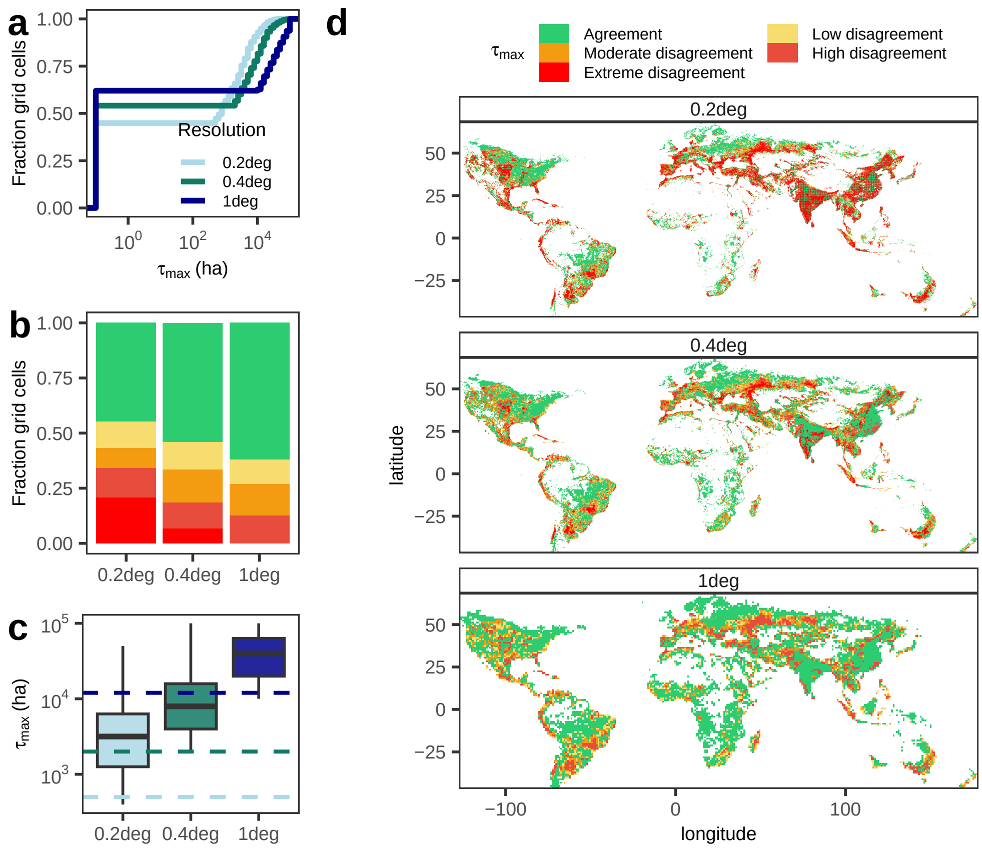
```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```



3.1 Composite figure

```
# MERGE PLOTS ####################################################################

plot_grid(top, plot_raster, rel_widths = c(0.3, 0.7), ncol = 2, labels = c("", "d"))
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```



3.1.1 Which datasets drive deep disagreement?

```
# WHICH DATASETS DRIVE DEEP DISAGREEMENT? ---------------------------------------

# Check leave one-out scenarios -------------------------------------------------

loo_stats_full <- tau_max_results[, .(
    median_tau = median(tau_max_ha, na.rm = TRUE),
    p75_tau = quantile(tau_max_ha, 0.75, na.rm = TRUE),
    p90_tau = quantile(tau_max_ha, 0.90, na.rm = TRUE),
    share_gt_1000 = mean(tau_max_ha >= 1000, na.rm = TRUE)),
    scenario] %>%
  .[order(-p90_tau)]

loo_stats_full
```

```
##      scenario median_tau   p75_tau  p90_tau share_gt_1000
##        <char>      <num>     <num>    <num>         <num>
## 1:        all   3981.072 10000.000 19952.62     0.8705251
## 2:    gaez_v4   3981.072 10000.000 19952.62     0.8700602
## 3:       giam   3981.072  7943.282 19952.62     0.8558969
## 4:       gmia   3981.072 10000.000 19952.62     0.8704727
## 5:      gripc   3981.072  7943.282 19952.62     0.8650565
```

```
##  6:     meier   3981.072 10000.000 19952.62     0.8693391
##  7: mirca2000   3981.072 10000.000 19952.62     0.8721156
##  8:  mirca_os   3981.072 10000.000 19952.62     0.8712516
##  9:   nagaraj   3981.072 10000.000 19952.62     0.8587307
## 10:      spam   3981.072 10000.000 19952.62     0.8702693
## 11:      luh2   3162.278  7943.282 15848.93     0.8537190
```

```r
# In half of all grid cells worldwide, at least one dataset reports >=1000 ha
# of irrigation while another reports none. Removing most datasets does not
# change anything. Removing GIAM, Nagaraj, LUH2 and GRIPC drops the median disagreement
# and the extreme values, as they particularly increase existential contradictions
# in the upper half of the distribution.

# In which regions is each dataset structurally responsible for
# deep existential contradictions? -----------------------------------------------

# Compute spatial delta -----------

tau_all <- tau_max_results[scenario == "all",.(lon, lat, resolution,
                                               tau_all = tau_max_ha)]

# Keep only datasets of interest--

tau_loo <- tau_max_results[scenario %in% c("giam", "luh2", "nagaraj", "gripc"),
                           .(lon, lat, country, continent, resolution, scenario,
                             tau_loo = tau_max_ha)]

# Merge ---------------------------

tau_delta <- merge(tau_loo, tau_all, by = c("lon", "lat", "resolution"),
                   all.x = TRUE)

# Compute difference --------------

tau_delta[, delta:= tau_all - tau_loo]

# Let us define "strong amplification" of differences --------------------------
# Differences larger than 1000 ha.

tau_delta[, strong:= delta >= 1000]

# Compute overlap ---------------------------------------------------------------

influence_matrix <- dcast(tau_delta[resolution == "0.2deg"],
                          lon + lat ~ scenario, value.var = "strong",
                          fill = FALSE)

influence_matrix[, n_datasets:= rowSums(.SD), .SDcols = c("giam", "luh2", "nagaraj", "gripc")]
```

```r
table(influence_matrix$n_datasets)
```

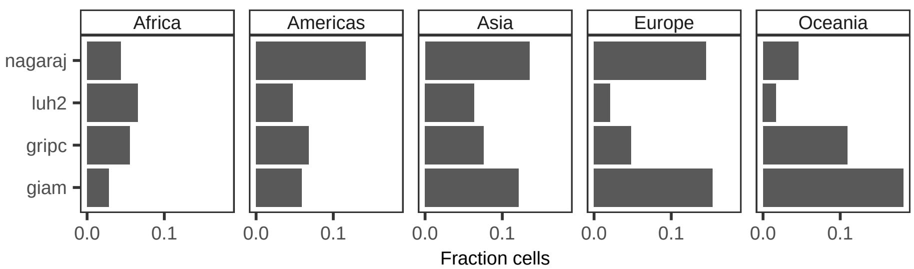
```
##
##     0     1
## 27483 18737
```

```r
# 0: deep disagreement in the cell not driven by these three.
# 1: dataset-specific amplification.
# 2: partial overlap.
# 3: same hotspot amplified by all three.

# Since there are zero cells where two datasets simultaneously aplify deep
# disagreement (> 1000 ha) or all three do, it means that their influence is
# spatially distinct. 16,511 cells show strong amplification but always due
# to exactly one dataset.

# Where does amplification concentrate? ----------------------------------------

regional_amp <- tau_delta[resolution == "0.2deg",
                          .(median_delta = median(delta, na.rm = TRUE),
                            share_strong = mean(delta >= 1000, na.rm = TRUE)),
                          .(continent, scenario)] %>%
  .[order(-median_delta)]


regional_amp
```

```
##     continent scenario median_delta share_strong
##        <char>   <char>        <num>        <num>
##  1:  Americas     giam            0   0.05904022
##  2:  Americas    gripc            0   0.06811812
##  3:  Americas     luh2            0   0.04731494
##  4:  Americas  nagaraj            0   0.14193931
##  5:    Europe     giam            0   0.15351445
##  6:    Europe    gripc            0   0.04808959
##  7:    Europe     luh2            0   0.02085566
##  8:    Europe  nagaraj            0   0.14524721
##  9:    Africa     giam            0   0.02829181
## 10:    Africa    gripc            0   0.05506249
## 11:    Africa     luh2            0   0.06565308
## 12:    Africa  nagaraj            0   0.04397835
## 13:      Asia     giam            0   0.12130263
## 14:      Asia    gripc            0   0.07585118
## 15:      Asia     luh2            0   0.06372931
## 16:      Asia  nagaraj            0   0.13518991
## 17:   Oceania     giam            0   0.18177718
## 18:   Oceania    gripc            0   0.10928320
## 19:   Oceania     luh2            0   0.01693667
```

```
## 20:   Oceania  nagaraj           0   0.04547121
##     continent scenario median_delta share_strong
##        <char>   <char>        <num>        <num>
```

```
# Table shows which continents each dataset most strongly affects:
# - Europe: GIAM dominates existential disagreement in Europe, LUH2 barely affects it.
# - Asia: all three datasets contribute, LUH2 contributes less existential disagreement
# - Ocenia: GIAM contributes the largest existential disagreement.
# - Americas: Nagaraj.
# - Africa: Overall amplification much lower, all contribute equally.
# So overall there is no global problem dataset! And since they all show
# median delta of zero everywhere, these datasets do not globally modify the
# disagreement but create region-specific deep contradictions.

ggplot(regional_amp, aes(scenario, share_strong)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_wrap(~continent, ncol = 5) +
  scale_y_continuous(breaks = breaks_pretty(n = 2)) +
  theme_AP() +
  labs(x = "", y = "Fraction cells")
```




```
# Which dataset drives the 22,615 cells? -------------------------------------

influence_matrix[, dominant := fifelse(giam, "GIAM",
                                       fifelse(luh2, "LUH2",
                                               fifelse(nagaraj, "Nagaraj",
                                                       fifelse(gripc, "GRIPC",
                                                       NA_character_))))]

table(influence_matrix$dominant)
```

```
##
##    GIAM   GRIPC    LUH2 Nagaraj
##    6527    3361    2754    8259
```

3.2 How many grid cells disagree?

```
# DETECTABILITY OF IRRIGATED AREA ACROSS MAPS ####################################
#################################################################################
```

```r
# UNWEIGHTED: ALL CELLS COUNT EQUALLY ##########################################
# QUESTION ANSWERED: HOW MANY GRID CELLS DISAGREE? -----------------------------

# Make dataset wide and retrieve only the 0.2deg resolution one ----------------

dt_wide <- dt[resolution == "0.2deg"] %>%
  dcast(., lon + lat + country + code + continent ~ dataset, value.var = "mha")

# Retrieve 0.2deg resolution from tau max dataset and remove unwanted cols -----

tau_max_results_clean <- tau_max_results[resolution == "0.2deg" & scenario == "all"]
tau_max_results_clean[, resolution:= NULL]
tau_max_results_clean[, scenario:= NULL]

# Merge ------------------------------------------------------------------------

dt2 <- merge(dt_wide, tau_max_results_clean[, .(lon, lat, tau_max_mha)],
             by = c("lon", "lat"), all.x = TRUE)

setnames(dt2, dataset_names, new_dataset_names)

# Convert to ha for easier interpretation --------------------------------------

dt2[, tau_max_ha:= tau_max_mha * 1e6]

# Which percentage of the world is in disagreement on whether there is
# or there is not irrigation? --------------------------------------------------

frac_cells_nonid <- dt2[, mean(!is.na(tau_max_mha))]
frac_cells_nonid
```

```
## [1] 0.5737804
```

```r
summary(dt2$tau_max_ha)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   398.1  1584.9  3162.3  5575.2  6309.6 50118.7   55250
```

```r
# Minimum = 398 ha, the detectability threshold for 0.2. No disagreement below it.
#
# Median disagreement (50%) is eight times the detectability threshold (400 ha)
# 75% of disagreeing cells exceed 2.5x detectability.
# Half exceed 8x detectability

# Even after imposing a conservative 1% rule some cells show contradictions at
# tens of thousands of hectares; these are large irrigation landscapes, not
# fragmented patches.

# Existential contradictions occur at scales that are agronomically and
```

```
# hydrologically meaningful and they are not confined to marginal lands. Include
# cells where irrigation is either extensive or completely absent depending on
# the dataset.Ddisagreement is not about "how much" irrigation exists but about
# whether large irrigation systems exist at all in certain locations.
```

3.3 How much irrigated land lies in disagreeement?

```
# WEIGHTED: CELLS WITH LARGER IRRIGATION COUNT MORE ############################
# QUESTION: HOW MUCH IRRIGATED LAND LIES IN DISAGREEMENT? ----------------------

# Keep only cells with existential disagreement -------------------------------
# (tau_max_ha non-NA)

dt2_disagree <- dt2[!is.na(tau_max_ha)]

# Sanity check: no NA tau_max_ha left
summary(dt2_disagree$tau_max_ha)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   398.1  1584.9  3162.3  5575.2  6309.6 50118.7
```

```
# should show NA's = 0

# Compute total and non-identifiable area per dataset -------------------------

area_nonid <- rbindlist(lapply(new_dataset_names, function(d) {

  # cells where this dataset has irrigation AND there is disagreement

  dt_d <- dt2[get(d) > 0, .(tau_max_ha, area = get(d))]

  # total irrigated area (for this dataset) in cells with disagreement

  total_area <- dt_d[, sum(area, na.rm = TRUE)]

  # irrigated area in cells where tau_max_ha is not NA; that is, in cells
  # that disagree

  nonid_area <- dt_d[!is.na(tau_max_ha), sum(area, na.rm = TRUE)]

  data.table(dataset = d, total_area_mha = total_area,
             nonid_area_mha = nonid_area, share_nonid = nonid_area / total_area)
}))

area_nonid
```

```
##        dataset total_area_mha nonid_area_mha share_nonid
##         <char>          <num>          <num>       <num>
```

```
## 1: GAEZ+2015      280.2879     154.70438  0.5519481
## 2:       GIAM      335.1898     204.36911  0.6097116
## 3:       GMIA      245.8296     131.80237  0.5361534
## 4:      GRIPC      305.9983     171.29686  0.5597967
## 5:       LUH2      262.3541      66.70482  0.2542549
## 6:      Meier      354.6320     209.34845  0.5903259
## 7: MIRCA 2000     210.5253     108.83424  0.5169652
## 8:    MIRCA OS     265.5109     140.13459  0.5277921
## 9:     Nagaraj     285.7831     186.01552  0.6508975
## 10:   SPAM2010    219.5544     112.55164  0.5126368
```

```
# Depending on dataset, between 50-70% of all global irrigated areas
# lie in areas where we do not even agree on whether there is irrigation at all.

# How much tolerance do we need to make datasets agree?-------------------------

tau_diag <- rbindlist(lapply(new_dataset_names, function(d) {

  dt_d <- dt2_disagree[get(d) > 0, .(tau_max_ha, area = get(d))]

  if (nrow(dt_d) == 0) {
    return(data.table(dataset = d,
                      tau_50 = NA_real_,
                      tau_80 = NA_real_,
                      tau_90 = NA_real_,
                      area_share_1000 = NA_real_  ))
  }

  # area-weighted empirical CDF of tau_max_ha (for this dataset in disagreeing cells)

  tmp <- dt_d[, .(area = sum(area)), by = tau_max_ha][order(tau_max_ha)]
  tmp[, cum_area  := cumsum(area)]
  tmp[, cum_share := cum_area / sum(area)]

  # tau_50: minimum tau required to make 50% of irrigated area identifiable
  # tau_80: tau required for 80%
  # tau_90: tau required for 90%

  tau_50 <- tmp[cum_share >= 0.5][1, tau_max_ha]
  tau_80 <- tmp[cum_share >= 0.8][1, tau_max_ha]
  tau_90 <- tmp[cum_share >= 0.9][1, tau_max_ha]

# share of area that is identifiable under tau_threshold_ha (e.g. 1000 ha)

  area_share_1000 <- tmp[tau_max_ha <= 1000, fifelse(.N == 0, 0,
                                                      max(cum_share, na.rm = TRUE))]

  data.table(dataset = d,
```

```r
              tau_50 = tau_50,
              tau_80 = tau_80,
              tau_90 = tau_90,
              area_share_1000 = area_share_1000)
}))

tau_diag
```

```
##        dataset    tau_50   tau_80   tau_90 area_share_1000
##         <char>     <num>    <num>    <num>           <num>
##  1:  GAEZ+2015 12589.254 25118.86 31622.78     0.024079566
##  2:       GIAM 15848.932 25118.86 31622.78     0.015560853
##  3:       GMIA 12589.254 25118.86 31622.78     0.020758906
##  4:      GRIPC 15848.932 25118.86 31622.78     0.017073267
##  5:       LUH2 10000.000 19952.62 25118.86     0.007732642
##  6:      Meier 12589.254 25118.86 31622.78     0.021551156
##  7: MIRCA 2000 12589.254 25118.86 31622.78     0.025574408
##  8:   MIRCA OS 12589.254 25118.86 31622.78     0.022653569
##  9:    Nagaraj  7943.282 19952.62 25118.86     0.018838929
## 10:   SPAM2010 15848.932 25118.86 31622.78     0.019254211
```

```r
# tau_50 ranges from 8,000 to 16,000 ha.
# tau_80 is 25,000-40,000 ha.
# tau90 is often 31,000-40,000 ha.
#
# At 0.2° resolution (40,000 ha per cell):
# tau_50 equals 12,000-16,000 ha; that is 30-40% of the grid cell.
# tau_90 equals 30,000-40,000 ha; most of the grid cell area.
#
# That means that Tto reconcile 90% of irrigated area in disagreeing cells one
# must tolerate differences approaching the size of an entire grid cell.

# area_share_1000: Under a tolerance of 1000 ha, only 0.6%-2.7% of irrigated area
# in disagreeing cells becomes identifiable.

# PARAGRAPH?
# Area-weighted quantiles of tau_maxreveal that resolving even half of irrigated
# area located in disagreeing cells would require tolerating discrepancies of
# 8,000-16,000 ha. resolving 90% would require accepting differences
# exceeding 30,000 ha in most datasets. Under a tolerance of 1,000 ha,
# less than 3% of irrigated area in disagreeing cells becomes identifiable.
# These values indicate that disagreement persists at scales corresponding to
# substantial irrigation systems rather than marginal patches.
```

4 Consequences for countries

4.1 Identification of irrigation hotspots

```r
# DOES THE DIFFERENT DATASETS IDENTIFY THE SAME TOP IRRIGATION HOTSPOTS? ########

# Identify top 10% hotspots per dataset ----------------------------------------

hotspots <- lapply(new_dataset_names, function(d) {

  tmp <- copy(dt2)
  tmp[, value:= get(d)]

  # rank cells---------------------------

  tmp <- tmp[order(-value)]
  n_hot <- ceiling(0.10 * nrow(tmp))

  tmp[, hotspot:= FALSE]
  tmp[1:n_hot, hotspot:= TRUE]

  tmp[, .(lon, lat, hotspot, dataset = d)]
})

# name the list elements by dataset --------------------------------------------

names(hotspots) <- sapply(hotspots, function(x) unique(x$dataset))

# extract hotspot cell IDs per dataset -----------------------------------------

hotspot_sets <- lapply(hotspots, function(dt) {
  dt[hotspot == TRUE, .(lon, lat)]})

# Country rankings per dataset -------------------------------------------------

country_totals <- rbindlist(lapply(new_dataset_names, function(d) {
  dt2[, .(irrigated_area = sum(get(d), na.rm = TRUE)), country] %>%
    .[, dataset:= d]}))

jaccard_cells <- function(dt1, dt2) {
  set1 <- paste(dt1$lon, dt1$lat)
  set2 <- paste(dt2$lon, dt2$lat)

  length(intersect(set1, set2)) / length(union(set1, set2))
}

# Pairwise jacqard similarity --------------------------------------------------
```

```r
datasets <- names(hotspot_sets)

jaccard_results <- rbindlist(combn(datasets, 2, simplify = FALSE,
                                   FUN = function(pair) {
    d1 <- pair[1]
    d2 <- pair[2]

    data.table(dataset_1 = d1, dataset_2 = d2,
               jaccard = jaccard_cells(hotspot_sets[[d1]], hotspot_sets[[d2]]))
  })
)

jaccard_results[order(jaccard)]
```

```
##      dataset_1  dataset_2   jaccard
##         <char>     <char>     <num>
##  1:       GIAM       LUH2 0.3018328
##  2:      GRIPC       LUH2 0.3175789
##  3:       LUH2    Nagaraj 0.3527785
##  4:       LUH2      Meier 0.3611592
##  5:  GAEZ+2015       LUH2 0.3622320
##  6:       LUH2   SPAM2010 0.3861947
##  7:       GMIA       LUH2 0.3943957
##  8:       LUH2   MIRCA OS 0.3944707
##  9:       GIAM    Nagaraj 0.4022392
## 10:       LUH2 MIRCA 2000 0.4062703
## 11:  GAEZ+2015       GIAM 0.4131691
## 12:      GRIPC    Nagaraj 0.4160249
## 13:       GIAM   SPAM2010 0.4306368
## 14:       GIAM       GMIA 0.4361844
## 15:       GIAM      GRIPC 0.4400133
## 16:      GRIPC   SPAM2010 0.4416148
## 17:  GAEZ+2015      GRIPC 0.4525183
## 18:       GIAM MIRCA 2000 0.4547189
## 19:       GIAM      Meier 0.4570899
## 20:      GRIPC      Meier 0.4571718
## 21:       GMIA      GRIPC 0.4574175
## 22:       GIAM   MIRCA OS 0.4621024
## 23:      GRIPC   MIRCA OS 0.4734868
## 24:      GRIPC MIRCA 2000 0.4804705
## 25:    Nagaraj   SPAM2010 0.4954144
## 26:  GAEZ+2015    Nagaraj 0.5058373
## 27:   MIRCA OS    Nagaraj 0.5133084
## 28: MIRCA 2000    Nagaraj 0.5171163
## 29:      Meier    Nagaraj 0.5343552
## 30:       GMIA    Nagaraj 0.5484680
## 31:  GAEZ+2015   SPAM2010 0.5702259
```

```
## 32:  GAEZ+2015 MIRCA 2000 0.5815287
## 33:      Meier   SPAM2010 0.5853972
## 34:      Meier MIRCA 2000 0.5887003
## 35:      Meier   MIRCA OS 0.5982000
## 36:  GAEZ+2015      Meier 0.6025467
## 37:  GAEZ+2015   MIRCA OS 0.6276996
## 38:  GAEZ+2015       GMIA 0.6362259
## 39: MIRCA 2000   SPAM2010 0.6371558
## 40:   MIRCA OS   SPAM2010 0.6599014
## 41: MIRCA 2000   MIRCA OS 0.6619231
## 42:       GMIA      Meier 0.6752391
## 43:       GMIA MIRCA 2000 0.6842721
## 44:       GMIA   SPAM2010 0.6969499
## 45:       GMIA   MIRCA OS 0.7250649
##      dataset_1  dataset_2   jaccard
##         <char>     <char>     <num>
```

```
jaccard_results[, .(min_jaccard = min(jaccard),
                    median_jaccard = median(jaccard),
                    mean_jaccard = mean(jaccard),
                    max_jaccard = max(jaccard))]
```

```
##    min_jaccard median_jaccard mean_jaccard max_jaccard
##          <num>          <num>        <num>       <num>
## 1:   0.3018328      0.4734868    0.5021623   0.7250649
```

```
# results:
# 1: both datasets identify the same hotspots
# 0: datasets disagree completely
# 0.5: half overlap, half disagreement

# No pair of global irrigation datasets agrees strongly on where
# irrigation hotspots are.

# Weak agreement (0.35 - 0.45):
# -----------------------------
# one dataset says "this is a core irrigated region",
# the other dataset says "no, it is not".

# Moderate agreement (0.5-0.65)
# -------------------------------
# one third to one half of hotspots do not overlap

# High agreement (0.7 - 0.8)
# -------------------------------
# High agreement only between specific subsets because of shared lineage
# (GMIA and Meier; MIRCA2000 and Meier, etc)
```

4.2 National rankings

```r
# NATIONAL RANKINGS USING ALL CELLS ############################################

# Country totals & ranks .-----------------------------------------------------

country_totals_all <- compute_country_totals(dt2, new_dataset_names)

# Pairwise rank correlations --------------------------------------------------

rank_corr_all <- compute_rank_correlations(country_totals_all)
rank_corr_all[order(-spearman)]
```

```
##      dataset_1  dataset_2   kendall  spearman
##         <char>     <char>     <num>     <num>
##  1: MIRCA 2000   MIRCA OS 0.8869269 0.9792934
##  2:       GMIA MIRCA 2000 0.8899580 0.9764568
##  3:       GMIA   MIRCA OS 0.8782725 0.9726811
##  4:      Meier MIRCA 2000 0.8613228 0.9701687
##  5:       GMIA      Meier 0.8605534 0.9678578
##  6:       GMIA       LUH2 0.8796674 0.9675411
##  7:      Meier   MIRCA OS 0.8402259 0.9608881
##  8:       GMIA    SPAM2010 0.8229857 0.9543192
##  9:       LUH2 MIRCA 2000 0.8326323 0.9521371
## 10:       LUH2   MIRCA OS 0.8306333 0.9516782
## 11:   MIRCA OS   SPAM2010 0.8140963 0.9469802
## 12:       LUH2   SPAM2010 0.8126740 0.9454728
## 13: MIRCA 2000   SPAM2010 0.8033644 0.9443709
## 14:  GAEZ+2015      Meier 0.8019412 0.9432197
## 15:  GAEZ+2015 MIRCA 2000 0.8032595 0.9423526
## 16:      Meier   SPAM2010 0.7935570 0.9393322
## 17:       LUH2      Meier 0.8078651 0.9361524
## 18:  GAEZ+2015   MIRCA OS 0.7958000 0.9351266
## 19:  GAEZ+2015       GMIA 0.7845397 0.9275645
## 20:      Meier    Nagaraj 0.7802484 0.9145597
## 21: MIRCA 2000    Nagaraj 0.7657801 0.9093070
## 22:       GMIA    Nagaraj 0.7570546 0.9069830
## 23:       GIAM      Meier 0.7434964 0.9066388
## 24:  GAEZ+2015       LUH2 0.7415172 0.9043953
## 25:   MIRCA OS    Nagaraj 0.7454631 0.9032900
## 26:  GAEZ+2015   SPAM2010 0.7309457 0.9002531
## 27:       GIAM MIRCA 2000 0.7256531 0.8932012
## 28:    Nagaraj   SPAM2010 0.7171736 0.8807849
## 29:       GIAM    Nagaraj 0.7057269 0.8777524
## 30:  GAEZ+2015    Nagaraj 0.7107326 0.8765867
## 31:       GIAM       GMIA 0.7023198 0.8763834
## 32:       GMIA      GRIPC 0.6971987 0.8749677
## 33:      GRIPC   MIRCA OS 0.6966780 0.8748337
```

```
## 34:      GRIPC MIRCA 2000 0.6958898 0.8692777
## 35:       GIAM   MIRCA OS 0.6917915 0.8674594
## 36:      GRIPC      Meier 0.6927134 0.8673210
## 37:       GIAM   SPAM2010 0.6834225 0.8653044
## 38:      GRIPC   SPAM2010 0.6796605 0.8649194
## 39:       LUH2    Nagaraj 0.7021109 0.8635624
## 40:  GAEZ+2015       GIAM 0.6798635 0.8568604
## 41:      GRIPC       LUH2 0.6783125 0.8541425
## 42:       GIAM       LUH2 0.6694764 0.8450648
## 43:      GRIPC    Nagaraj 0.6514905 0.8300450
## 44:  GAEZ+2015      GRIPC 0.6442363 0.8237471
## 45:       GIAM      GRIPC 0.6289032 0.8063916
##      dataset_1  dataset_2   kendall  spearman
##         <char>     <char>     <num>     <num>
```

```
# Pairwise top-N overlap (e.g. top 20 irrigators) ------------------------------

topN_overlap_all <- compute_topN_overlap(country_totals_all,
                                         dataset_names = new_dataset_names,
                                         N = 20)
topN_overlap_all[order(-jaccard_topN)]
```

```
##      dataset_1  dataset_2     N intersection_N jaccard_topN
##         <char>     <char> <num>          <int>        <num>
##  1:       GMIA       LUH2    20             19    0.9047619
##  2:       GMIA MIRCA 2000    20             19    0.9047619
##  3:       LUH2      Meier    20             19    0.9047619
##  4:       GIAM MIRCA 2000    20             18    0.8181818
##  5:       GMIA      Meier    20             18    0.8181818
##  6:       LUH2 MIRCA 2000    20             18    0.8181818
##  7:      Meier MIRCA 2000    20             18    0.8181818
##  8:      Meier   MIRCA OS    20             18    0.8181818
##  9:       GIAM       GMIA    20             17    0.7391304
## 10:       GIAM       LUH2    20             17    0.7391304
## 11:       GIAM      Meier    20             17    0.7391304
## 12:       GMIA   MIRCA OS    20             17    0.7391304
## 13:       LUH2   MIRCA OS    20             17    0.7391304
## 14:      Meier    Nagaraj    20             17    0.7391304
## 15:      Meier   SPAM2010    20             17    0.7391304
## 16: MIRCA 2000   MIRCA OS    20             17    0.7391304
## 17: MIRCA 2000    Nagaraj    20             17    0.7391304
## 18:   MIRCA OS   SPAM2010    20             17    0.7391304
## 19:  GAEZ+2015      GRIPC    20             16    0.6666667
## 20:       GIAM      GRIPC    20             16    0.6666667
## 21:       GIAM   MIRCA OS    20             16    0.6666667
## 22:       GIAM    Nagaraj    20             16    0.6666667
## 23:       GIAM   SPAM2010    20             16    0.6666667
## 24:       GMIA    Nagaraj    20             16    0.6666667
```

```
## 25:      GRIPC MIRCA 2000    20              16    0.6666667
## 26:       LUH2    Nagaraj    20              16    0.6666667
## 27:       LUH2   SPAM2010    20              16    0.6666667
## 28: MIRCA 2000   SPAM2010    20              16    0.6666667
## 29:   MIRCA OS    Nagaraj    20              16    0.6666667
## 30:    Nagaraj   SPAM2010    20              16    0.6666667
## 31:  GAEZ+2015 MIRCA 2000    20              15    0.6000000
## 32:       GMIA      GRIPC    20              15    0.6000000
## 33:       GMIA   SPAM2010    20              15    0.6000000
## 34:      GRIPC       LUH2    20              15    0.6000000
## 35:      GRIPC      Meier    20              15    0.6000000
## 36:      GRIPC   MIRCA OS    20              15    0.6000000
## 37:  GAEZ+2015       GIAM    20              14    0.5384615
## 38:  GAEZ+2015       GMIA    20              14    0.5384615
## 39:  GAEZ+2015   MIRCA OS    20              14    0.5384615
## 40:  GAEZ+2015    Nagaraj    20              14    0.5384615
## 41:      GRIPC   SPAM2010    20              14    0.5384615
## 42:  GAEZ+2015       LUH2    20              13    0.4814815
## 43:  GAEZ+2015      Meier    20              13    0.4814815
## 44:      GRIPC    Nagaraj    20              13    0.4814815
## 45:  GAEZ+2015   SPAM2010    20              12    0.4285714
##      dataset_1  dataset_2     N intersection_N jaccard_topN
##         <char>     <char> <num>          <int>        <num>
```

```
# NATIONAL RANKINGS USING ONLY "IDENTIFIABLE" CELLS ############################

# Only areas that agree -------------------------------------------------------

dt_ident <- dt2[is.na(tau_max_ha)]

# Country totals & ranks on identifiable cells only ----------------------------

country_totals_ident <- compute_country_totals(dt_ident, new_dataset_names)

# Pairwise rank correlations (identifiable-only) -------------------------------

rank_corr_ident <- compute_rank_correlations(country_totals_ident)
rank_corr_ident[order(-spearman)]
```

```
##      dataset_1  dataset_2   kendall  spearman
##         <char>     <char>     <num>     <num>
##   1:      GMIA   MIRCA OS 0.8835329 0.9783609
##   2:      GMIA MIRCA 2000 0.8814086 0.9765933
##   3:      GMIA   SPAM2010 0.8545396 0.9657292
##   4: MIRCA 2000   MIRCA OS 0.8595946 0.9652307
##   5:     Meier MIRCA 2000 0.8459998 0.9635799
##   6:      GMIA      Meier 0.8436327 0.9596853
##   7:  MIRCA OS   SPAM2010 0.8333674 0.9594505
```

```
##  8:       LUH2    Nagaraj 0.8977533 0.9561767
##  9: MIRCA 2000   SPAM2010 0.8250296 0.9536111
## 10:  GAEZ+2015 MIRCA 2000 0.8187728 0.9535817
## 11:  GAEZ+2015   MIRCA OS 0.8204898 0.9521117
## 12:  GAEZ+2015       GMIA 0.8146970 0.9500532
## 13:      Meier   SPAM2010 0.8096173 0.9466166
## 14:      Meier   MIRCA OS 0.8119047 0.9429633
## 15:  GAEZ+2015      Meier 0.7973006 0.9353605
## 16:  GAEZ+2015   SPAM2010 0.7717534 0.9270248
## 17:       GIAM MIRCA 2000 0.7717169 0.9150786
## 18:       GIAM   MIRCA OS 0.7601485 0.9139648
## 19:       GIAM   SPAM2010 0.7645711 0.9139506
## 20:       GIAM      Meier 0.7714894 0.9111299
## 21:       GIAM       GMIA 0.7609975 0.9067002
## 22:       GIAM    Nagaraj 0.8031987 0.9044408
## 23:      Meier    Nagaraj 0.7968289 0.8993616
## 24:  GAEZ+2015       GIAM 0.7402983 0.8991512
## 25:    Nagaraj   SPAM2010 0.7761437 0.8971564
## 26:       GMIA    Nagaraj 0.7818313 0.8961145
## 27:   MIRCA OS    Nagaraj 0.7756057 0.8950570
## 28: MIRCA 2000    Nagaraj 0.7814858 0.8935640
## 29:       GMIA       LUH2 0.7711712 0.8901905
## 30:       LUH2 MIRCA 2000 0.7701480 0.8869588
## 31:      GRIPC    Nagaraj 0.7656340 0.8864691
## 32:       LUH2      Meier 0.7698723 0.8856886
## 33:       LUH2   MIRCA OS 0.7580659 0.8823827
## 34:      GRIPC   SPAM2010 0.7197618 0.8816127
## 35:       GMIA      GRIPC 0.7205550 0.8809232
## 36:       LUH2   SPAM2010 0.7535114 0.8787780
## 37:  GAEZ+2015    Nagaraj 0.7444534 0.8763956
## 38:      GRIPC      Meier 0.7150236 0.8731797
## 39:      GRIPC MIRCA 2000 0.7172403 0.8706369
## 40:       GIAM      GRIPC 0.7091910 0.8680663
## 41:      GRIPC       LUH2 0.7301817 0.8674414
## 42:       GIAM       LUH2 0.7445265 0.8669448
## 43:      GRIPC   MIRCA OS 0.7031575 0.8664374
## 44:  GAEZ+2015       LUH2 0.7097899 0.8530376
## 45:  GAEZ+2015      GRIPC 0.6858624 0.8455006
##      dataset_1  dataset_2   kendall  spearman
##         <char>     <char>     <num>     <num>
```

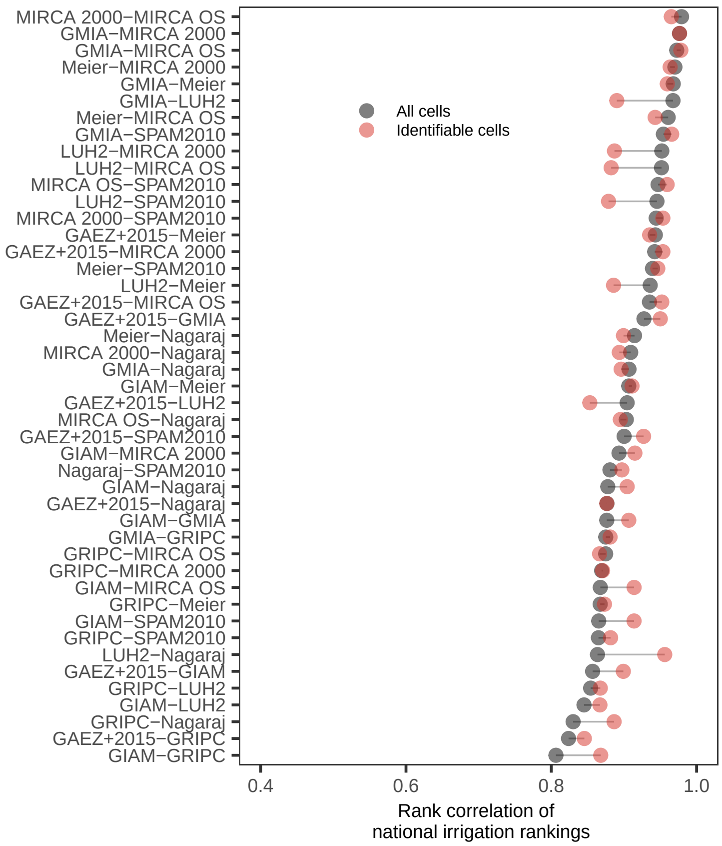
```
# Pairwise top-N overlap (identifiable-only) ----------------------------------

topN_overlap_ident <- compute_topN_overlap(country_totals_ident,
                                           dataset_names = new_dataset_names,
                                           N = 20)
topN_overlap_ident[order(-jaccard_topN)]
```

```
##       dataset_1  dataset_2     N intersection_N jaccard_topN
##          <char>     <char> <num>          <int>        <num>
##  1:        GIAM       GMIA    20             19    0.9047619
##  2:        GIAM      Meier    20             19    0.9047619
##  3:        GIAM    Nagaraj    20             19    0.9047619
##  4:        GMIA MIRCA 2000    20             19    0.9047619
##  5:        GMIA    Nagaraj    20             19    0.9047619
##  6:       Meier   MIRCA OS    20             19    0.9047619
##  7:     Nagaraj   SPAM2010    20             19    0.9047619
##  8:   GAEZ+2015   MIRCA OS    20             18    0.8181818
##  9:        GIAM       LUH2    20             18    0.8181818
## 10:        GIAM MIRCA 2000    20             18    0.8181818
## 11:        GIAM   MIRCA OS    20             18    0.8181818
## 12:        GIAM   SPAM2010    20             18    0.8181818
## 13:        GMIA       LUH2    20             18    0.8181818
## 14:        GMIA      Meier    20             18    0.8181818
## 15:        GMIA   MIRCA OS    20             18    0.8181818
## 16:        GMIA   SPAM2010    20             18    0.8181818
## 17:       GRIPC       LUH2    20             18    0.8181818
## 18:        LUH2    Nagaraj    20             18    0.8181818
## 19:        LUH2   SPAM2010    20             18    0.8181818
## 20:       Meier MIRCA 2000    20             18    0.8181818
## 21:       Meier    Nagaraj    20             18    0.8181818
## 22:       Meier   SPAM2010    20             18    0.8181818
## 23:  MIRCA 2000   MIRCA OS    20             18    0.8181818
## 24:  MIRCA 2000    Nagaraj    20             18    0.8181818
## 25:    MIRCA OS    Nagaraj    20             18    0.8181818
## 26:    MIRCA OS   SPAM2010    20             18    0.8181818
## 27:   GAEZ+2015      Meier    20             17    0.7391304
## 28:   GAEZ+2015    Nagaraj    20             17    0.7391304
## 29:   GAEZ+2015   SPAM2010    20             17    0.7391304
## 30:        GIAM      GRIPC    20             17    0.7391304
## 31:        GMIA      GRIPC    20             17    0.7391304
## 32:       GRIPC      Meier    20             17    0.7391304
## 33:       GRIPC MIRCA 2000    20             17    0.7391304
## 34:       GRIPC    Nagaraj    20             17    0.7391304
## 35:       GRIPC   SPAM2010    20             17    0.7391304
## 36:        LUH2      Meier    20             17    0.7391304
## 37:        LUH2 MIRCA 2000    20             17    0.7391304
## 38:  MIRCA 2000   SPAM2010    20             17    0.7391304
## 39:   GAEZ+2015       GIAM    20             16    0.6666667
## 40:   GAEZ+2015       GMIA    20             16    0.6666667
## 41:   GAEZ+2015      GRIPC    20             16    0.6666667
## 42:   GAEZ+2015 MIRCA 2000    20             16    0.6666667
## 43:       GRIPC   MIRCA OS    20             16    0.6666667
## 44:        LUH2   MIRCA OS    20             16    0.6666667
## 45:   GAEZ+2015       LUH2    20             15    0.6000000
##       dataset_1  dataset_2     N intersection_N jaccard_topN
```

```
##          <char>     <char> <num>          <int>        <num>
```

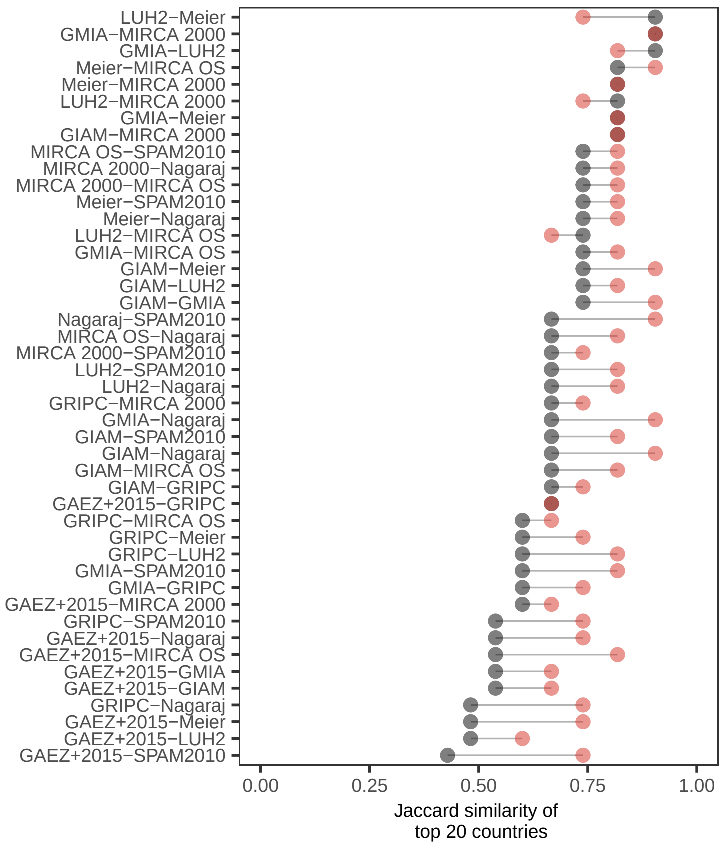
```r
# HOW DO RANKINGS CHANGE WHEN NON-IDENTIFIABLE CELLS ARE MASKED? ################

# How much does each country's rank shift for each dataset? --------------------

rank_change <- merge(
  country_totals_all[, .(country, dataset, rank_all = rank)],
  country_totals_ident[, .(country, dataset, rank_ident = rank)],
  by = c("country", "dataset"),
  all.x = TRUE) %>%
  .[, rank_shift := rank_ident - rank_all]

rank_change

# PLOT RESULTS #################################################################

# Arrange data -----------------------------------------------------------------

rank_corr_all[, pair:= paste(dataset_1, dataset_2, sep = "-")]
rank_corr_ident[, pair:= paste(dataset_1, dataset_2, sep = "-")]

rank_corr_long <- rbind(rank_corr_all[, .(pair, spearman, mode = "All cells")],
                        rank_corr_ident[, .(pair, spearman, mode = "Identifiable cells")])

# order pairs by Spearman (all cells) so the y-axis is readable ----------------

pair_order <- rank_corr_all[order(spearman), pair]
rank_corr_long[, pair := factor(pair, levels = pair_order)]

# Plot rank correlation --------------------------------------------------------

p_rankcorr <- ggplot(rank_corr_long, aes(spearman, pair)) +
  geom_segment(data = dcast(rank_corr_long, pair ~ mode, value.var = "spearman"),
               aes(x = `All cells`, xend = `Identifiable cells`,
                   y = pair, yend = pair),
               inherit.aes = FALSE,
               colour = "grey70", linewidth = 0.3) +
  geom_point(aes(colour = mode), size = 2, alpha = 0.5) +
  scale_x_continuous(limits = c(0.4, 1.0)) +
  scale_color_manual(values = c("Identifiable cells" = "#d73027",
                                 "All cells" = "black")) +
  labs(x = "Rank correlation of \n national irrigation rankings",
       y = "",
       colour = NULL) +
  theme_AP() +
  theme(legend.position = c(0.4, 0.85))
```

```r
p_rankcorr
```



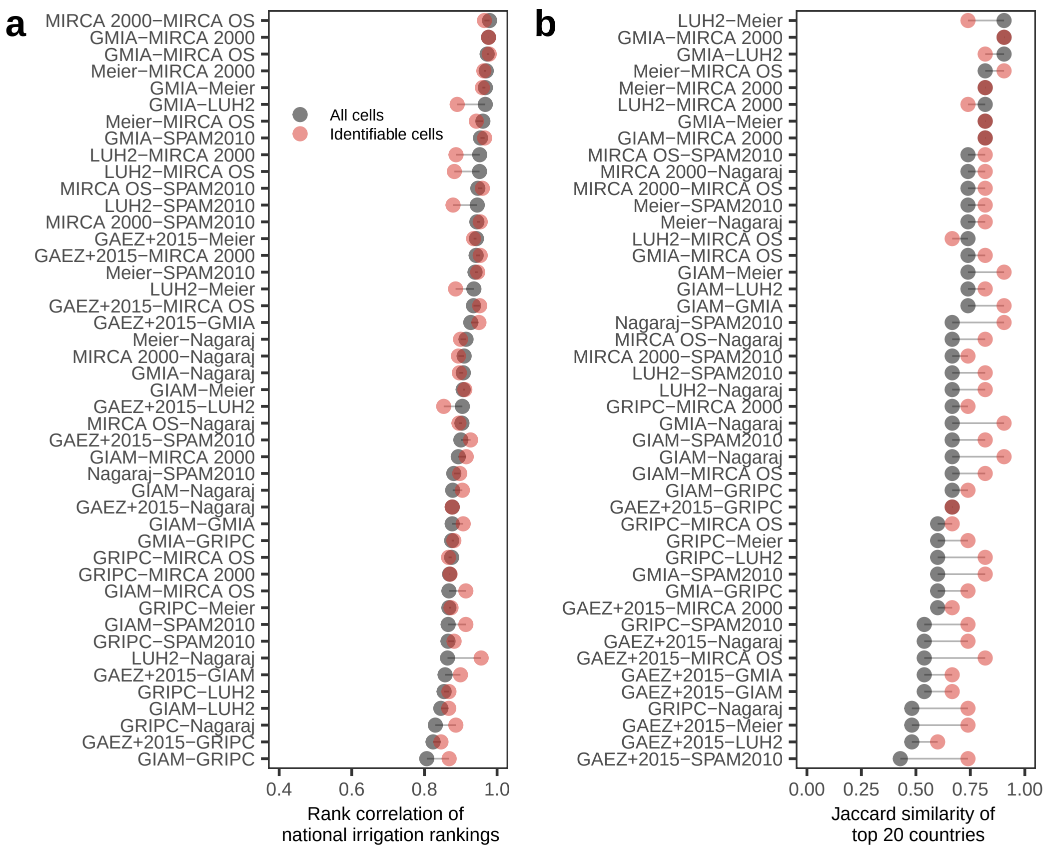

```r
# Only for top N countries -------------------------------------------------

topN_overlap_all[, pair:= paste(dataset_1, dataset_2, sep = "-")]
topN_overlap_ident[, pair:= paste(dataset_1, dataset_2, sep = "-")]

topN_long <- rbind(topN_overlap_all[, .(pair, jaccard_topN, mode = "All cells")],
                   topN_overlap_ident[, .(pair, jaccard_topN, mode = "Identifiable cells")])

# order by All-cells Jaccard for consistency -----------------------------------

pair_order_topN <- topN_overlap_all[order(jaccard_topN), pair]
topN_long[, pair:= factor(pair, levels = pair_order_topN)]

# Plot -----------------------------------------------------------------------

p_topN <- ggplot(topN_long, aes(x = jaccard_topN, y = pair)) +
  geom_segment(data = dcast(topN_long, pair ~ mode, value.var = "jaccard_topN"),
               aes(x = `All cells`, xend = `Identifiable cells`,
                   y = pair, yend = pair), inherit.aes = FALSE, colour = "grey70",
               linewidth = 0.3) +
  geom_point(aes(colour = mode), size = 2, alpha = 0.5) +
  scale_x_continuous(limits = c(0, 1)) +
```

```
  scale_color_manual(values = c("Identifiable cells" = "#d73027",
                                "All cells" = "black")) +
  labs(x = "Jaccard similarity of \n top 20 countries",
       y = "", colour = NULL) +
  theme_AP() +
  theme(legend.position = "none")

p_topN
```




```
# MERGE ######################################################################

# (a) Spearman rank correlation of national total irrigated area rankings for all
# pairwise combinations of datasets. Black points show correlations computed across
# all grid cells, while red points show correlations restricted to identifiable
# cells (i.e., excluding cells where datasets fundamentally disagree on the
# presence of irrigation). Correlations are generally high (0.7-1.0),
# indicating broad agreement in country-level rankings. In some cases removing
# disputed cells increases the correlation (red dot to the right of black dot),
# while in others it does not (red dot to the left of black dot)
#
# (b) Jaccard similarity of the top 20 irrigated countries for each dataset pair.
# Black points again denote results using all cells and red points identifiable cells
# only. Most of the time focusing only on identifiable cells increases the
```

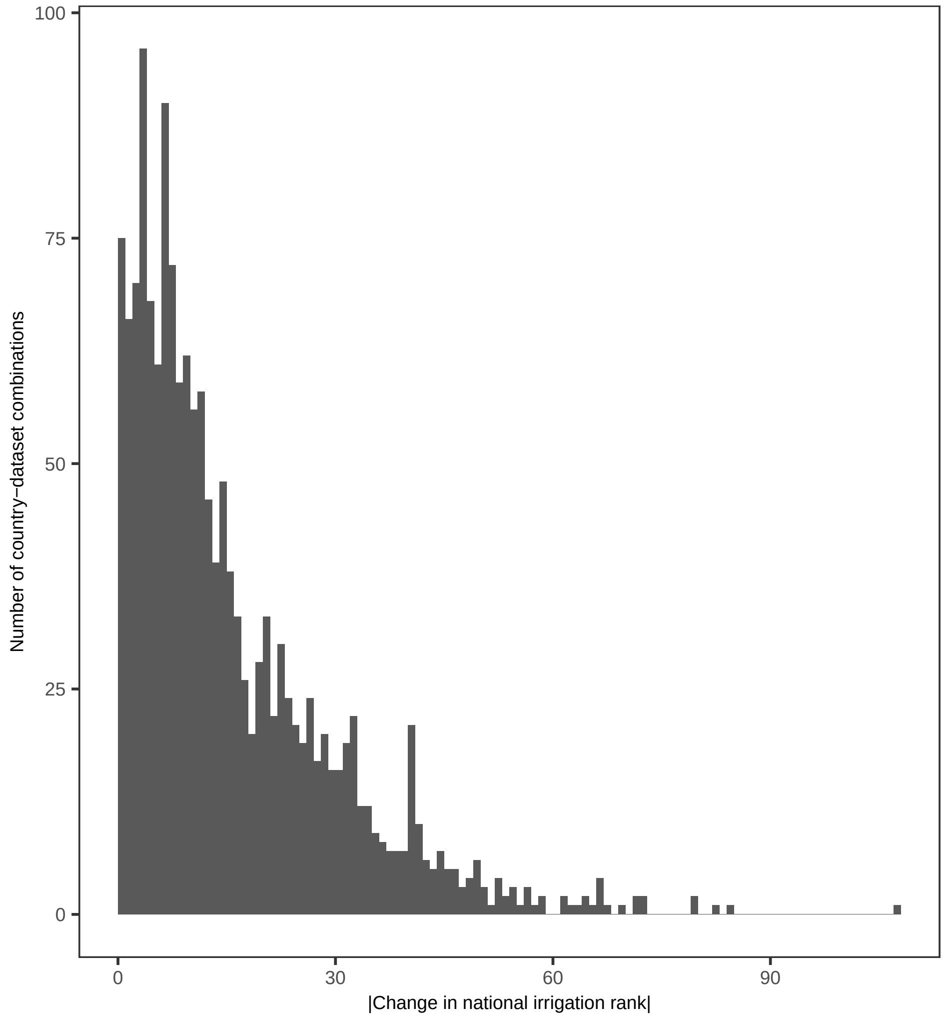
```r
# similarity; meaning that the top 20 become more similar across datasets. In other
# words, once disputed cells are removed there is a more coherent "core" of
# globally dominant irrigation countries. Possibly this is because the strongest
# disagreement disproportionately affect marginal or mid-rank countries that
# occassionally enter or leave the top 20 depending on dataset.

# SO: removing disagreeing cells stabilizes who belongs in the top twenty, but not
# necessarily the full ranking hierarchy across al coutnries.

plot_grid(p_rankcorr, p_topN, ncol = 2, labels = "auto")
```



4.3 Shift in country ranks

```r
# SHIFT IN COUNTRY RANKS ##########################################################

# Total rank shift ----------------------------------------------------------------

rank_change[, abs_shift:= abs(rank_shift)]
rank_change_clean <- rank_change[!is.na(abs_shift)]

# Histogram of rank shift ---------------------------------------------------------

p_rankshift_hist <- ggplot(rank_change_clean, aes(abs_shift)) +
```

```
  geom_histogram(binwidth = 1, boundary = 0, closed = "left") +
  labs(x = "|Change in national irrigation rank|",
       y = "Number of country-dataset combinations") +
  theme_AP()

p_rankshift_hist
```




```
## Cumulative share ------------------------------------------------------------

rank_shift_stats <- rank_change_clean[, .(share_ge_1  = mean(abs_shift >= 1),
                                          share_ge_5  = mean(abs_shift >= 5),
                                          share_ge_10 = mean(abs_shift >= 10),
                                          share_ge_20 = mean(abs_shift >= 20),
                                          share_ge_50 = mean(abs_shift >= 50))]

rank_shift_stats
```

```
##    share_ge_1 share_ge_5 share_ge_10 share_ge_20 share_ge_50
##         <num>      <num>       <num>       <num>       <num>
```

```
## 1:  0.9522293  0.7611465   0.5420382   0.2923567  0.02675159
```

```r
# 98.4% of country-dataset combinations change rank by at least 1 position.
# No country keeps the same rank once we enforce identifiability.
#
# 90% of countries move by at least 5 rank positions
#
# 71% of country-dataset combinations move by 10 ranks or more

# FOCUS ON THE SO-CALLED "TOP IRRIGATORS" ####################################

number_countries <- 20

# Countries consistently classified as top irrigators across datasets ----------

top10_all <- country_totals_all[order(rank), .SD[1:number_countries], dataset] %>%
  .[, .N, by = country] %>%
  .[!country == ""] %>%
  .[N > 5]

top10_all
```

```
##           country     N
##            <char> <int>
##  1:         China    10
##  2:         India    10
##  3: United States    10
##  4:      Pakistan    10
##  5:          Iran    10
##  6:        Mexico    10
##  7:        Brazil    10
##  8:         Egypt     8
##  9:        Turkey    10
## 10:    Uzbekistan    10
## 11:         Italy     8
## 12:         Spain    10
## 13:        Russia     6
## 14:    Kazakhstan     8
## 15:      Thailand     8
## 16:    Bangladesh     8
## 17:       Vietnam     9
## 18:     Indonesia     8
## 19:          Iraq     6
```

```r
# Also with the median ---------------------------------------------------------

top10_median <- country_totals_all[, .(median_rank = median(rank)), country] %>%
  .[order(median_rank)] %>%
  .[1:number_countries]
```

```r
top10_median
```

```
##              country median_rank
##               <char>       <num>
##  1:            China         1.0
##  2:            India         2.0
##  3:    United States         3.0
##  4:         Pakistan         4.0
##  5:             Iran         5.0
##  6:           Mexico         7.0
##  7:        Indonesia         9.0
##  8:         Thailand         9.0
##  9:       Bangladesh        11.0
## 10:           Brazil        11.5
## 11:          Vietnam        12.0
## 12:           Turkey        12.5
## 13:       Uzbekistan        12.5
## 14:            Spain        13.5
## 15:            Egypt        15.0
## 16:            Italy        15.5
## 17:           Russia        15.5
## 18:       Kazakhstan        18.5
## 19:             Iraq        20.0
## 20: Myanmar (Burma)        20.0
##              country median_rank
##               <char>       <num>
```

```r
# See change of ranks in countries when only confirmed irrigation is
# accounted for ----------------------------------------------------------

rank_focus <- merge(country_totals_all[, .(country, dataset, rank_all = rank)],
                    country_totals_ident[, .(country, dataset, rank_ident = rank)],
                    by = c("country", "dataset"), all = TRUE)

rank_focus[, rank_shift:= rank_ident - rank_all]
rank_focus
```

```
## Key: <country, dataset>
##              country    dataset rank_all rank_ident rank_shift
##               <char>     <char>    <num>      <num>      <num>
##    1:    Afghanistan  GAEZ+2015     23.0         29          6
##    2:    Afghanistan       GIAM     37.0         30         -7
##    3:    Afghanistan       GMIA     24.0         27          3
##    4:    Afghanistan      GRIPC     44.0         31        -13
##    5:    Afghanistan       LUH2     21.0         27          6
##   ---
## 1626: Åland Islands MIRCA 2000    161.5         NA         NA
## 1627: Åland Islands   MIRCA OS    161.0         NA         NA
```

```
## 1628: Åland Islands      Meier    150.0          NA          NA
## 1629: Åland Islands    Nagaraj    136.0          NA          NA
## 1630: Åland Islands   SPAM2010    150.0          NA          NA
```

```
# Summarize -------------------------------------------------------------------

rank_focus_summary <- rank_focus[!is.na(rank_ident), .(median_shift = median(rank_shift, na.rm
                                     max_drop = max(rank_shift, na.rm = TRUE),
                                     max_rise = min(rank_shift, na.rm = TRUE),
                                     share_drop5 = mean(rank_shift >= 5, na.rm = TRUE),
                                     share_drop10 = mean(rank_shift >= 10, na.rm = TRUE)),
                                 country] %>%
  .[order(-median_shift)]

# Focus on top 20 irrigation hotspots ------------------------------------------

focus_countries <- top10_median$country
rank_focus_summary[country %in% focus_countries]
```

```
##              country median_shift max_drop max_rise share_drop5 share_drop10
##               <char>        <num>    <num>    <num>       <num>        <num>
##  1:           Brazil         12.5       21        6         1.0          0.9
##  2:           Russia         10.0       21        1         0.8          0.5
##  3:        Indonesia          5.5       12       -2         0.8          0.1
##  4:           Mexico          3.5        7       -1         0.4          0.0
##  5:           Turkey          3.0        9       -3         0.4          0.0
##  6:             Iran          2.5        5       -8         0.1          0.0
##  7:       Kazakhstan          1.0       12       -4         0.3          0.1
##  8:            China          0.0        0        0         0.0          0.0
##  9:            India          0.0        0        0         0.0          0.0
## 10: Myanmar (Burma)          0.0        4       -4         0.0          0.0
## 11:         Pakistan          0.0        0       -1         0.0          0.0
## 12:    United States          0.0        1        0         0.0          0.0
## 13:         Thailand         -1.0        1       -5         0.0          0.0
## 14:            Spain         -1.5        3       -6         0.0          0.0
## 15:            Italy         -3.0       -1       -8         0.0          0.0
## 16:          Vietnam         -3.0        1       -9         0.0          0.0
## 17:       Bangladesh         -5.0       -3      -14         0.0          0.0
## 18:             Iraq         -5.5        6       -9         0.1          0.0
## 19:       Uzbekistan         -5.5       -4      -11         0.0          0.0
## 20:            Egypt         -7.0       -3      -15         0.0          0.0
##              country median_shift max_drop max_rise share_drop5 share_drop10
##               <char>        <num>    <num>    <num>       <num>        <num>
```

```
# Positive rank shift: worse rank if only confirmed cells are accounted for.
# Negative rank shift: better rank if only confirmed cells are accounted for.

# - median_shift: median of rank_shift across datasets.
# - max_drop: maximum worsening (largest positive rank_shift).
```

```r
# - max_rise: maximum improvement (most negative rank_shift).
# - share_drop5: share of dataset cases with rank_shift >= 5.
# - share_drop10: share with rank_shift >= 10.

# VIETNAM, BANGLADESH, THAILAND, INDONESIA: in 90% of cases, Vietnam's rank gets
# at least 10 places worse. Median drop is c. 40 places. Bangladesh, Indonesia,
# Thailand show similar pattern; median shifts of 26-27 ranks and 90% of cases
# with drops larger than 90 ranks.

# CHINA, INDIA, PAKISTAN, USA: Global rank is invariant to whether we use all cells
# or only agreement cells. Irrigated area dominance is robust.

# ITALY, EGYPT, UZBEKISTAN, IRAQ, SPAIN, KAZAKHSTAN: they
# improve their rank when we restrict to confirmed cells; because when we remove
# ambiguous cells in other countries, these countries become more prominent.

# Focus on the countries whose ranking drops the most -------------------------

rank_focus_summary[order(-median_shift)][1:number_countries]
```

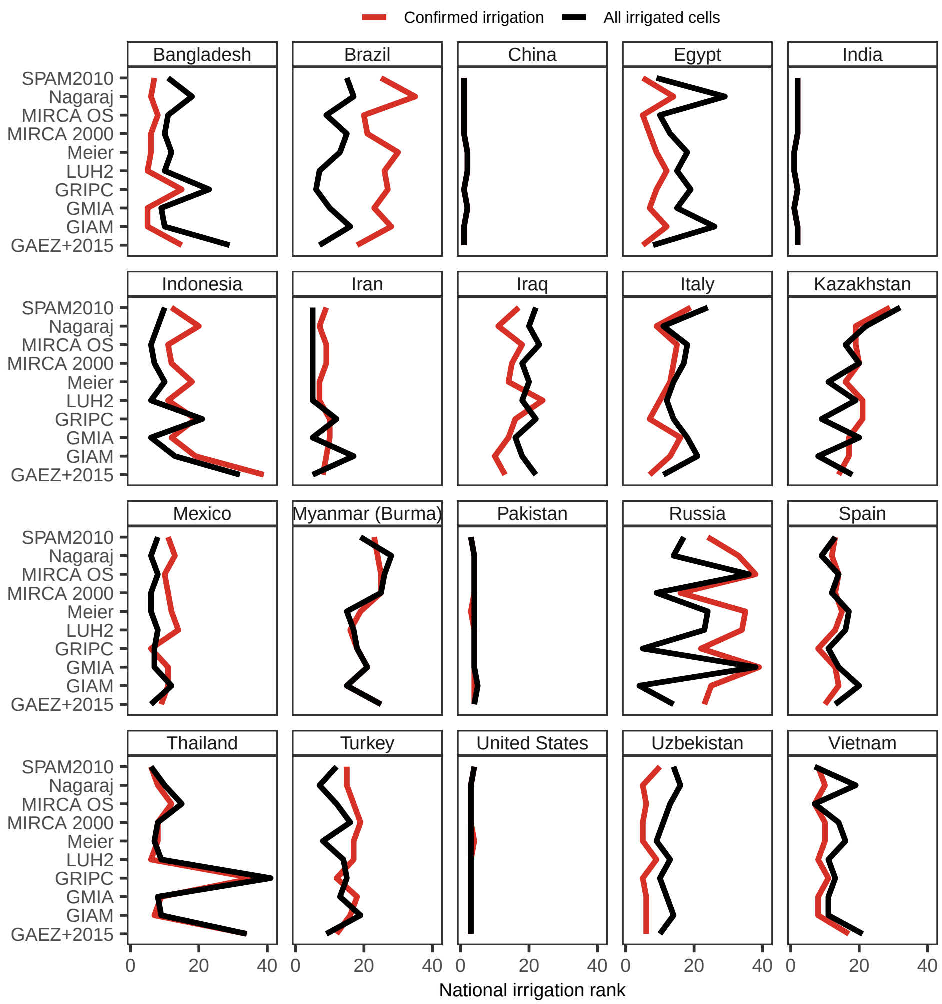
```
##                  country median_shift max_drop max_rise share_drop5
##                   <char>        <num>    <num>    <num>       <num>
##  1:          Netherlands        64.50    108.0     23.0         1.0
##  2:              Hungary        48.50     62.5     10.0         1.0
##  3:            Guatemala        41.25     72.5    -28.0         0.9
##  4: United Arab Emirates        40.50     79.0    -24.5         0.8
##  5:             Eswatini        39.00     71.0     -3.5         0.9
##  6:           Madagascar        39.00     79.5      8.0         1.0
##  7:                 Mali        36.00     57.5    -13.0         0.7
##  8:              Germany        30.50     67.5     18.0         1.0
##  9:             Slovakia        30.50     52.0      1.0         0.9
## 10:             Bulgaria        27.00     52.0    -42.0         0.9
## 11:              Jamaica        27.00     40.0      3.5         0.9
## 12:           Costa Rica        26.50     49.0      7.0         1.0
## 13:            Venezuela        26.50     40.0     11.0         1.0
## 14:             Ethiopia        24.00     71.5      1.0         0.9
## 15:              Nigeria        23.50     65.5      5.0         1.0
## 16:              Ecuador        21.00     33.0     10.0         1.0
## 17:          New Zealand        21.00     31.0    -19.5         0.9
## 18:              Somalia        19.50     66.0    -33.5         0.9
## 19:               Cyprus        19.00     52.0      7.5         1.0
## 20:          Timor-Leste        17.75     48.0     -7.0         0.7
##                  country median_shift max_drop max_rise share_drop5
##                   <char>        <num>    <num>    <num>       <num>
##     share_drop10
##            <num>
##  1:          1.0
```

```
##  2:           1.0
##  3:           0.9
##  4:           0.8
##  5:           0.8
##  6:           0.9
##  7:           0.7
##  8:           1.0
##  9:           0.8
## 10:           0.9
## 11:           0.8
## 12:           0.9
## 13:           1.0
## 14:           0.9
## 15:           0.8
## 16:           1.0
## 17:           0.8
## 18:           0.9
## 19:           0.8
## 20:           0.7
##     share_drop10
##            <num>
```

```r
# Plot the results --------------------------------------------------------------

plot_shift_ranks <- rank_focus[country %in% focus_countries] %>%
  .[, .(country, dataset,
        `All irrigated cells` = rank_all,
        `Confirmed irrigation` = rank_ident)] %>%
  melt(., measure.vars = c("Confirmed irrigation", "All irrigated cells")) %>%
  ggplot(., aes(dataset, value, color = variable,
                group = interaction(country, variable))) +
  geom_line(linewidth = 1) +
  scale_y_reverse() +
  scale_color_manual(values = c("Confirmed irrigation" = "#d73027",
                                "All irrigated cells" = "black"),
                     name = "") +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  facet_wrap(~country) +
  labs(y = "National irrigation rank", x = NULL) +
  coord_flip() +
  theme_AP() +
  theme(legend.position = "top")
```

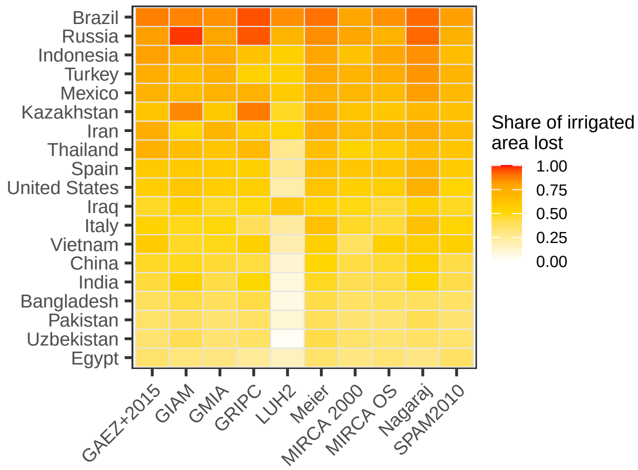
```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```

```r
plot_shift_ranks
```




```
# Dashed (red) line much lower than solid (black)
# - country drops in rank when ambiguity is removed
# - it looks large only because of non-identifiable cells
#
# Red equals black
# - rank is robust to identifiability
# - irrigation signal is spatially coherent
#
# Red above black
# - country rises in rank when ambiguity is removed
# - it was previously masked by others' ambiguous irrigation

# If Red and black lines cross across datasets:
# The ordering of countries changes
# Rankings are dataset-dependent
# There is no stable hierarchy
# Crossings are visual proof of rank instability.
```

```r
# COMPUTE AREA RETAINED AND LOST OF TOP 20 COUNTRIES ############################

country_loss <- rbindlist(lapply(new_dataset_names, function(d) {

  dt_country <- dt2[country %in% top10_all$country]

  dt_country[, .(dataset = d,
               total_area = sum(get(d), na.rm = TRUE),
               retained_area = sum(get(d)[is.na(tau_max_ha)], na.rm = TRUE),
               lost_area = sum(get(d)[!is.na(tau_max_ha)], na.rm = TRUE)),
             country]
}))

country_loss[, share_lost:= lost_area / total_area]

# Compute mean loss per country across datasets to reorder ---------------------

country_order <- country_loss[, .(mean_share_lost = mean(share_lost, na.rm = TRUE)),
  by = country][order(-mean_share_lost)]  # decreasing order

# Set factor levels in that order-----------------------------------------------

country_loss[, country_f:= factor(country, levels = rev(country_order$country))]

# Plot tileplot ----------------------------------------------------------------

plot_tileplot <- ggplot(country_loss, aes(dataset, country_f, fill = share_lost)) +
  geom_tile(color = "grey90") +
  scale_fill_gradientn(colours = c("white", "gold", "orange", "red"),
                       values  = c(0, 0.5, 0.8, 1),
                       limits  = c(0, 1),
                       name = "Share of irrigated\narea lost") +
  scale_y_discrete(name = NULL) +
  theme_AP() +
  labs(x = "", y = "") +
  theme(axis.text.x  = element_text(angle = 45, hjust = 1),
        legend.position = "right")

plot_tileplot
```



4.4 Fraction of irrigated areas lost

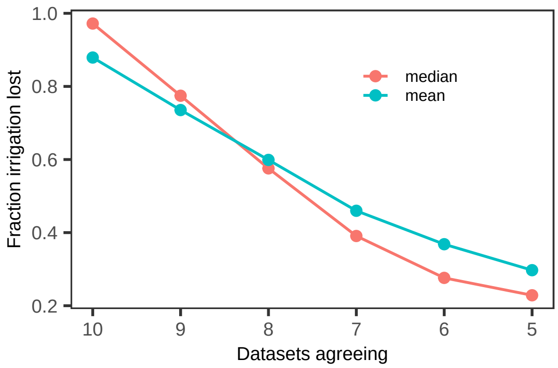
```r
# COLLAPSE ASSESSMENT ACROS A CONTINUUM OF AGREEMENT ############################

dt_share_lost <- dcast(dt, lon + lat + country + code + continent + resolution ~ dataset,
      value.var = "mha")

# Define resolution specific thresholds ----------------------------------------

zero_tol_mha_fun <- function(res) {
  switch(res,
    "0.2deg" = 400  / 1e6, # 0.0004 Mha
    "0.4deg" = 1600 / 1e6, # 0.0016 Mha
    "1deg" = 10000 / 1e6,  # 0.01 Mha
    stop("Unknown resolution: ", res)
  )
}

# Make copy just in case -------------------------------------------------------

dt_pres <- copy(dt_share_lost)

# Apply resolution-specific thresholds -----------------------------------------

dt_pres[, (dataset_names) := {
    tol_mha <- zero_tol_mha_fun(resolution[1])
    lapply(.SD, function(x) as.integer(x >= tol_mha))
    },
    resolution,
    .SDcols = dataset_names]

# How many datasets say "irrigated" per cell -----------------------------------
```

```r
dt_pres[, n_pos:= rowSums(.SD, na.rm = TRUE), .SDcols = dataset_names]

n_datasets <- length(dataset_names)  # should be 10

# Melt for k-of-n agreement ---------------------------------------------------

grid_long <- melt(dt_pres, id.vars = c("lon","lat","country","code",
                    "continent","resolution","n_pos"),
                  measure.vars = dataset_names,
                  variable.name = "dataset",
                  value.name = "present")

# Run calculations for multiple k (10 down to 5) ------------------------------

ks <- n_datasets:5
loss_by_k <- rbindlist(lapply(ks, compute_loss_for_k), use.names = TRUE)

# Summarize results ------------------------------------------------------------

country_loss_by_k <- loss_by_k[, .(min  = min(share_lost,  na.rm = TRUE),
                                   max  = max(share_lost,  na.rm = TRUE),
                                   mean = mean(share_lost, na.rm = TRUE)),
                               .(country, k)] %>%
  .[, allowed_disagree:= 10 - k]

# PLOT RESULTS: THE GLOBAL COLLAPSE CURVE ######################################

global_k <- country_loss_by_k[, .(mean_loss   = mean(mean, na.rm = TRUE),
                                  median_loss = median(mean, na.rm = TRUE)), k] %>%
  .[order(-k)] %>%
  .[, allowed_disagree:= 10 - k]

print(global_k)
```

```
##        k mean_loss median_loss allowed_disagree
##    <int>     <num>       <num>            <num>
## 1:    10 0.8789739   0.9719816                0
## 2:     9 0.7353161   0.7745389                1
## 3:     8 0.5989455   0.5757149                2
## 4:     7 0.4597504   0.3907818                3
## 5:     6 0.3682308   0.2761470                4
## 6:     5 0.2972035   0.2285847                5
```

```r
plot_global_k <- global_k %>%
  melt(., measure.vars = c("median_loss", "mean_loss")) %>%
  ggplot(., aes(k, value, color = variable)) +
  geom_line() +
  geom_point()  +
```

```r
  scale_color_discrete(labels = c(median_loss = "median", mean_loss = "mean"),
                       name = "") +
  labs(x = "Datasets agreeing", y = "Fraction irrigation lost") +
  theme_AP() +
  scale_x_reverse() +
  theme(legend.position = c(0.7, 0.8))

plot_global_k
```




```r
# COUNTRY-LEVEL COLLAPSE PROFILES ##############################################

# Optional: keep only countries with mean loss < 1 at k=5

collapse_profiles <- country_loss_by_k[country %in% top10_all$country] %>%
  .[, allowed_disagree:= 10 - k]

# Under unanimity
tmp <- collapse_profiles[k == 10 & country %in% top10_all$country] %>%
  .[order(-mean)]

tmp
```

```
##           country     k        min       max      mean allowed_disagree
##            <char> <int>      <num>     <num>     <num>            <num>
##  1:        Brazil    10 0.88064269 0.9533493 0.9275510                0
##  2:        Russia    10 0.75986842 0.9877311 0.9062287                0
##  3:     Indonesia    10 0.69328494 0.9186718 0.8443967                0
##  4:        Mexico    10 0.73804971 0.8904438 0.8298258                0
##  5:    Kazakhstan    10 0.65354331 0.9279574 0.7943901                0
##  6:        Turkey    10 0.61011905 0.8732463 0.7883040                0
##  7: United States    10 0.47315809 0.8526610 0.7551067                0
##  8:          Iran    10 0.53748783 0.8364325 0.7399565                0
##  9:      Thailand    10 0.42953020 0.7826087 0.6993478                0
## 10:         Spain    10 0.36942675 0.7899576 0.6815685                0
## 11:         Italy    10 0.46445498 0.7618546 0.6523922                0
## 12:       Vietnam    10 0.38235294 0.7169118 0.6246655                0
## 13:          Iraq    10 0.45000000 0.6653144 0.5841619                0
```

```
## 14:          China    10 0.30208040 0.6298919 0.5560644                0
## 15:          India    10 0.15740741 0.6092448 0.5245148                0
## 16:       Pakistan    10 0.37068063 0.5587372 0.4671962                0
## 17:          Egypt    10 0.28909953 0.5575221 0.4548616                0
## 18:     Uzbekistan    10 0.07220217 0.4711934 0.3867142                0
## 19:     Bangladesh    10 0.08823529 0.4213333 0.3637888                0
```

```
tmp[, .(n_above_90 = sum(mean > 0.9),
        n_above_80 = sum(mean > 0.8),
        n_above_50 = sum(mean > 0.5),
        pct_above_90 = 100 * mean(mean > 0.9),
        pct_above_80 = 100 * mean(mean > 0.8),
        pct_above_50 = 100 * mean(mean > 0.5))]
```

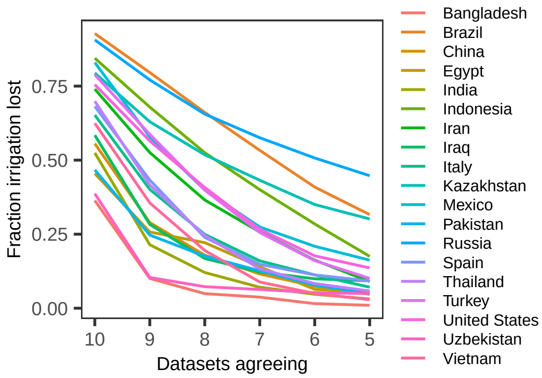
```
##    n_above_90 n_above_80 n_above_50 pct_above_90 pct_above_80 pct_above_50
##         <int>      <int>      <int>        <num>        <num>        <num>
## 1:          2          4         15     10.52632     21.05263     78.94737
```

```
# Under the most permissive criteria (5/10)
collapse_profiles[k == 5 & country %in% top10_all$country] %>%
  .[order(-mean)]
```

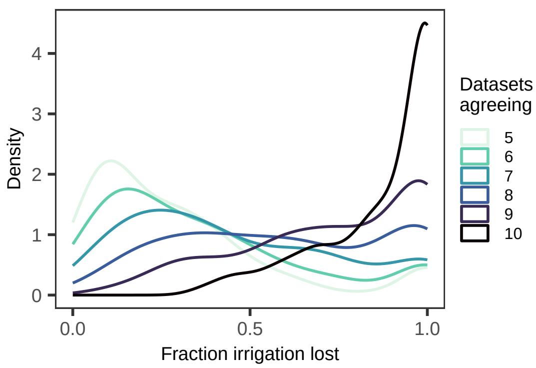
```
##            country     k         min        max        mean allowed_disagree
##             <char> <int>       <num>      <num>       <num>            <num>
##  1:         Russia     5 0.028199566 0.88873950 0.447484937                5
##  2:         Brazil     5 0.069400631 0.62105969 0.316117863                5
##  3:     Kazakhstan     5 0.017716535 0.77691363 0.301456424                5
##  4:      Indonesia     5 0.040892193 0.30895091 0.175197788                5
##  5:         Mexico     5 0.031192661 0.35785686 0.162330836                5
##  6: United States     5 0.035277325 0.32532615 0.136061138                5
##  7:         Turkey     5 0.013750955 0.22883406 0.100755304                5
##  8:           Iraq     5 0.000000000 0.32962138 0.092350998                5
##  9:          Spain     5 0.005928854 0.25742574 0.091921916                5
## 10:           Iran     5 0.041157294 0.17183196 0.090127081                5
## 11:          Italy     5 0.001572327 0.20021075 0.071055620                5
## 12:       Thailand     5 0.007352941 0.12020460 0.058142125                5
## 13:          Egypt     5 0.004484305 0.16853933 0.055262476                5
## 14:       Pakistan     5 0.004280822 0.11967695 0.049999775                5
## 15:     Uzbekistan     5 0.000000000 0.11728395 0.049347680                5
## 16:          China     5 0.003233049 0.08483041 0.047069619                5
## 17:          India     5 0.003975265 0.08171255 0.031221534                5
## 18:        Vietnam     5 0.002083333 0.07843137 0.028810410                5
## 19:     Bangladesh     5 0.000000000 0.02840909 0.009946725                5
```

```
plot_collapse_profiles <- ggplot(collapse_profiles, aes(k, mean, group = country,
                                                        color = country)) +
  scale_color_discrete(name = "") +
  geom_line() +
  scale_x_reverse() +
  labs(x = "Datasets agreeing", y = "Fraction irrigation lost") +
```

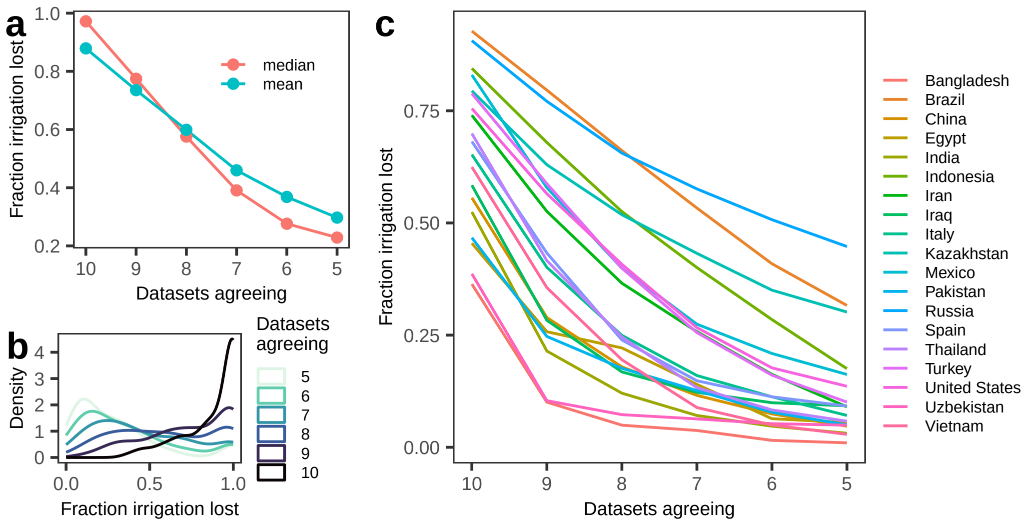
```r
  theme_AP()

plot_collapse_profiles
```




```r
# Global distribution by k ------------------------------------------------

plot_curves_country <- ggplot(country_loss_by_k, aes(mean, color = factor(k), group = k)) +
  geom_density() +
  labs(x = "Fraction irrigation lost", y = "Density", color = "Datasets \nagreeing") +
  theme_AP() +
  theme(legend.position = "right") +
  scale_x_continuous(breaks = scales::breaks_pretty(n = 3)) +
  scale_color_viridis_d(option = "mako", direction = -1)

plot_curves_country
```




```r
# MERGE ##################################################################

left <- plot_grid(plot_global_k, plot_curves_country, ncol = 1, labels = "auto",
                  rel_heights = c(0.6, 0.4))
plot_grid(left, plot_collapse_profiles, ncol = 2, labels = c("", "c"),
          rel_widths = c(0.35, 0.65))
```




```r
# PLOTS SHOWING K-OF-10 AGREEMENT ##############################################

# Prepare data ---------------------------------------------------------------

dt_plot <- dt_pres[, .N, .(resolution, n_pos)]
dt_plot[, frac:= N / sum(N), by = resolution]

# Fraction of cells in the consensus core (>=9/10) ----------------------------

dt_plot[order(n_pos)] %>%
  .[n_pos >= 9] %>%
  .[, sum(frac), resolution]
```

```
##    resolution        V1
##        <char>     <num>
## 1:       1deg 0.2196331
## 2:     0.4deg 0.1945520
## 3:     0.2deg 0.1620483
```

```r
# Fraction of cells irrigated by less than half of the datasets .---------------

dt_plot[order(n_pos)] %>%
  .[n_pos < 5 & n_pos != 0] %>%
  .[, sum(frac), resolution]
```

```
##    resolution        V1
##        <char>     <num>
## 1:     0.2deg 0.3057287
## 2:     0.4deg 0.2735299
## 3:       1deg 0.2205007
```

```r
# Compute statistics ---------------------------------------------------------

dt_summary <- dt_plot[, .(total = sum(N),
```

```r
                             absence = sum(N[n_pos == 0]),
                             contested = sum(N[n_pos >=1 & n_pos <=8]),
                             fewer_than_half = sum(N[n_pos < 5 & n_pos > 0]),
                             presence = sum(N[n_pos >=9])),
                         resolution]

dt_summary[, `:=`(frac_absence = absence / total,
                  frac_contested = contested / total,
                  frac_fewer_than_half = fewer_than_half / total,
                  frac_presence = presence / total)]

dt_summary
```

```
##    resolution  total absence contested fewer_than_half presence frac_absence
##        <char>  <int>   <int>     <int>           <int>    <int>        <num>
## 1:       1deg   8068    3473      2823            1779     1772    0.4304660
## 2:     0.2deg 129628   43077     65545           39631    21006    0.3323125
## 3:     0.4deg  39758   14514     17509           10875     7735    0.3650586
##    frac_contested frac_fewer_than_half frac_presence
##             <num>                <num>         <num>
## 1:      0.3499008            0.2205007     0.2196331
## 2:      0.5056392            0.3057287     0.1620483
## 3:      0.4403894            0.2735299     0.1945520
```

```r
# Probability that a dataset detects irrigation in a given cell ----------------

dt_k <- dt_pres[, .N, .(resolution, n_pos)]

dt_prob <- dt_k[, .(mean_k = weighted.mean(n_pos, N),
                    detection_prob = weighted.mean(n_pos, N) / 10),
                resolution]

dt_prob
```

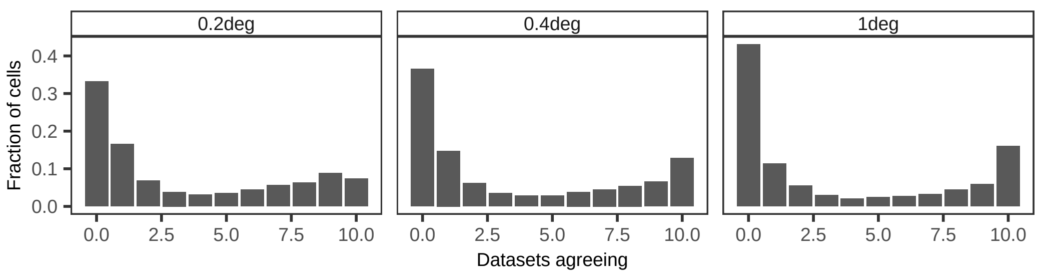
```
##    resolution   mean_k detection_prob
##        <char>    <num>          <num>
## 1:       1deg 3.410387      0.3410387
## 2:     0.2deg 3.426945      0.3426945
## 3:     0.4deg 3.498114      0.3498114
```

```r
# In how many cells do fewer than half of the datasets detect irrigation? ------

dt_weak <- dt_k[, .(total_cells = sum(N),
  no_irrigation = sum(N[n_pos == 0]),
  any_irrigation = sum(N[n_pos >= 1]),
  weak_detection = sum(N[n_pos >= 1 & n_pos <= 5])), resolution]

dt_weak[, `:=`(frac_weak_all = weak_detection / total_cells,
               frac_weak_given_any = weak_detection / any_irrigation)]
```

```r
dt_weak
```

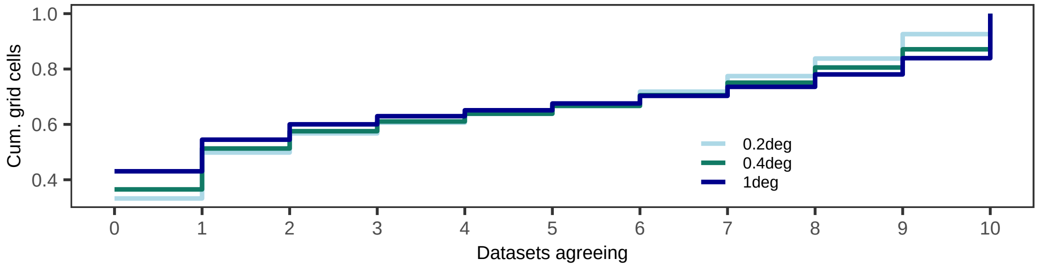
```
##    resolution total_cells no_irrigation any_irrigation weak_detection
##        <char>       <int>         <int>          <int>          <int>
## 1:       1deg        8068          3473           4595           1977
## 2:     0.2deg      129628         43077          86551          44255
## 3:     0.4deg       39758         14514          25244          11993
##    frac_weak_all frac_weak_given_any
##            <num>               <num>
## 1:     0.2450421           0.4302503
## 2:     0.3414000           0.5113170
## 3:     0.3016500           0.4750832
```

```r
# Plot distribution -----------------------------------------------------------

p1 <- plot_agreement <- ggplot(dt_plot, aes(n_pos, frac)) +
  geom_col() +
  facet_wrap(~resolution) +
  labs(x = "Datasets agreeing",
       y = "Fraction of cells") +
  theme_AP()

p1
```

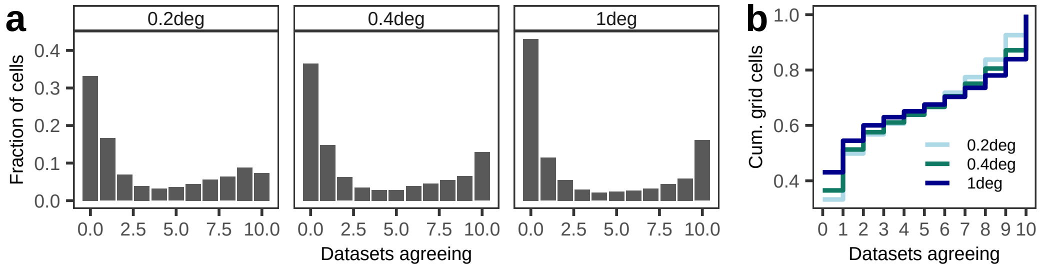



```r
# Prepare data for cumulative plot --------------------------------------------

dt_ecdf <- dt_pres[, .N, .(resolution, n_pos)]
dt_ecdf[, frac:= N / sum(N), resolution]
setorder(dt_ecdf, resolution, n_pos)
dt_ecdf[, cdf:= cumsum(frac), resolution]

dt_ecdf[, resolution:= factor(resolution, levels = c("0.2deg","0.4deg","1deg"))]

# Plot ------------------------------------------------------------------------

p2 <- ggplot(dt_ecdf, aes(n_pos, cdf, colour = resolution)) +
  geom_step(linewidth = 0.8) +
  scale_x_continuous(breaks = 0:10) +
  scale_color_manual(values = res_palette, name = "") +
  labs(x = "Datasets agreeing", y = "Cum. grid cells",
```

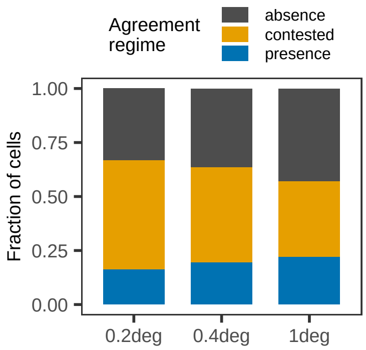
```r
      colour = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.7, 0.3))

p2
```




```r
plot_grid(p1, p2, ncol = 2, rel_widths = c(0.7, 0.3), labels = "auto")
```




```r
# Plot: consensus regime plot -----------------------------------------------

dt_regime <- dt_k[, .(absence = sum(N[n_pos == 0]),
                      contested = sum(N[n_pos >= 1 & n_pos <= 8]),
                      presence = sum(N[n_pos >= 9])), resolution]

dt_regime <- melt(dt_regime, id.vars = "resolution", variable.name = "regime",
                  value.name = "N")

dt_regime[, frac:= N / sum(N), by = resolution]

plot_regime <- ggplot(dt_regime, aes(resolution, frac, fill = regime)) +
  geom_col(width = 0.7) +
  scale_fill_manual(values = c(absence = "#4D4D4D",
                               contested = "#E69F00",
                               presence = "#0072B2"),
                    guide = guide_legend(nrow = 3),
                    name = "Agreement \nregime") +
  labs(x = NULL, y = "Fraction of cells") +
  theme_AP() +
  theme(legend.position = "top")


plot_regime
```


Agreement regime
absence
contested
presence
Fraction of cells
1.00
0.75
0.50
0.25
0.00
0.2deg
0.4deg
1deg

5 Consequences for crop production, water and climate

```r
# GLOBAL CROP PRODUCTION UDNER DIFFERENT AGREEMENT RULES ########################
#################################################################################

# Read crop datasets ------------------------------------------------------------

gaez_crops_02_aligned <- fread("./datasets/crops/gaez2015_crops_irrigated_0p2deg_aligned.csv")
gaez_crops_04_aligned <- fread("./datasets/crops/gaez2015_crops_irrigated_0p4deg_aligned.csv")
gaez_crops_1_aligned <- fread("./datasets/crops/gaez2015_crops_irrigated_1deg_aligned.csv")

# Extract crop cols-------------------------------------------------------------

crop_cols <- setdiff(names(gaez_crops_02_aligned),
                     c("lon", "lat", "country", "code", "continent"))

# Run for all grids -------------------------------------------------------------

res_list <- list("0.2deg" = gaez_crops_02_aligned,
                 "0.4deg" = gaez_crops_04_aligned,
                 "1deg"   = gaez_crops_1_aligned)

results <- lapply(names(res_list), function(r) {
  run_prod_analysis(r, res_list[[r]], dt_pres, crop_cols)
})
```

```
## Warning in melt.data.table(prod_agreement_wide, id.vars = "k", variable.name =
## "crop", : 'measure.vars' [banana, barley, cassava, cotton, ...] are not all of
## the same type. By order of hierarchy, the molten data value column will be of
## type 'double'. All measure variables not of type 'double' will be coerced too.
## Check DETAILS in ?melt.data.table for more on coercion.
## Warning in melt.data.table(prod_agreement_wide, id.vars = "k", variable.name =
## "crop", : 'measure.vars' [banana, barley, cassava, cotton, ...] are not all of
## the same type. By order of hierarchy, the molten data value column will be of
## type 'double'. All measure variables not of type 'double' will be coerced too.
## Check DETAILS in ?melt.data.table for more on coercion.
## Warning in melt.data.table(prod_agreement_wide, id.vars = "k", variable.name =
## "crop", : 'measure.vars' [banana, barley, cassava, cotton, ...] are not all of
## the same type. By order of hierarchy, the molten data value column will be of
## type 'double'. All measure variables not of type 'double' will be coerced too.
## Check DETAILS in ?melt.data.table for more on coercion.
```

```r
crop_all  <- rbindlist(lapply(results, `[[`, "crop"))
total_all <- rbindlist(lapply(results, `[[`, "total"))

total_all
```

```
##          k total_production frac_vs_5  loss_vs_5 resolution
```

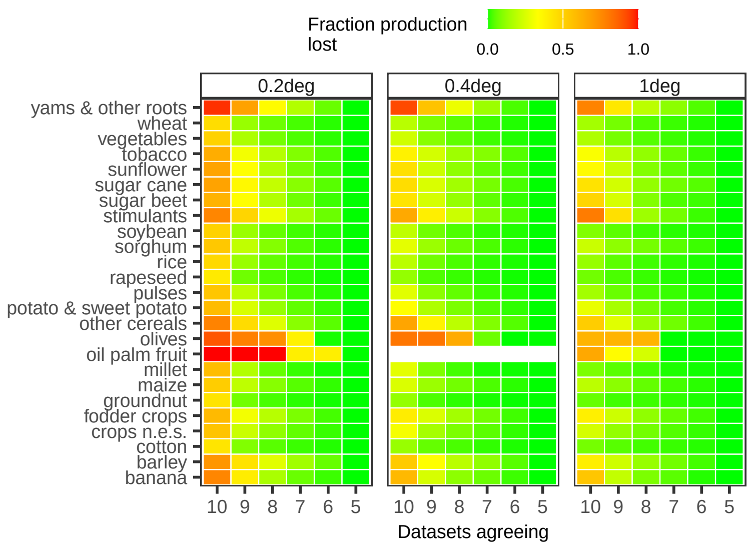
```
##     <int>           <num>     <num>      <num>     <char>
##  1:    10         1593770 0.4360333 0.56396667     0.2deg
##  2:     9         2791062 0.7635958 0.23640417     0.2deg
##  3:     8         3188880 0.8724333 0.12756672     0.2deg
##  4:     7         3434924 0.9397475 0.06025253     0.2deg
##  5:     6         3572541 0.9773975 0.02260246     0.2deg
##  6:     5         3655156 1.0000000 0.00000000     0.2deg
##  7:    10         2509734 0.6846610 0.31533904     0.4deg
##  8:     9         3072563 0.8382021 0.16179789     0.4deg
##  9:     8         3338060 0.9106301 0.08936985     0.4deg
## 10:     7         3501154 0.9551226 0.04487735     0.4deg
## 11:     6         3606887 0.9839669 0.01603314     0.4deg
## 12:     5         3665659 1.0000000 0.00000000     0.4deg
## 13:    10         2765987 0.7272890 0.27271097       1deg
## 14:     9         3261876 0.8576783 0.14232170       1deg
## 15:     8         3527925 0.9276333 0.07236671       1deg
## 16:     7         3652546 0.9604011 0.03959886       1deg
## 17:     6         3728283 0.9803155 0.01968445       1deg
## 18:     5         3803146 1.0000000 0.00000000       1deg
```

```r
total_all[, .(mean = mean(loss_vs_5)), k]
```

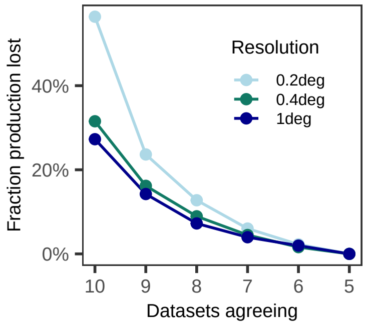
```
##        k       mean
##    <int>      <num>
## 1:    10 0.38400556
## 2:     9 0.18017459
## 3:     8 0.09643443
## 4:     7 0.04824291
## 5:     6 0.01944002
## 6:     5 0.00000000
```

```r
# PLOT #########################################################################

# Heatmap ----------------------------------------------------------------------

plot_dt <- crop_all[!is.na(frac_vs_5)]

# Fix merged crop names only ---------------------------------------------------

plot_dt[, crop:= gsub("potatoandsweetpotato", "potato & sweet potato", crop)]
plot_dt[, crop:= gsub("yamsandotherroots", "yams & other roots", crop)]
plot_dt[, crop:= gsub("oilpalmfruit", "oil palm fruit", crop)]
plot_dt[, crop:= gsub("sugarbeet", "sugar beet", crop)]
plot_dt[, crop:= gsub("sugarcane", "sugar cane", crop)]
plot_dt[, crop:= gsub("othercereals", "other cereals", crop)]
plot_dt[, crop:= gsub("foddercrops", "fodder crops", crop)]
plot_dt[, crop:= gsub("cropsnes", "crops n.e.s.", crop)]

plot_dt[, k:= factor(k, levels = 10:5)]
```

```r
plot_heatmap <- ggplot(plot_dt, aes(x = k, y = crop, fill = loss_vs_5)) +
  geom_tile(color = "white") +
  scale_fill_gradientn(colours = c("green", "yellow", "orange", "red"),
                       values = c(0, 0.5, 1),
                       breaks = c(0, 0.5, 1),
                       name     = "Fraction production \nlost") +
  facet_wrap(~resolution, ncol = 3) +
  labs(x = "Datasets agreeing",
       y = NULL) +
  theme_AP() +
  theme(axis.text.y = element_text(size = 7),
        legend.position = "top")

plot_heatmap
```




```r
# Global production -------------------------------------------------------------

plot_crop_total <- ggplot(total_all,
       aes(x = k, y = loss_vs_5, group = resolution, color = resolution)) +
  geom_line() +
  geom_point() +
  scale_color_manual(values = res_palette) +
  scale_x_reverse(breaks = 10:5) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(x = "Datasets agreeing", y = "Fraction production lost",
       color = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.7, 0.7))

plot_crop_total
```



5.1 Effect on global breadbaskets

```r
# COMPUTE BREADBASKETS ##################################################################
##################################################################################

# LAND SEA MASK -----------------------------------------------------------------

grain_crops <- intersect(c("wheat", "rice", "maize", "barley", "millet",
                           "sorghum", "othercereals"), names(gaez_crops_02_aligned))

lon_min <- min(gaez_crops_02_aligned$lon)
lon_max <- max(gaez_crops_02_aligned$lon)
lat_min <- min(gaez_crops_02_aligned$lat)
lat_max <- max(gaez_crops_02_aligned$lat)

template <- rast(xmin = lon_min, xmax = lon_max,
                 ymin = lat_min, ymax = lat_max,
                 resolution = 0.2,
                 crs = "EPSG:4326")

world <- vect(spData::world)
land_mask <- rasterize(world, template, field = 1, background = NA)

# Focus regions ----------------------------------------------------------------

# PREVIOUS LAND MASK EXCLUDES WATER BODIES IN THE AREAS ########################

# India: Indus Valley (Pakistan) and the Ganges Delta (Bangladesh).
# China: Hebei-Henan-Shandong axis
# USA: Ogallala Box; the Nebraska-to-Texas irrigation corridor.
# Egypt: Egyptian Nile
# Mediterranean: Spain, Italy, Greece, Turkey and North Africa

regions <- list("Indo-Gangetic Plain" = ext(68, 91, 21.5, 33),
                "North China Plain" = ext(113, 121, 34, 41),
                "US High Plains" = ext(-104, -96, 32, 44),
                "Nile Delta/Basin" = ext(29.5, 33, 24, 31.5),
```

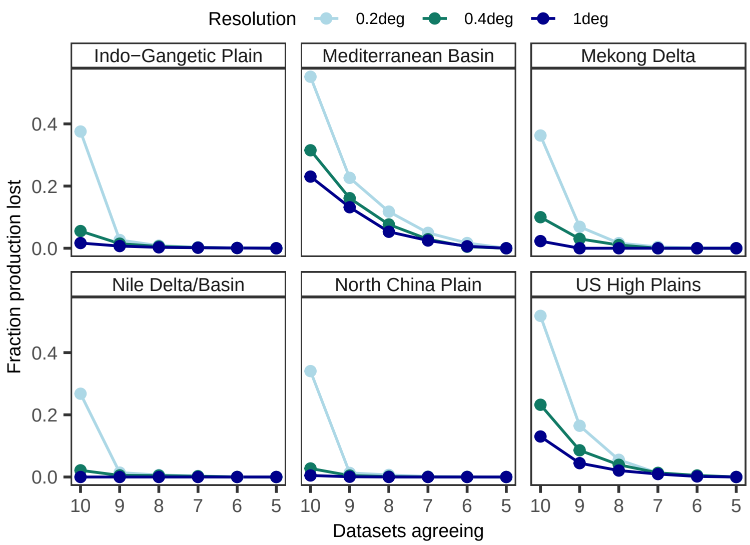
```r
                "Mekong Delta" = ext(104.5, 106.5, 8.5, 11.5),
                "Mediterranean Basin" = ext(-6, 36, 30, 46))

# Run for 0.2, 0.4, 1------------------------------------------------------------

res_list <- list("0.2deg" = gaez_crops_02_aligned,
                 "0.4deg" = gaez_crops_04_aligned,
                 "1deg"   = gaez_crops_1_aligned)

results_regions_all <- rbindlist(
  lapply(names(res_list), function(r) {
    compute_breadbaskets(
      res_tag = r,
      crops_aligned = res_list[[r]],
      dt_pres = dt_pres,
      regions = regions,
      grain_crops = grain_crops,
      land_mask = land_mask
    )
  }),
  use.names = TRUE,
  fill = TRUE
)

# Print
results_regions_all[k %in% 8:10]
```

```
##                   region resolution     k total_prod frac_vs_5    loss_vs_5
##                   <char>     <char> <int>      <num>     <num>        <num>
##  1: Indo-Gangetic Plain     0.2deg    10  152495.01 0.6245379 0.3754620851
##  2: Indo-Gangetic Plain     0.2deg     9  237714.86 0.9735528 0.0264472379
##  3: Indo-Gangetic Plain     0.2deg     8  242132.56 0.9916453 0.0083546813
##  4:   North China Plain     0.2deg    10   76569.73 0.6594939 0.3405061368
##  5:   North China Plain     0.2deg     9  114582.60 0.9868981 0.0131018740
##  6:   North China Plain     0.2deg     8  115328.71 0.9933243 0.0066756566
##  7:      US High Plains     0.2deg    10   30405.85 0.4817647 0.5182352530
##  8:      US High Plains     0.2deg     9   52715.43 0.8352483 0.1647517457
##  9:      US High Plains     0.2deg     8   59616.06 0.9445851 0.0554149384
## 10:    Nile Delta/Basin     0.2deg    10   15938.59 0.7322171 0.2677828672
## 11:    Nile Delta/Basin     0.2deg     9   21459.46 0.9858457 0.0141542582
## 12:    Nile Delta/Basin     0.2deg     8   21628.49 0.9936107 0.0063893457
## 13:        Mekong Delta     0.2deg    10   10528.15 0.6373951 0.3626048634
## 14:        Mekong Delta     0.2deg     9   15377.19 0.9309657 0.0690342500
## 15:        Mekong Delta     0.2deg     8   16244.31 0.9834627 0.0165372549
## 16: Mediterranean Basin     0.2deg    10   22224.66 0.4484119 0.5515881058
## 17: Mediterranean Basin     0.2deg     9   38343.30 0.7736269 0.2263730645
## 18: Mediterranean Basin     0.2deg     8   43722.71 0.8821637 0.1178363312
```

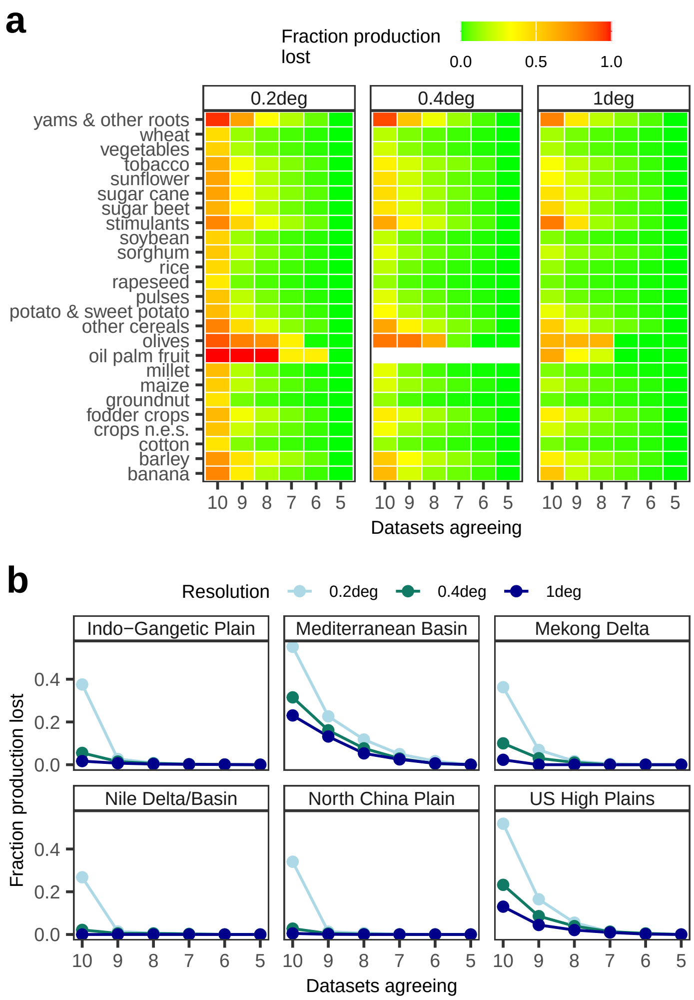
```
## 19: Indo-Gangetic Plain     0.4deg    10  231359.13 0.9447325 0.0552674899
## 20: Indo-Gangetic Plain     0.4deg     9  241268.36 0.9851959 0.0148040985
## 21: Indo-Gangetic Plain     0.4deg     8  243495.11 0.9942886 0.0057113975
## 22:   North China Plain     0.4deg    10  115448.77 0.9724987 0.0275012503
## 23:   North China Plain     0.4deg     9  118154.72 0.9952927 0.0047072861
## 24:   North China Plain     0.4deg     8  118537.90 0.9985205 0.0014795430
## 25:      US High Plains     0.4deg    10   48594.40 0.7677196 0.2322803545
## 26:      US High Plains     0.4deg     9   57860.58 0.9141116 0.0858884088
## 27:      US High Plains     0.4deg     8   60810.78 0.9607203 0.0392797499
## 28:    Nile Delta/Basin     0.4deg    10   21101.49 0.9785229 0.0214770555
## 29:    Nile Delta/Basin     0.4deg     9   21452.91 0.9948191 0.0051809031
## 30:    Nile Delta/Basin     0.4deg     8   21465.73 0.9954136 0.0045864071
## 31:        Mekong Delta     0.4deg    10   14801.62 0.9000954 0.0999045759
## 32:        Mekong Delta     0.4deg     9   15948.76 0.9698536 0.0301463545
## 33:        Mekong Delta     0.4deg     8   16272.27 0.9895269 0.0104730879
## 34: Mediterranean Basin     0.4deg    10   33920.85 0.6848763 0.3151236655
## 35: Mediterranean Basin     0.4deg     9   41551.12 0.8389347 0.1610652881
## 36: Mediterranean Basin     0.4deg     8   45736.74 0.9234441 0.0765558599
## 37: Indo-Gangetic Plain       1deg    10  243237.89 0.9832300 0.0167699846
## 38: Indo-Gangetic Plain       1deg     9  245662.06 0.9930291 0.0069708686
## 39: Indo-Gangetic Plain       1deg     8  246673.77 0.9971187 0.0028813002
## 40:   North China Plain       1deg    10  115152.37 0.9950748 0.0049252292
## 41:   North China Plain       1deg     9  115633.52 0.9992326 0.0007673787
## 42:   North China Plain       1deg     8  115722.33 1.0000000 0.0000000000
## 43:      US High Plains       1deg    10   55115.99 0.8697789 0.1302211437
## 44:      US High Plains       1deg     9   60547.28 0.9554894 0.0445105927
## 45:      US High Plains       1deg     8   62052.48 0.9792428 0.0207572056
## 46:    Nile Delta/Basin       1deg    10   17575.89 1.0000000 0.0000000000
## 47:    Nile Delta/Basin       1deg     9   17575.89 1.0000000 0.0000000000
## 48:    Nile Delta/Basin       1deg     8   17575.89 1.0000000 0.0000000000
## 49:        Mekong Delta       1deg    10   17437.57 0.9770672 0.0229327793
## 50:        Mekong Delta       1deg     9   17846.84 1.0000000 0.0000000000
## 51:        Mekong Delta       1deg     8   17846.84 1.0000000 0.0000000000
## 52: Mediterranean Basin       1deg    10   50894.12 0.7695525 0.2304475498
## 53: Mediterranean Basin       1deg     9   57415.03 0.8681528 0.1318472087
## 54: Mediterranean Basin       1deg     8   62630.96 0.9470211 0.0529789104
##                  region resolution     k total_prod frac_vs_5    loss_vs_5
##                  <char>     <char> <int>      <num>     <num>        <num>
```

```r
# PLOT: FRACTION OF GRAIN PRODUCTION LOST ######################################

plot_fraction_lost <- ggplot(results_regions_all, aes(k, loss_vs_5,
                                                       color = resolution,
                                                       group = resolution)) +
  geom_line() +
  geom_point() +
  scale_x_reverse(breaks = 10:5) +
  scale_color_manual(values = res_palette) +
```

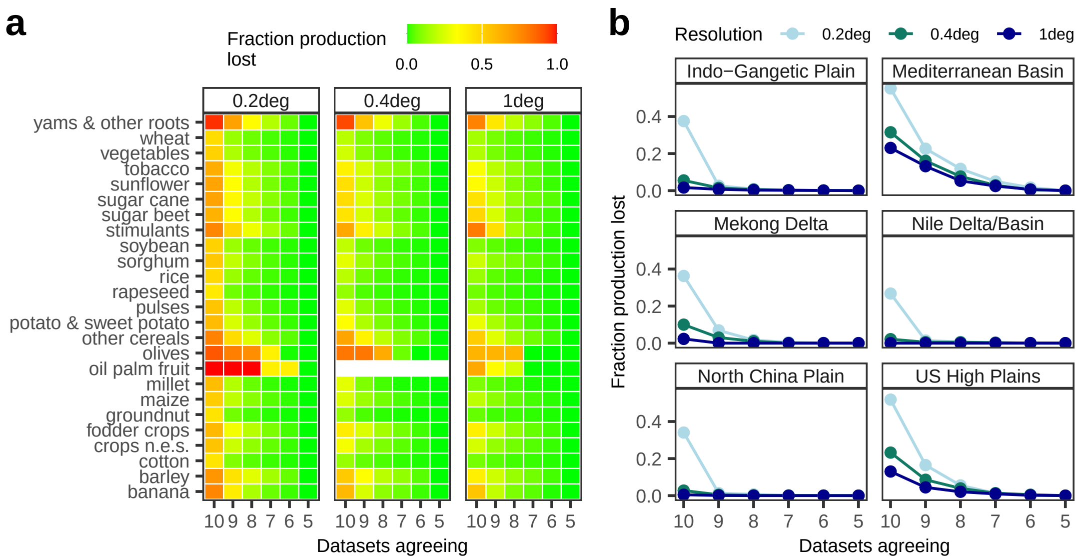
```r
  labs(x = "Datasets agreeing",
       y = "Fraction production lost",
       color = "Resolution") +
  facet_wrap(~region, ncol = 3) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  theme(legend.position = "top")

plot_fraction_lost
```




```r
# MERGE ##########################################################################

plot_grid(plot_heatmap, plot_fraction_lost, ncol = 1, labels = "auto", rel_heights = c(0.55, 0
```




```
# MERGE ######################################################################

plot_fraction_lost <- plot_fraction_lost +
  facet_wrap(~region, ncol = 2)

top <- plot_grid(plot_heatmap, plot_fraction_lost, ncol = 2, labels = "auto",
                  rel_widths =  c(0.55, 0.45))

top
```



5.2 Irrigation-induced evapotranspiration

```r
# LOAD THE HARMONIZED GRIDS ###########################################################

tmpl_02 <- rast("./mha_stack_ll_02_aligned_NEAR.tif")[[1]]
tmpl_04 <- rast("./mha_stack_ll_04_aligned_NEAR.tif")[[1]]
tmpl_1 <- rast("./mha_stack_ll_10_aligned_NEAR.tif")[[1]]

templates <- list("0.2deg" = tmpl_02, "0.4deg" = tmpl_04, "1deg" = tmpl_1)

# Enforce lon/lat WGS84 on templates if needed---------------------------------

templates <- lapply(templates, function(t) {
  if (is.na(crs(t)) || crs(t) == "") crs(t) <- "EPSG:4326"
  t
})

# LOAD THE FILES ###############################################################
# Files come from the Yao et al 2025 paper (Nature Water)

# Year window ------------------------------------------------------------------

win <- "1985_2014_timmean"

# Base directory ---------------------------------------------------------------

base_dir <- "./yao_et_al_2025_datasets/water_fluxes/CESM2"

# Paste ------------------------------------------------------------------------

irr_files <- paste0(base_dir, "/CESM2_IRR0", 1:3, "_1901_2014_QFLX_EVAP_TOT_yearmean_", win)
```

```r
noi_files <- paste0(base_dir, "/CESM2_NOI0", 1:3, "_1901_2014_QFLX_EVAP_TOT_yearmean_", win)

# BUILD ET STACKS + ENSEMBLE MEAN PER RESOLUTION ################################

d_list <- Map(function(irr_f, noi_f) delta_pair_resampled(irr_f, noi_f, templates),
              irr_files, noi_files)

# d_list length 3; each element is list("0.2deg"=rast, "0.4deg"=rast, "1deg"=rast)
dET_mean_rasters <- lapply(names(templates), function(res_name) {

  d_rasters <- lapply(d_list, function(one_pair) one_pair[[res_name]])
  s <- terra::rast(d_rasters) # 3-layer stack
  names(s) <- paste0("dET_IRR0", 1:3, "_mmday")

  # ensemble mean: cell-by-cell mean across 3 members
  m <- terra::app(s, fun = mean, na.rm = TRUE)
  names(m) <- paste0("CESM2_dET_mean_", win, "_", res_name, "_mmday")

  list(stack = s, mean = m)
})
names(dET_mean_rasters) <- names(templates)

# EXPORT TO DATA TABLE ##########################################################

dET_dt <- lapply(names(templates), function(res_name) {
  tmpl <- templates[[res_name]]
  out <- dET_mean_rasters[[res_name]]

  dt_mean <- rast_to_dt_with_cell(out$mean, tmpl, value_name = "dET_mean_mmday")
  dt_mean[, resolution:= res_name]

  dt_members <- rbindlist(lapply(1:terra::nlyr(out$stack), function(i) {
    rr <- out$stack[[i]]
    dt <- rast_to_dt_with_cell(rr, tmpl, value_name = "dET_mmday")
    dt[, member:= names(rr)] # dET_IRR01_mmday / dET_IRR02_mmday / dET_IRR03_mmday
    dt[, resolution:= res_name]
    dt[]
  }))

  list(mean = dt_mean, members = dt_members)
})
names(dET_dt) <- names(templates)

# COMPUTE K-OF-10 AGREEMENT WITH CELL JOINS ####################################

resolutions <- c("0.2deg", "0.4deg", "1deg")
ks <- 5:10
```

```r
# Compute -----------------------------------------------------------------------

out_k_all <- rbindlist(lapply(resolutions, function(res) {

  tmpl <- templates[[res]]

  # irrigation presence-------------

  dt_irrig_raw <- copy(dt_pres)[resolution == res]
  dt_irrig <- pres_to_cell(dt_irrig_raw, tmpl)

  # add cell area------------------

  dt_area <- cell_area_dt(tmpl)
  setkey(dt_irrig, cell)
  setkey(dt_area,  cell)
  dt_irrig <- dt_irrig[dt_area]  # add area_m2

  # increase ET mean table ----------

  dt_et <- copy(dET_dt[[res]]$mean)
  valcol <- setdiff(names(dt_et), c("cell","lon","lat","resolution"))[1]

  # Join: we keep all ET cells and bring irrigation info when available:
  # to summarise ONLY cells that exist in the irrigation table:
  #   dt_join <- dt_et[dt_irrig, on="cell"]
  setkey(dt_et, cell)
  dt_join <- dt_irrig[dt_et]

  rbindlist(lapply(ks, function(k) {

    dt_sub <- dt_join[n_pos >= k]

    # restrict weights to cells with non-missing ET and non-missing area
    ok <- !is.na(dt_sub[[valcol]]) & !is.na(dt_sub$area_m2)
    w  <- dt_sub$area_m2[ok]
    x  <- dt_sub[[valcol]][ok]

    # area-weighted mean increase ET (mm/day)
    mean_aw <- if (length(x) == 0L) NA_real_ else sum(x * w) / sum(w)

    # area-weighted aggregate: mm/day * m^2 -> m^3/day
    # 1 mm = 0.001 m
    sum_m3day <- if (length(x) == 0L) NA_real_ else sum(x * w) * 1e-3

    data.table(resolution = res,
               k = k,
```

```r
                n_cells_irrig_k = sum(dt_irrig$n_pos >= k, na.rm = TRUE),
                n_cells_join_k = nrow(dt_sub),
                n_cells_with_dET_k = sum(!is.na(dt_sub[[valcol]])),
                mean_dET_mmday = mean(dt_sub[[valcol]], na.rm = TRUE),
                sum_dET_mmday_unweighted = sum(dt_sub[[valcol]], na.rm = TRUE),
                mean_dET_mmday_areaweighted = mean_aw,
                sum_dET_m3day_areaweighted = sum_m3day,
                area_m2_with_dET_k = if (length(w) == 0L) 0 else sum(w)
    )
  }))
}))

out_k_all
```

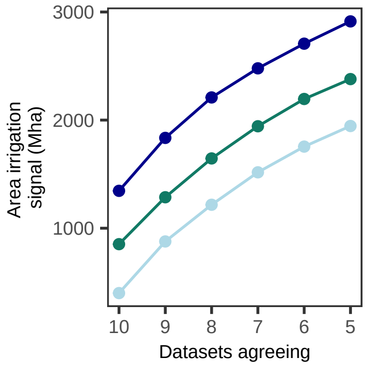
```
##     resolution     k n_cells_irrig_k n_cells_join_k n_cells_with_dET_k
##         <char> <int>           <int>          <int>              <int>
##   1:      0.2deg     5           46920          46920              46723
##   2:      0.2deg     6           42296          42296              42144
##   3:      0.2deg     7           36510          36510              36396
##   4:      0.2deg     8           29235          29235              29158
##   5:      0.2deg     9           21006          21006              20964
##   6:      0.2deg    10            9590           9590               9575
##   7:      0.4deg     5           14369          14369              14312
##   8:      0.4deg     6           13251          13251              13200
##   9:      0.4deg     7           11715          11715              11675
##  10:      0.4deg     8            9910           9910               9878
##  11:      0.4deg     9            7735           7735               7718
##  12:      0.4deg    10            5118           5118               5112
##  13:        1deg     5            2816           2816               2811
##  14:        1deg     6            2618           2618               2613
##  15:        1deg     7            2396           2396               2391
##  16:        1deg     8            2133           2133               2129
##  17:        1deg     9            1772           1772               1769
##  18:        1deg    10            1298           1298               1297
##     mean_dET_mmday sum_dET_mmday_unweighted mean_dET_mmday_areaweighted
##              <num>                    <num>                       <num>
##   1:     0.04012408               1874.71758                  0.04011012
##   2:     0.04332301               1825.80479                  0.04337120
##   3:     0.04821179               1754.71632                  0.04832421
##   4:     0.05598796               1632.49706                  0.05619227
##   5:     0.06818188               1429.36490                  0.06853504
##   6:     0.08271255                791.97271                  0.08345616
##   7:     0.03678079                526.40665                  0.03666735
##   8:     0.03922764                517.80487                  0.03914187
##   9:     0.04295393                501.48715                  0.04292040
##  10:     0.04846748                478.76176                  0.04850110
##  11:     0.05689614                439.12441                  0.05710005
```

```
## 12:     0.07209324              368.54062                 0.07260780
## 13:     0.03344324               94.00895                 0.03322915
## 14:     0.03568753               93.25152                 0.03550908
## 15:     0.03845227               91.93937                 0.03830816
## 16:     0.04232406               90.10792                 0.04219702
## 17:     0.04798235               84.88078                 0.04799651
## 18:     0.05863885               76.05459                 0.05891090
##     sum_dET_m3day_areaweighted area_m2_with_dET_k
##                          <num>              <num>
##  1:                  780279411       1.945343e+13
##  2:                  761085387       1.754817e+13
##  3:                  733047410       1.516936e+13
##  4:                  683881025       1.217038e+13
##  5:                  601363946       8.774547e+12
##  6:                  333696792       3.998468e+12
##  7:                  872410097       2.379256e+13
##  8:                  859248031       2.195214e+13
##  9:                  833991565       1.943112e+13
## 10:                  798030563       1.645387e+13
## 11:                  734575092       1.286470e+13
## 12:                  618966902       8.524799e+12
## 13:                  968022615       2.913173e+13
## 14:                  961187451       2.706878e+13
## 15:                  949610682       2.478873e+13
## 16:                  932414251       2.209669e+13
## 17:                  880927203       1.835398e+13
## 18:                  792833208       1.345818e+13
```

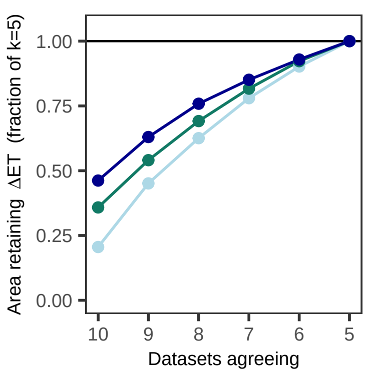
```r
# PLOT ##########################################################################

# Condition ifesle -------------------------------------------------------------

total_col <- if ("sum_dET_m3day_areaweighted" %in% names(out_k_all)) {
  "sum_dET_m3day_areaweighted"
} else if ("sum_dET_mmday_unweighted" %in% names(out_k_all)) {
  "sum_dET_mmday_unweighted"
} else {
  stop("out_k_all has neither sum_dET_m3day_areaweighted nor sum_dET_mmday_unweighted")
}

# Convert m3/day to km3/day ----------------------------------------------------

out_plot <- copy(out_k_all)

if (total_col == "sum_dET_m3day_areaweighted") {
  out_plot[, total_km3day := get(total_col) / 1e9]
  total_ylab <- "Total irrigation-\ninduced ET (km³ d-¹)"
```

```r
} else {

  out_plot[, total_km3day := get(total_col)]
  total_ylab <- "Total irrigation-induced \nET signal (sum of mm/day)"
}

# Keep only k = 5, ..., 10-----------------------------------------------------

out_plot <- out_plot[k %in% 5:10]

# Plot mean increase in irrigation-induced ET ----------------------------------
# (average irrigation-induced evapotranspiration change per grid cell)

plot_mean_ET <- ggplot(out_plot, aes(k, mean_dET_mmday, colour = resolution,
                                     group = resolution)) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing",
       y = expression(atop("Mean " * Delta * ET[irr], "(mm d"^{-1} * ")"))) +
  scale_x_reverse(breaks = 10:5) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

# Plot total irrigation induced ET ---------------------------------------------
# (global irrigation-induced evapotranspiration signal, obtained by integrating
# over all retained cells; e.g., total amount of water transferred to the
# atmosphere per day because of irrigation)

plot_total_ET <- ggplot(out_plot, aes(k, total_km3day, colour = resolution, group = resolution
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing", y = total_ylab) +
  scale_x_reverse(breaks = 10:5) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

# Plot area with irrigation-induced ET retained under k-------------------------

out_plot <- copy(out_k_all)[k %in% 5:10]

# convert m2 to Mha ------------------------------------
out_plot[, area_Mha:= area_m2_with_dET_k / 1e10]

# also show contraction relative to k=5 within each resolution
out_plot[, area_frac_vs_k5:= area_m2_with_dET_k / area_m2_with_dET_k[k == 5], by = resolution]
```

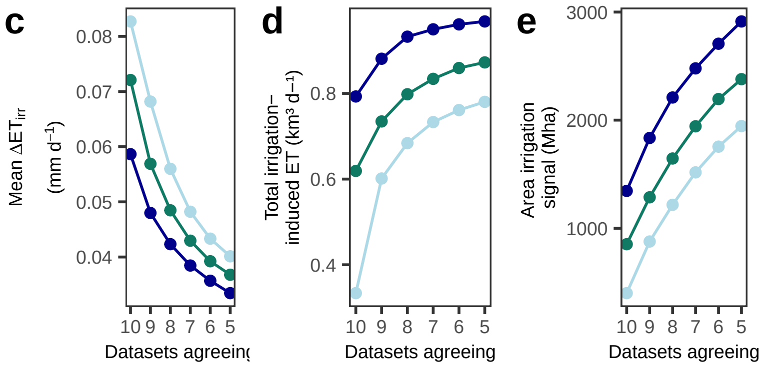
```r
# Total land area of grid cells with reported irrigated areas
plot_area <- ggplot(out_plot, aes(k, area_Mha, colour = resolution, group = resolution)) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing", y = "Area irrigation \nsignal (Mha)") +
  scale_x_reverse(breaks = 10:5) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

plot_area
```



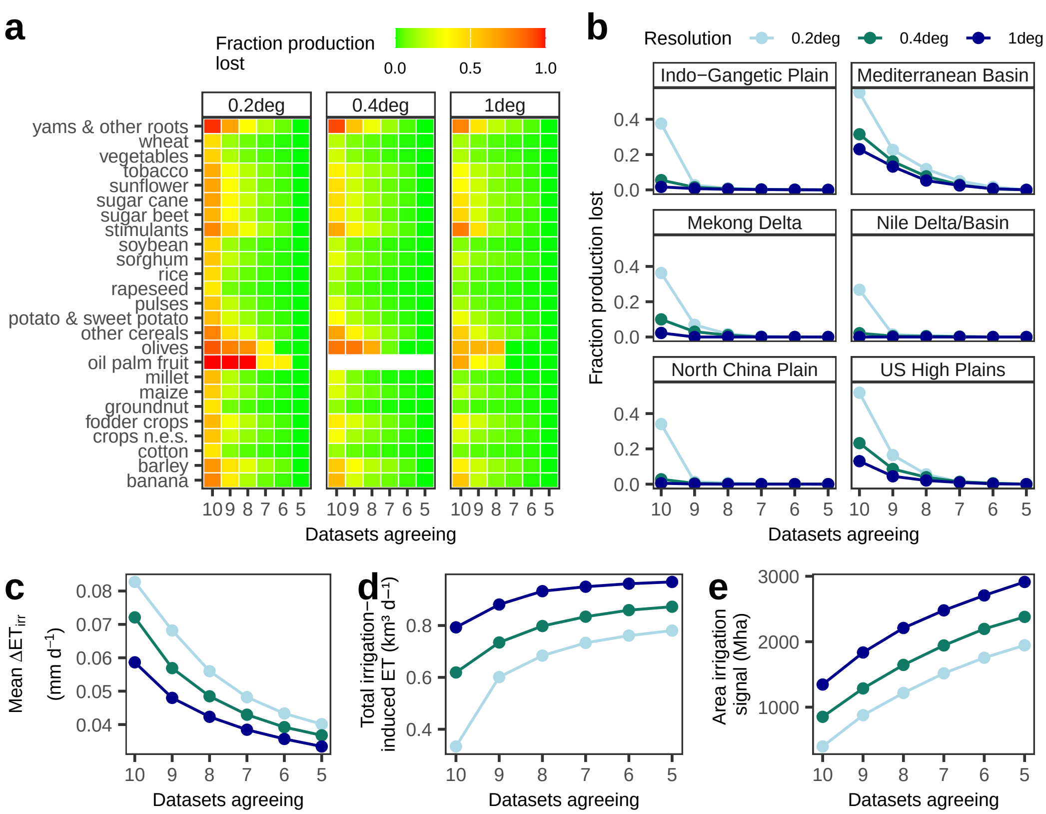

```r
# Also in case: fraction of area retained vs k=5
plot_area_fraction <- ggplot(out_plot, aes(k, area_frac_vs_k5, colour = resolution, group = res
  geom_hline(yintercept = 1, linewidth = 0.4) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing",
       y = expression("Area retaining "~Delta*ET~" (fraction of k=5)")) +
  scale_x_reverse(breaks = 10:5) +
  scale_y_continuous(limits = c(0, 1.05)) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

plot_area_fraction
```



```
bottom <- plot_grid(plot_mean_ET, plot_total_ET, plot_area, labels = c("c", "d", "e"),
                    ncol = 3)
bottom
```




```
plot_grid(top, bottom, ncol = 1, rel_heights = c(0.685, 0.325))
```



6 Final figures

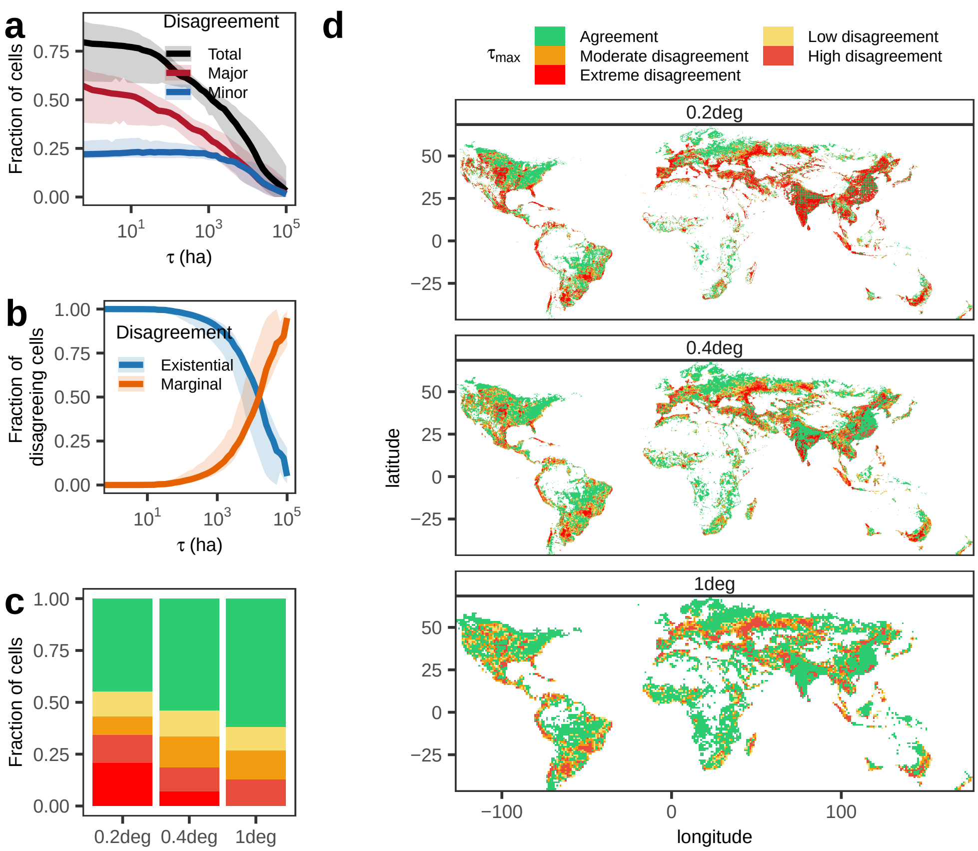
```
left <- plot_grid(plot_uncertainty1, plot_agreement1,
                  plot_stacked +
                    labs(y = "Fraction of cells"), ncol = 1, labels = "auto")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
plot_grid(left, plot_raster, ncol = 2, rel_widths = c(0.3, 0.7), labels = c("", "d"))
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

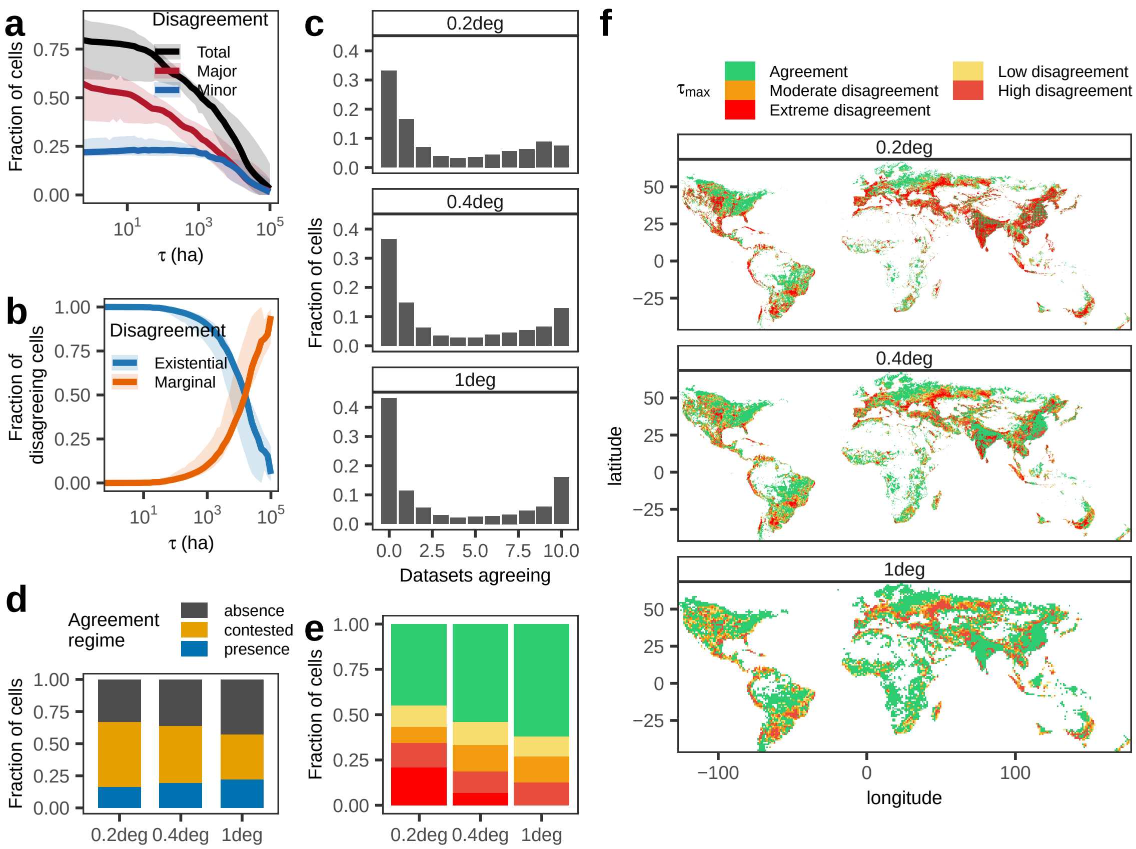



```
left <- plot_grid(plot_uncertainty1, plot_agreement1,
                  plot_regime, ncol = 1, labels = c("a", "b", "d"))
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```r
middle <- plot_grid(p1 + facet_wrap(~resolution, ncol = 1),
                    plot_stacked +
                      labs(y = "Fraction of cells"),
                    ncol = 1, rel_heights = c(0.7, 0.3), labels = c("c", "e"))
plot_grid(left, middle, plot_raster, ncol = 3, rel_widths = c(0.26, 0.26, 0.48), labels = c(""
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.

## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

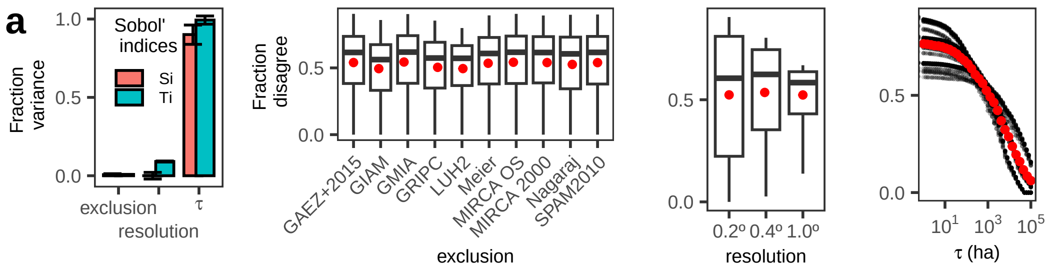



```r
top <- plot_grid(plot_indices1 +
                   labs(x = "", y = "Fraction \n variance") +
                   scale_fill_discrete(name = "Sobol' \n indices") +
                   theme(legend.position = c(0.4, 0.7)),
                 a1 + labs(x = "exclusion", y = "Fraction \n disagree"),
                 b1, c1, ncol = 4, labels = c("a", "", "", ""),
                 rel_widths = c(0.23, 0.37, 0.2, 0.2))
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced i
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

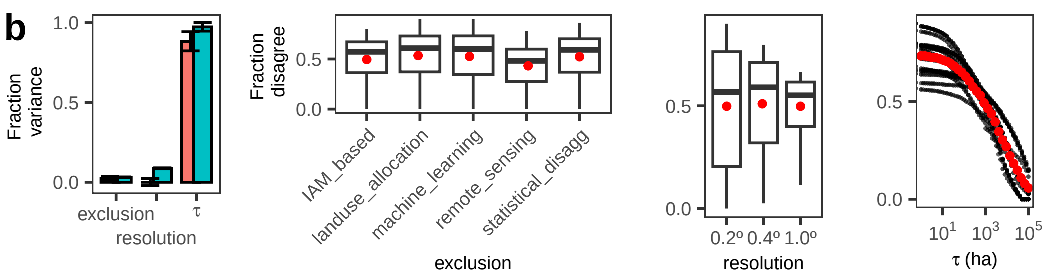
```
top
```




```
middle <- plot_grid(plot_indices2 +
                      labs(x = "", y = "Fraction \n variance") +
                      scale_fill_discrete(name = "") +
                      theme(legend.position = "none"),
                    a2 + labs(x = "exclusion", y = "Fraction \n disagree"),
                    b2, c2, ncol = 4, labels = c("b", "", "", ""),
                    rel_widths = c(0.23, 0.37, 0.2, 0.2))
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
middle
```

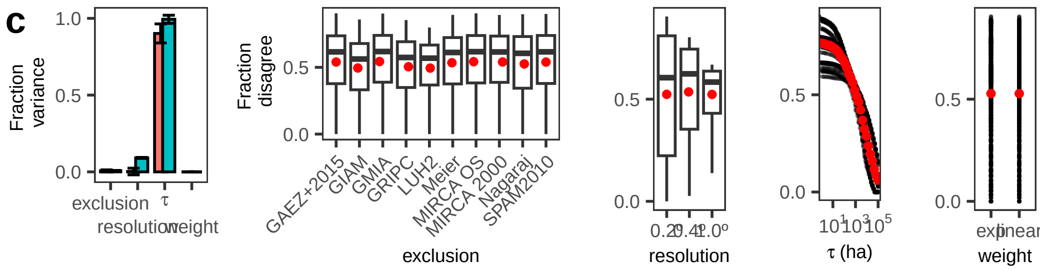



```
bottom <- plot_grid(plot_indices3 +
                      labs(x = "", y = "Fraction \n variance") +
                      scale_fill_discrete(name = "") +
                      theme(legend.position = "none"),
                    a3 + labs(x = "exclusion", y = "Fraction \n disagree"),
                    b3, c3, d3, ncol = 5, labels = c("c", "", "", "", ""),
                    rel_widths = c(0.22, 0.34, 0.16, 0.15, 0.15))
```

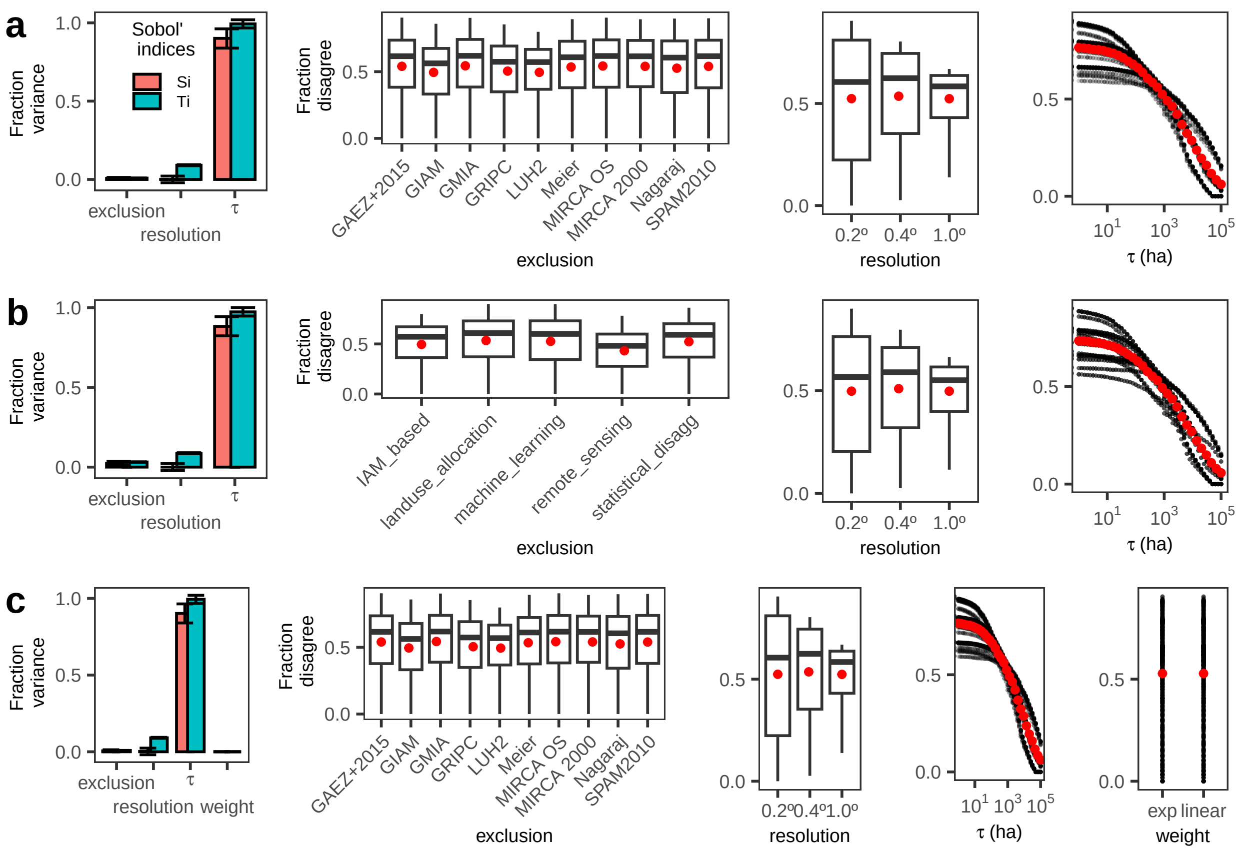
```
## Scale for fill is already present.
```

```
## Adding another scale for fill, which will replace the existing scale.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
bottom
```



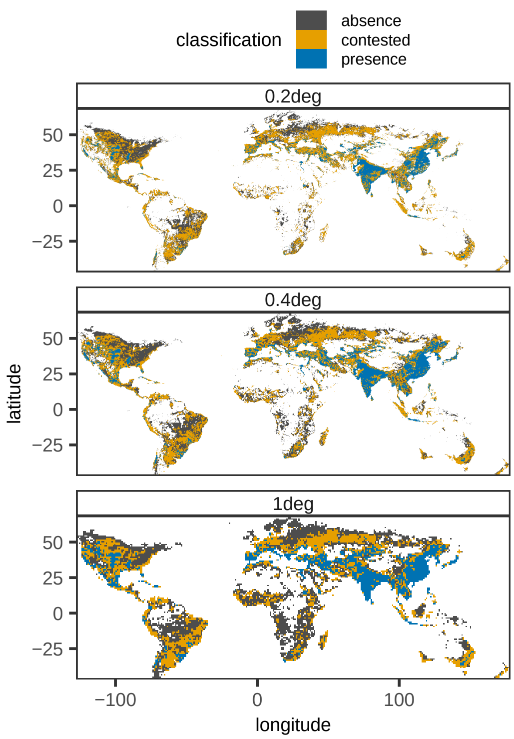

```
plot_grid(top, middle, bottom, ncol = 1)
```




```
dt_pres[, classification := fifelse(
  n_pos == 0, "absence",
  fifelse(n_pos <= 8, "contested", "presence")
)]
```

```r
plot_classification <- ggplot(dt_pres, aes(lon, lat, fill = classification)) +
  geom_raster() +
  coord_equal(expand = FALSE) +
  facet_wrap(~ resolution, ncol = 1) +
  scale_fill_manual(values = c(absence = "#4D4D4D",
                               contested = "#E69F00",
                              presence = "#0072B2")) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  theme_AP() +
  theme(panel.grid = element_blank(),
        strip.text = element_text(size = 11),
        legend.position = "top") +
  labs(x = "longitude", y = "latitude") +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

plot_classification
```

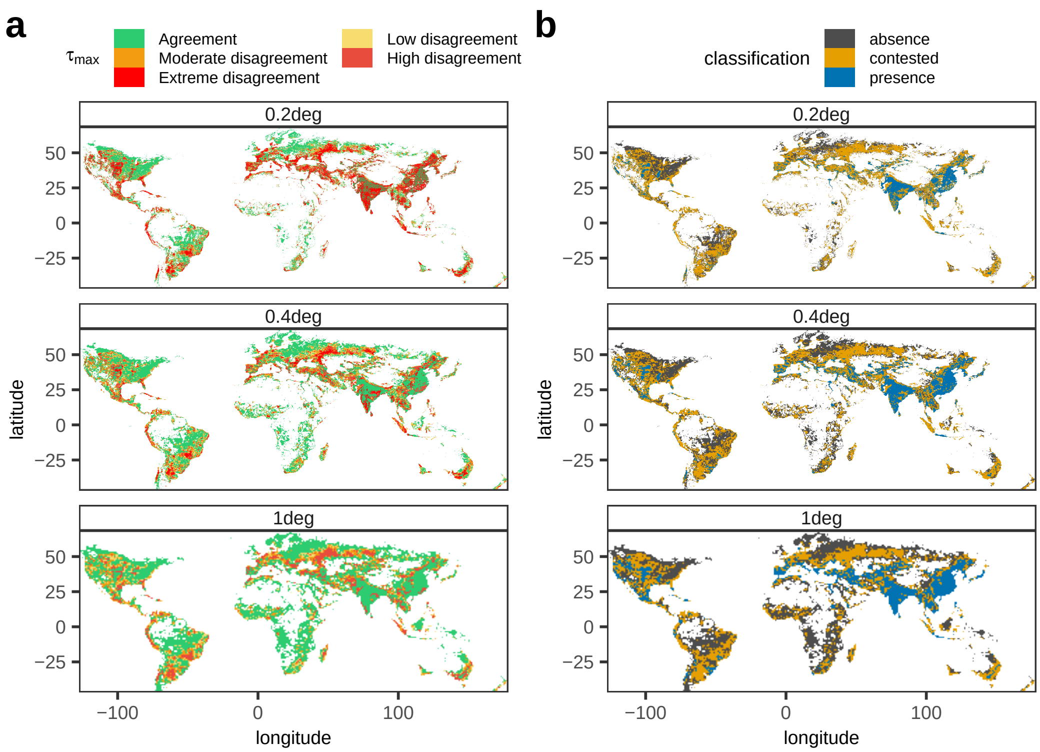
```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```




```r
plot_grid(plot_raster, plot_classification, ncol = 2, labels = "auto")
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

```
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```



7 Is disagreement caused by different irrigation conceptualizations?

7.1 Different definitions

```
# STUDY OF DEFINITION HETEROGENEITY #####################################

dt_definitions <- fread("./llm_irrigated_definitions/outputs/irrigation_definition_table.csv")

# Create crosswalk table -------------------------------------------------------

crosswalk <- data.table(
  source_file = c(
    "Grogan et al. - 2022 - Global gridded crop harvested area, production, yield, and monthly
    "Hurtt et al. - 2020 - Harmonization of global land use change and management for the peri
    "Meier et al_2018_A global approach to estimate irrigated areas.pdf",
    "Nagaraj et al_2021_A new dataset of global irrigation areas from 2001 to 2015.pdf",
    "Portmann et al. - 2010 - MIRCA2000-Global monthly irrigated and rainfed cro.pdf",
    "Salmon et al. - 2015 - Global rain-fed, irrigated, and paddy croplands A new high resoluti
    "Siebert et al_2013_Update of the digital global map of irrigation areas to version 5.pdf"
```

```r
    "Thenkabail et al_2009_Global irrigated area map (GIAM), derived from remote sensing, for t
    "Yu et al. - 2020 - A cultivated planet in 2010 - Part 2 The global gridded agricultural-pr
    "s41597-024-04313-w-1.pdf"
  ),
  dataset = c(
    "gaez_v4",
    "luh2",
    "meier",
    "nagaraj",
    "mirca2000",
    "gripc",
    "gmia",
    "giam",
    "spam",
    "mirca_os"
  )
)

# Keep the columns needed for the audit ----------------------------------------

vars_def <- c("source_file",
              "paper_title",
              "irrigation_target",
              "representation_unit",
              "temporal_concept",
              "paddy_treatment",
              "permanent_crops_treatment",
              "irrigation_basis")

dt_definitions <- dt_definitions[, ..vars_def]

# Merge with dataset column ----------------------------------------------------

dt_definitions <- merge(dt_definitions, crosswalk, by = "source_file", all.x = TRUE)

# Convert empty strings to NA, then label as "unclear" ------------------------

for (v in setdiff(vars_def, c("source_file", "paper_title"))) {
  dt_definitions[get(v) == "" | is.na(get(v)), (v):= "unclear"]
}

# Make frequency tables --------------------------------------------------------

freq_list <- lapply(
  setdiff(vars_def, c("source_file", "paper_title")),
  function(v) make_freq_table(dt_definitions, v)
)
```

```r
freq_dt <- rbindlist(freq_list, use.names = TRUE, fill = TRUE)
freq_dt
```

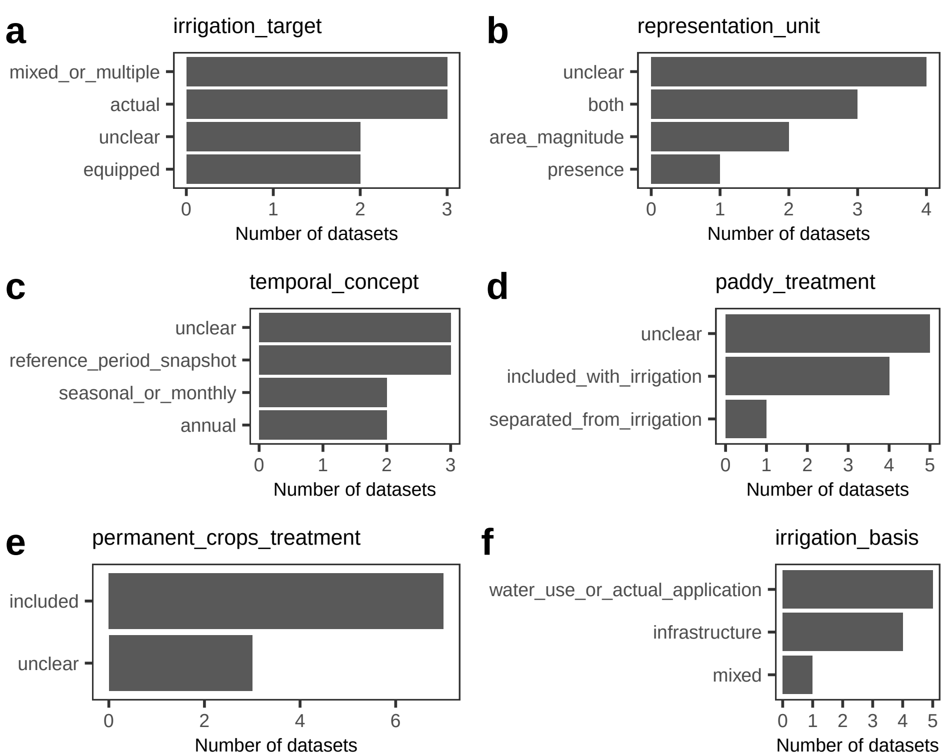
```
##                              category     N                 variable share
##                                <char> <int>                   <char> <num>
##  1:                            actual     3        irrigation_target   0.3
##  2:                 mixed_or_multiple     3        irrigation_target   0.3
##  3:                          equipped     2        irrigation_target   0.2
##  4:                           unclear     2        irrigation_target   0.2
##  5:                           unclear     4      representation_unit   0.4
##  6:                              both     3      representation_unit   0.3
##  7:                    area_magnitude     2      representation_unit   0.2
##  8:                          presence     1      representation_unit   0.1
##  9:                           unclear     3         temporal_concept   0.3
## 10:         reference_period_snapshot     3         temporal_concept   0.3
## 11:                            annual     2         temporal_concept   0.2
## 12:               seasonal_or_monthly     2         temporal_concept   0.2
## 13:                           unclear     5          paddy_treatment   0.5
## 14:          included_with_irrigation     4          paddy_treatment   0.4
## 15:        separated_from_irrigation     1          paddy_treatment   0.1
## 16:                          included     7 permanent_crops_treatment   0.7
## 17:                           unclear     3 permanent_crops_treatment   0.3
## 18: water_use_or_actual_application     5         irrigation_basis   0.5
## 19:                    infrastructure     4         irrigation_basis   0.4
## 20:                             mixed     1         irrigation_basis   0.1
##                              category     N                 variable share
##                                <char> <int>                   <char> <num>
```

```r
# Share of mixed / unclear datasets------------------------------------------

summary_mixed_unclear <- rbindlist(lapply(
  setdiff(vars_def, c("source_file", "paper_title")),
  function(v) {
    tmp <- dt_definitions[, .(n_total = .N,
                              n_unclear = sum(get(v) == "unclear", na.rm = TRUE),
                              n_mixed = sum(get(v) %in% c("mixed", "mixed_or_multiple"), na.rm
    tmp[, variable:= v]
    tmp[, share_unclear:= n_unclear / n_total]
    tmp[, share_mixed:= n_mixed / n_total]
    tmp[]
  }), fill = TRUE)

print(summary_mixed_unclear)
```

```
##    n_total n_unclear n_mixed                variable share_unclear
##      <int>     <int>   <int>                  <char>         <num>
## 1:      10         2       3       irrigation_target           0.2
## 2:      10         4       0     representation_unit           0.4
```

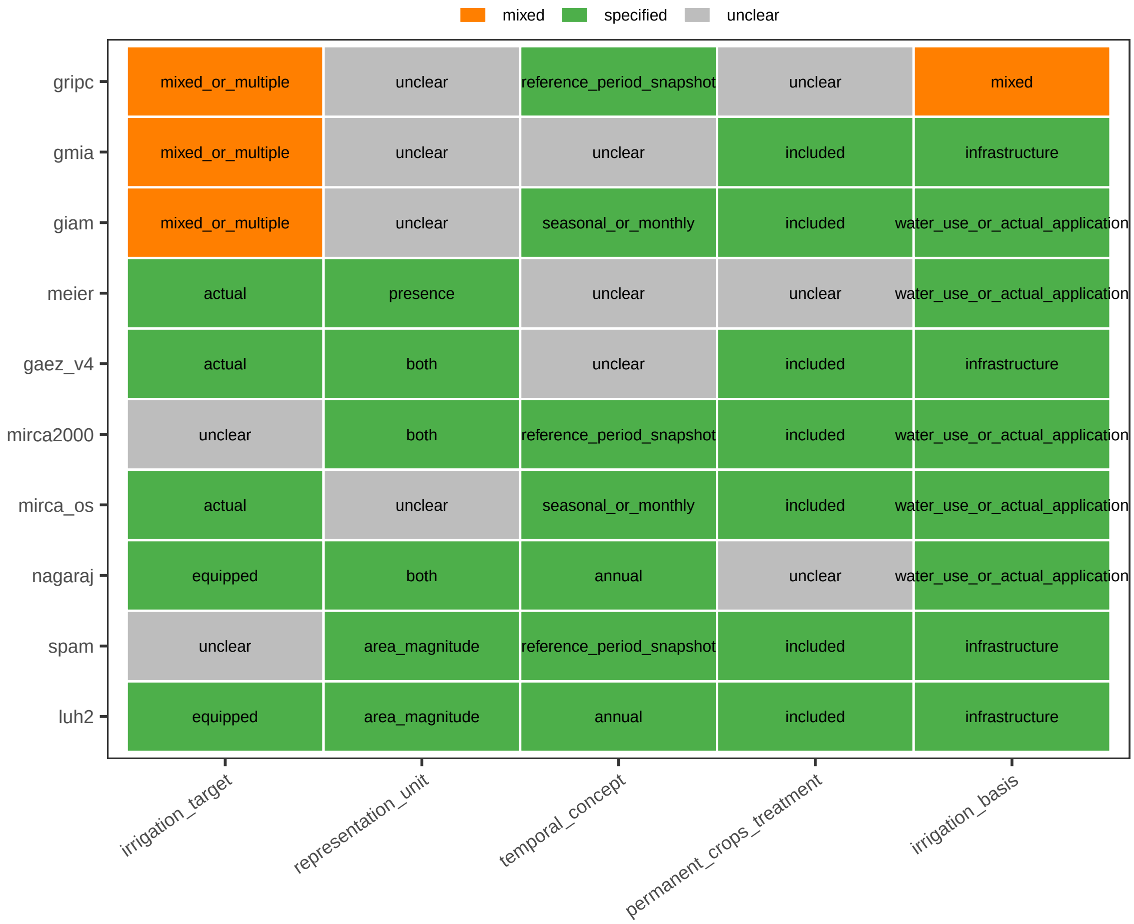
```
## 3:       10          3       0         temporal_concept          0.3
## 4:       10          5       0          paddy_treatment          0.5
## 5:       10          3       0 permanent_crops_treatment          0.3
## 6:       10          0       1         irrigation_basis          0.0
##    share_mixed
##          <num>
## 1:         0.3
## 2:         0.0
## 3:         0.0
## 4:         0.0
## 5:         0.0
## 6:         0.1
```

```r
# Barplots of definitional heterogeneity ---------------------------------------

p1 <- plot_bar_var(freq_dt, "irrigation_target")
p2 <- plot_bar_var(freq_dt, "representation_unit")
p3 <- plot_bar_var(freq_dt, "temporal_concept")
p4 <- plot_bar_var(freq_dt, "paddy_treatment")
p5 <- plot_bar_var(freq_dt, "permanent_crops_treatment")
p6 <- plot_bar_var(freq_dt, "irrigation_basis")

plot_grid(p1, p2, p3, p4, p5, p6, ncol = 2, labels = "auto")
```




```r
# Dataset × definition heatmap panel -------------------------------------------

dt_long <- melt(dt_definitions, id.vars = c("dataset"),
```

```r
                measure.vars = c("irrigation_target",
                                 "representation_unit",
                                 "temporal_concept",
                                 "permanent_crops_treatment",
                                 "irrigation_basis"),
  variable.name = "definition_variable",
  value.name = "definition_value")

dataset_order <- dt_long[, .(
  score = sum(definition_value %in% c("unclear", "mixed", "mixed_or_multiple"))),
  by = dataset][order(-score, dataset)]$dataset

dt_long[, dataset:= factor(dataset, levels = rev(dataset_order))]

dt_long[, class_type:= fifelse(
  definition_value == "unclear", "unclear",
  fifelse(definition_value %in% c("mixed", "mixed_or_multiple"), "mixed", "specified")
)]

# Plot heatmap: Cells report the definition assigned to each dataset;
# colour highlights whether the definition is specified, mixed, or unclear

p_heat <- ggplot(dt_long, aes(definition_variable, dataset, fill = class_type)) +
  geom_tile(color = "white", linewidth = 0.4) +
  geom_text(aes(label = definition_value), size = 2) +
  scale_fill_manual(values = c("specified" = "#4daf4a", "mixed" = "#ff7f00",
                               "unclear" = "#bdbdbd")) +
  labs(x = NULL, y = NULL, fill = NULL) +
  theme_AP() +
  theme(panel.grid = element_blank(),
        axis.text.x = element_text(angle = 35, hjust = 1),
        legend.position = "top")

p_heat
```




```r
# Summary metrics---------------------------------------------------------------

n_datasets <- uniqueN(dt_definitions$dataset)

summary_text_dt <- rbindlist(lapply(
  c("irrigation_target",
    "representation_unit",
    "temporal_concept",
    "paddy_treatment",
    "permanent_crops_treatment",
    "irrigation_basis"),
  function(v) {
    tmp <- dt_definitions[, .(
      n_unique_categories = uniqueN(get(v)),
      n_unclear = sum(get(v) == "unclear"),
      n_mixed = sum(get(v) %in% c("mixed", "mixed_or_multiple"))
    )]
    tmp[, variable:= v]
    tmp[, share_unclear:= round(n_unclear / n_datasets, 3)]
    tmp[, share_mixed:= round(n_mixed / n_datasets, 3)]
    tmp[]
```

```
  }
), fill = TRUE)

print(summary_text_dt)
```

```
##    n_unique_categories n_unclear n_mixed                  variable
##                  <int>     <int>   <int>                    <char>
## 1:                   4         2       3         irrigation_target
## 2:                   4         4       0       representation_unit
## 3:                   4         3       0          temporal_concept
## 4:                   3         5       0           paddy_treatment
## 5:                   2         3       0 permanent_crops_treatment
## 6:                   3         0       1          irrigation_basis
##    share_unclear share_mixed
##            <num>       <num>
## 1:           0.2         0.3
## 2:           0.4         0.0
## 3:           0.3         0.0
## 4:           0.5         0.0
## 5:           0.3         0.0
## 6:           0.0         0.1
```

```
# The literature audit reveals that global irrigation maps do not represent a single
# concept of irrigated area. Across products, irrigation is variously defined as
# equipped, actual, or harvested irrigation, while treatment of paddy rice, temporal
# reference, and irrigation basis also varies. Several datasets are mixed or
# conceptually unclear, indicating that definitional heterogeneity is real and must
# be explicitly accounted for before assessing whether it can explain the observed
# detectability crisis.
```

7.2 Tests: same definition versus different definition

```
# TEST: PAIRWISE SAME DEFINITION VERSUS DIFFERENT DEFINITION TESTS ############

# We focus on 0.2 resolution----------------------------------------------------

resolution_sel <- "0.2deg"

# PREPARE PRESENCE DATA --------------------------------------------------------

dtp <- copy(dt_pres)[resolution == resolution_sel]

# Dataset columns available in dt_pres -----------------------------------------

id_cols <- c("lon", "lat", "country", "code", "continent", "resolution", "n_pos")
dataset_cols <- setdiff(names(dtp), id_cols)

# Keep only datasets that are both in def_dt and dt_pres ------------------------
```

```r
common_datasets <- intersect(dt_definitions$dataset, dataset_cols)

dt_definitions <- dt_definitions[dataset %in% common_datasets]
dtp <- dtp[, c(setdiff(id_cols, "n_pos"), common_datasets), with = FALSE]

cat("\nDatasets used in pairwise test:\n")
```

```
##
## Datasets used in pairwise test:
```

```r
print(common_datasets)
```

```
##  [1] "gaez_v4"   "luh2"      "meier"     "nagaraj"   "mirca2000" "gripc"
##  [7] "gmia"      "giam"      "spam"      "mirca_os"
```

```r
# RUN PAIRWISE DISAGREEMENT FUNCTION ###########################################

pair_dt <- compute_pairwise_disagreement(dtp, common_datasets)

# ATTACH DEFINITIONAL ATTRIBUTES TO EACH PAIR ---------------------------------

# Attributes for dataset i -----------------------------------------------------

def_i <- copy(dt_definitions)
setnames(def_i,
         old = c("dataset", "irrigation_target", "representation_unit",
                 "temporal_concept", "paddy_treatment",
                 "permanent_crops_treatment", "irrigation_basis"),
         new = c("dataset_i", "irrigation_target_i", "representation_unit_i",
                 "temporal_concept_i", "paddy_treatment_i",
                 "permanent_crops_treatment_i", "irrigation_basis_i"))

# Attributes for dataset j------------------------------------------------------

def_j <- copy(dt_definitions)
setnames(def_j,
         old = c("dataset", "irrigation_target", "representation_unit",
                 "temporal_concept", "paddy_treatment",
                 "permanent_crops_treatment", "irrigation_basis"),
         new = c("dataset_j", "irrigation_target_j", "representation_unit_j",
                 "temporal_concept_j", "paddy_treatment_j",
                 "permanent_crops_treatment_j", "irrigation_basis_j"))

pair_dt <- merge(pair_dt, def_i[, .(
  dataset_i,
  irrigation_target_i,
  representation_unit_i,
  temporal_concept_i,
```

```
  paddy_treatment_i,
  permanent_crops_treatment_i,
  irrigation_basis_i
)], by = "dataset_i", all.x = TRUE)

pair_dt <- merge(pair_dt, def_j[, .(
  dataset_j,
  irrigation_target_j,
  representation_unit_j,
  temporal_concept_j,
  paddy_treatment_j,
  permanent_crops_treatment_j,
  irrigation_basis_j
)], by = "dataset_j", all.x = TRUE)

# same-definition indicators---------------------------------------------------

pair_dt[, same_irrigation_target:= irrigation_target_i == irrigation_target_j]
pair_dt[, same_representation_unit:= representation_unit_i == representation_unit_j]
pair_dt[, same_temporal_concept:= temporal_concept_i == temporal_concept_j]
pair_dt[, same_paddy_treatment:= paddy_treatment_i == paddy_treatment_j]
pair_dt[, same_permanent_crops_treatment:=permanent_crops_treatment_i == permanent_crops_treatm
pair_dt[, same_irrigation_basis:= irrigation_basis_i == irrigation_basis_j]

# stricter indicators ----------------------------------------------------------

pair_dt[, same_core_definition:= same_irrigation_target & same_representation_unit &
          same_temporal_concept]

pair_dt[, same_extended_definition:= same_irrigation_target & same_representation_unit &
          same_temporal_concept & same_paddy_treatment & same_irrigation_basis]

# optional: exclude unclear/mixed comparisons-----------------------------------

pair_dt[, clear_irrigation_target:=
          !(irrigation_target_i %in% c("unclear", "mixed_or_multiple")) &
          !(irrigation_target_j %in% c("unclear", "mixed_or_multiple"))]

pair_dt[, clear_core_definition :=
          !(irrigation_target_i %in% c("unclear", "mixed_or_multiple")) &
          !(irrigation_target_j %in% c("unclear", "mixed_or_multiple")) &
          !(representation_unit_i %in% c("unclear", "mixed")) &
          !(representation_unit_j %in% c("unclear", "mixed")) &
          !(temporal_concept_i %in% c("unclear", "mixed")) &
          !(temporal_concept_j %in% c("unclear", "mixed"))]
```

```r
# DESCRIPTIVE COMPARISONS -----------------------------------------------

# Mean disagreement by same/different target
desc_target <- pair_dt[, .(
  n_pairs = .N,
  mean_disagree_all = mean(share_disagree_all, na.rm = TRUE),
  median_disagree_all = median(share_disagree_all, na.rm = TRUE),
  mean_disagree_union = mean(share_disagree_union, na.rm = TRUE),
  median_disagree_union = median(share_disagree_union, na.rm = TRUE),
  mean_jaccard_dissimilarity = mean(jaccard_dissimilarity, na.rm = TRUE)
), by = same_irrigation_target]

print(desc_target)
```

```
##    same_irrigation_target n_pairs mean_disagree_all median_disagree_all
##                    <lgcl>   <int>             <num>               <num>
## 1:                  FALSE      37         0.2057572           0.2308683
## 2:                   TRUE       8         0.1892357           0.1783141
##    mean_disagree_union median_disagree_union mean_jaccard_dissimilarity
##                  <num>                 <num>                      <num>
## 1:           0.4642512             0.5291120                  0.4642512
## 2:           0.4200700             0.4052928                  0.4200700
```

```r
# Mean disagreement by same/different extended definition
desc_extended <- pair_dt[, .(
  n_pairs = .N,
  mean_disagree_all = mean(share_disagree_all, na.rm = TRUE),
  median_disagree_all = median(share_disagree_all, na.rm = TRUE),
  mean_disagree_union = mean(share_disagree_union, na.rm = TRUE),
  median_disagree_union = median(share_disagree_union, na.rm = TRUE),
  mean_jaccard_dissimilarity = mean(jaccard_dissimilarity, na.rm = TRUE)
), by = same_extended_definition]

print(desc_extended)
```

```
##    same_extended_definition n_pairs mean_disagree_all median_disagree_all
##                      <lgcl>   <int>             <num>               <num>
## 1:                    FALSE      45           0.20282           0.2308683
##    mean_disagree_union median_disagree_union mean_jaccard_dissimilarity
##                  <num>                 <num>                      <num>
## 1:           0.4563968             0.5244522                  0.4563968
```

```r
# WILCOXON TESTS ---------------------------------------------------------

# same vs different irrigation target -----------------
# Are the disagreement values in the two groups statistically different?
# It compares two groups of pairs of datasets:

# Group 1: pairs where same_irrigation_target == TRUE
```

```r
# (both datasets claim to represent the same type of irrigation (e.g. both "actual"))
#
# Group 2: pairs where same_irrigation_target == FALSE
# (datasets represent different concepts (e.g. "actual" vs "equipped"))

# What is being compared?
# The variable: share_disagree_union
# (the fraction of cells where the two datasets disagree,
# conditional on at least one detecting irrigation.)
#
# What the test asks: Are the disagreement values in the two groups statistically
# different?

w_target <- wilcox.test(share_disagree_union ~ same_irrigation_target,
                        data = pair_dt)

print(w_target)
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  share_disagree_union by same_irrigation_target
## W = 178, p-value = 0.3883
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Dataset pairs representing the same irrigation concept do not exhibit lower
# disagreement than those representing different concepts (Wilcoxon test, $p = 0.97$).
# This indicates that definitional alignment does not reduce disagreement.

# same vs different core definition-------------------

pair_dt[, core_similarity_score:= same_irrigation_target + same_representation_unit +
          same_temporal_concept]

cor.test(pair_dt$core_similarity_score, pair_dt$share_disagree_union, method = "spearman")
```

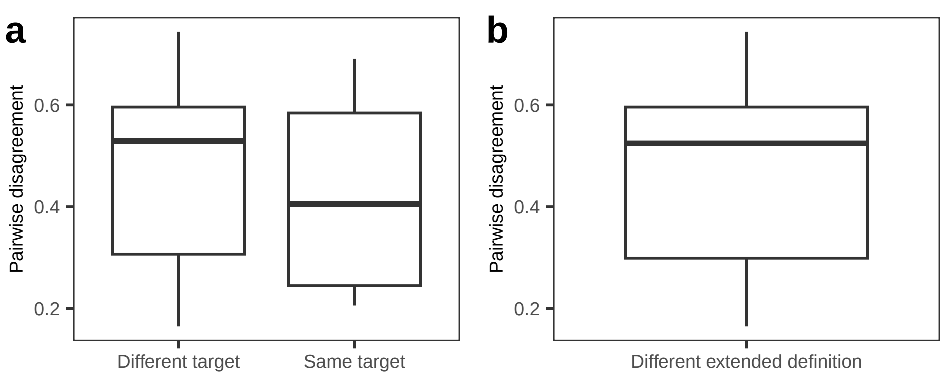
```
## Warning in cor.test.default(pair_dt$core_similarity_score,
## pair_dt$share_disagree_union, : Cannot compute exact p-value with ties
```

```
##
##  Spearman's rank correlation rho
##
## data:  pair_dt$core_similarity_score and pair_dt$share_disagree_union
## S = 18311, p-value = 0.174
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.2062814
```

```r
# Increasing definitional similarity does not reduce disagreement
# Even when datasets match across multiple definitional dimensions, they do not
# agree more spatially.

# PARAGRAPH: Disagreement does not decline with increasing definitional similarity
# (Spearman's $\rho = -0.04$, $p = 0.78$), indicating that even datasets aligned
# across multiple conceptual dimensions fail to identify irrigation consistently.

# REGRESSION MODEL---------------------------------------------------------------

# Simple OLS on pairwise disagreement
# Dependent variable: disagreement conditional on at least one map detecting irrigation
mod1 <- lm(share_disagree_union ~
             same_irrigation_target +
             same_representation_unit +
             same_temporal_concept +
             same_paddy_treatment +
             same_irrigation_basis,
           data = pair_dt)

summary(mod1)
```

```
##
## Call:
## lm(formula = share_disagree_union ~ same_irrigation_target +
##     same_representation_unit + same_temporal_concept + same_paddy_treatment +
##     same_irrigation_basis, data = pair_dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32529 -0.14896  0.03409  0.13944  0.30826
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   0.48009    0.04791  10.020 2.42e-12 ***
## same_irrigation_targetTRUE   -0.05099    0.07654  -0.666    0.509
## same_representation_unitTRUE  0.01027    0.06621   0.155    0.878
## same_temporal_conceptTRUE    -0.04662    0.07596  -0.614    0.543
## same_paddy_treatmentTRUE      0.02035    0.05872   0.347    0.731
## same_irrigation_basisTRUE    -0.04461    0.05757  -0.775    0.443
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1796 on 39 degrees of freedom
## Multiple R-squared:  0.04198,    Adjusted R-squared:  -0.08084
## F-statistic: 0.3418 on 5 and 39 DF,  p-value: 0.8843
```

```r
#
# None of the definitional dimensions significantly explains pairwise
# disagreement (all $p > 0.18$), and together they account for less than 8% of
# its variance. Definitional differences therefore fail to explain the widespread
# inconsistencies across irrigation maps.

# Matching definitions across five dimensions does not systematically reduce disagreement.

# PLOTS -------------------------------------------------------------------------

# Boxplot: same vs different irrigation target
# title = "Pairwise disagreement for same-definition vs different-definition pairs",
# subtitle = "Comparison by irrigation target"
p_target <- ggplot(pair_dt, aes(x = factor(same_irrigation_target,
                                           levels = c(FALSE, TRUE),
                                           labels = c("Different target",
                                                      "Same target")),
                                y = share_disagree_union)) +
  geom_boxplot() +
  labs(x = NULL, y = "Pairwise disagreement") +
  theme_AP()

# Boxplot: same vs different extended definition
# title = "Pairwise disagreement for same-definition vs different-definition pairs",
# subtitle = "Comparison by extended definition"
p_ext <- ggplot(pair_dt, aes(x = factor(same_extended_definition,
                                        levels = c(FALSE, TRUE),
                                        labels = c("Different extended definition",
                                                   "Same extended definition")),
                             y = share_disagree_union)) +
  geom_boxplot() +
  labs(x = NULL, y = "Pairwise disagreement") +
  theme_AP()

plot_grid(p_target, p_ext, ncol = 2, labels = "auto")
```



```r
# Global irrigation datasets do not represent a single, consistent concept of irrigated area.
# Across products, irrigation is variously defined as actual, equipped or mixed irrigation,
# and differs in representation unit (presence versus area), temporal reference
# (annual, seasonal or reference-period snapshots), and treatment of paddy rice and
# permanent crops (Fig. X). Several datasets combine multiple concepts or do not
# explicitly state their definition, indicating substantial conceptual heterogeneity
# across the ensemble.
#
# If definitional differences were the primary source of disagreement, datasets
# sharing the same conceptual definition should exhibit higher spatial agreement.
# However, this is not the case. Pairwise comparisons show that disagreement does
# not differ between dataset pairs that share the same irrigation target and those
# that do not (Wilcoxon test, p = 0.97), indicating that datasets aligned along
# several conceptual attributes do not agree more on where irrigation exists.
#
# We further tested whether definitional harmonisation reduces disagreement at the
# ensemble level by restricting the analysis to subsets of datasets sharing similar
# definitions. Although disagreement appears lower within these subsets, this
# reduction is entirely explained by the smaller number of datasets involved. When
# compared against randomly sampled subsets of equal size, definition-consistent
# subsets do not exhibit lower disagreement (Fig. X), demonstrating that the
# apparent reduction is a statistical artefact rather than a consequence of
# conceptual alignment.
#
# A multivariate regression including all definitional dimensions confirms these
# results. None of the variables significantly explains pairwise disagreement (all p > 0.18)
# and together they account for less than 8% of its variance (adjusted R2<0). Thus,
# even when considered jointly, differences in irrigation definitions fail to explain
# the widespread inconsistencies across datasets.
#
# Taken together, these results show that the detectability crisis cannot be attributed
# to definitional heterogeneity. Datasets do not disagree because they represent
# different types of irrigation; they disagree because they do not identify the
# same locations as irrigated.
```

8 Assessment of non-identifiability

```r
# INSTABILITY UNDER SMALL PERTURBATIONS ########################################

# Fron long to wide dataset ---------------------------------------------------

dt_wide <- dcast(dt, lon + lat + country + code + continent +
                   resolution ~ dataset, value.var = "mha")

# Run analysis -----------------------------------------------------------------
# 1% and 2% of the grid cell
```

```r
inst_res <- run_instability_analysis(dt = dt_wide, dataset_names = dataset_names,
                                      tau_vals = c(1, 2), include_full = TRUE,
                                      include_loo = TRUE)

instability_dt <- inst_res$instability
classification_dt <- inst_res$classifications

instability_sum_all <- summarize_instability(instability_dt)
instability_sum_rel <- summarize_instability_relevant(instability_dt)

print(instability_sum_all)
```

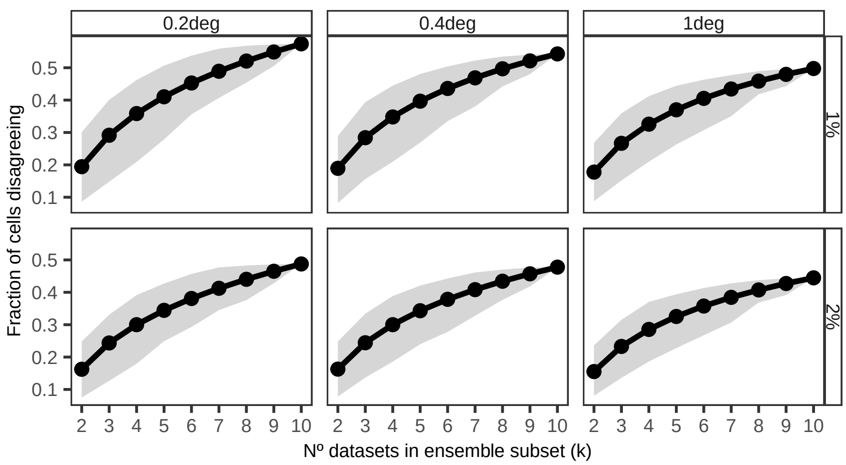
```
##    resolution n_cells frac_always_absent frac_always_present frac_unstable
##        <char>   <int>              <num>               <num>         <num>
## 1:     0.2deg  129628          0.3548770           0.3902243     0.2548986
## 2:     0.4deg   39758          0.3156597           0.4543237     0.2300166
## 3:       1deg    8068          0.2791274           0.5249132     0.1959593
##    mean_flip_rate median_flip_rate q05_flip_rate q25_flip_rate q75_flip_rate
##             <num>            <num>         <num>         <num>         <num>
## 1:     0.05682169                0             0             0    0.04545455
## 2:     0.05049070                0             0             0    0.00000000
## 3:     0.04826024                0             0             0    0.00000000
##    q95_flip_rate
##            <num>
## 1:     0.4545455
## 2:     0.4545455
## 3:     0.4545455
```

```r
print(instability_sum_rel)
```

```
##    resolution n_cells frac_unstable mean_flip_rate median_flip_rate
##        <char>   <int>         <num>          <num>            <num>
## 1:     0.2deg   83626     0.3951164     0.08807885                0
## 2:     0.4deg   27208     0.3361144     0.07378010                0
## 3:       1deg    5816     0.2718363     0.06694698                0
##    q05_flip_rate q25_flip_rate q75_flip_rate q95_flip_rate
##            <num>         <num>         <num>         <num>
## 1:             0             0    0.09090909     0.4545455
## 2:             0             0    0.04545455     0.4545455
## 3:             0             0    0.04545455     0.5000000
```

```r
# CONVERGENCE TEST #############################################################

# Run on exact combinations ----------------------------------------------------

tau_vals <- c(1, 2)

res_exact_all <- rbindlist(lapply(tau_vals, function(tt) {
```

```
  run_by_resolution(dt = dt_wide, dataset_names = dataset_names,
                    tau_percent = tt, mode = "exact")
}), fill = TRUE)

res_exact_all_sum <- summarize_agreement_vs_information(res_exact_all)

# Prettier labels-------------------------------------------------------------

res_exact_all_sum[, tau_percent:= paste(tau_percent, "%", sep = "")]

# Plot results ----------------------------------------------------------------

p_disagree_exact <- ggplot(res_exact_all_sum, aes(k, frac_disagree_mean)) +
  geom_ribbon(aes(ymin = frac_disagree_q05, ymax = frac_disagree_q95), alpha = 0.2) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_grid(tau_percent~ resolution) +
  scale_x_continuous(breaks = 2:length(dataset_names)) +
  labs(x = "Nº datasets in ensemble subset (k)",
       y = "Fraction of cells disagreeing") +
  theme_AP()

p_disagree_exact
```



9 Session information

```r
# SESSION INFORMATION ##########################################################

sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framew
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] parallel  stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] benchmarkme_1.0.8   wesanderson_0.3.7   countrycode_1.6.1
##  [4] sf_1.1-0            rnaturalearth_1.1.0 terra_1.8-93
##  [7] here_1.0.2          sensobol_1.1.6      lubridate_1.9.4
## [10] forcats_1.0.1       stringr_1.6.0       dplyr_1.1.4
## [13] purrr_1.2.1         readr_2.2.0         tidyr_1.3.2
## [16] tibble_3.3.1        ggplot2_4.0.1.9000  tidyverse_2.0.0
## [19] cowplot_1.2.0.9000  scales_1.4.0        data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1       viridisLite_0.4.2      farver_2.1.2
##  [4] S7_0.2.1               fastmap_1.2.0          digest_0.6.39
##  [7] timechange_0.3.0       lifecycle_1.0.5        magrittr_2.0.4
## [10] compiler_4.5.2         rlang_1.1.7            tools_4.5.2
## [13] yaml_2.3.12            knitr_1.51             labeling_0.4.3
## [16] sp_2.2-1               classInt_0.4-11        RColorBrewer_1.1-3
## [19] KernSmooth_2.23-26     withr_3.0.2            grid_4.5.2
## [22] e1071_1.7-17           iterators_1.0.14       tinytex_0.58
## [25] cli_3.6.5              rmarkdown_2.30         generics_0.1.4
## [28] otel_0.2.0             RcppParallel_5.1.11-1  rstudioapi_0.17.1
## [31] httr_1.4.8             tzdb_0.5.0             DBI_1.3.0
## [34] proxy_0.4-29           s2_1.1.9               vctrs_0.7.1
## [37] boot_1.3-32            Matrix_1.7-4           spData_2.3.4
## [40] hms_1.1.4              foreach_1.5.2          units_1.0-0
```

```
## [43] glue_1.8.0              benchmarkmeData_1.0.4   codetools_0.2-20
## [46] rngWELL_0.10-10         stringi_1.8.7           gtable_0.3.6
## [49] randtoolbox_2.0.5       pillar_1.11.1           htmltools_0.5.9
## [52] rnaturalearthdata_1.0.0 R6_2.6.1                zigg_0.0.2
## [55] wk_0.9.5                Rdpack_2.6.4            doParallel_1.0.17
## [58] rprojroot_2.1.1         evaluate_1.0.5          lattice_0.22-7
## [61] rbibutils_2.4           Rfast_2.1.5.2           class_7.3-23
## [64] Rcpp_1.1.1              xfun_0.55               pkgconfig_2.0.3
```

```r
## Return the machine CPU ---------------------------------------------------

cat("Machine:     "); print(get_cpu()$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```r
## Return number of true cores ----------------------------------------------

cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```r
## Return number of threads -------------------------------------------------

cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```